LOCOMOTOR SYSTEM 1

Liu, ili

LOCOMOTOR SYSTEM

- The abaxial part includes the anatomical structures of the extremities: bones (including diaphyses, epiphyses and apophyses, growth plates, metaphyses, sesamoid bones), joints (including menisci), tendons, ligaments, and muscles. This part of the locomotor system provides support and protection, and offers the possibility of standing and moving around.

- The axial part includes the vertebral bodies, vertebral joints, intervertebral discs, ligaments, tendons, and muscles. The vertebral column protects the spinal cord and forms a connection with the caudal extremities via the iliosacral joints.

- History
- Observation (stance and gait)
- Physical examination(ROM, muscle volume)
- General palpation
- Orthopedic examination
- Image
- Joint fluid examination

骨外科學檢查表

病例號 _____ 畜種_____品種_____.
名字_____ 年齡_____性別_____.

主訴：_____ 原因_____.

一般狀態：精神(有•無) 食慾(有•無) 發燒(有•無) 嘔吐(有•無) 下痢(有•無) BCS(　　)

症狀的經過：急性 • 慢性 • 間斷性 • 緩慢進行性 • 其他(　　)

現在用藥：NSAID(　　) •類固醇(　　) •軟骨保護劑(　　) •其他(　　)

跛腳發現的時間：經常 • 坐下欲站立時 • 早晨 • 運動後 • 休息後 • 其他(　　)

疼痛指數（Visual Analog Scale: VAS）



關節可動區域(Range of Motion: ROM)

前肢	ROM				後肢	ROM			
	伸展		彎曲			伸展		彎曲	
	L	R	L	R		L	R	L	R
肩關節					股關節				
肘關節					膝關節				
腕關節					跗關節				

記入角度
or
O：正常
↑：增大
↓：減少

觸診

前肢

所見	L	R
肢端的傷口•皮膚炎•化膿	有•無	有•無
腕關節的可動性異常	有•無	有•無
腕關節的可動區域限制	有•無	有•無
前腕的變形	有•無	有•無
前腕的疼痛	有•無	有•無
肘關節內側側副韌帶損傷	有•無	有•無
肘關節外側側副韌帶損傷	有•無	有•無
肘關節伸展時疼痛	有•無	有•無
內側喙狀突起的壓痛	有•無	有•無
肘關節的關節可動區域限制	有•無	有•無
二頭筋腱試驗	+ • −	+ • −
肩關節伸展時的疼痛	有•無	有•無
肩關節外轉的角度(>50°)	+ • −	+ • −

（其他發現）

後肢

所見	L	R
肢端的傷口•皮膚炎•化膿	有•無	有•無
跗關節的可動性異常	有•無	有•無
跗關節的可動區域限制	有•無	有•無
阿基里斯腱損傷•移位	有•無	有•無
膝蓋骨脫臼	有•無	有•無
膝蓋骨脫臼的方向	內•外	內•外
膝蓋骨脫臼的Grade	(　)	(　)
膝關節外側側副韌帶損傷	有•無	有•無
膝關節內側側副韌帶損傷	有•無	有•無
脛骨Cranial Drawer Test	+ • −	+ • −
脛骨Tibial Compression Test	+ • −	+ • −
膝關節屈伸時的click音	+ • −	+ • −
Barden Test	+ • −	+ • −
Ortolani Test	+ • −	+ • −
股關節伸展時的疼痛	有•無	有•無
Thumb Pinch Test •三角試驗	+ • −	+ • −

HISTORY , OBSERVATION AND PHYSICAL EXAMINATION OF THE LOCOMOTOR SYSTEM

- A functional diagnosis (mechanical limitation, painful process)
- An anatomical diagnosis (location of the disease process)
- Insight into the pathogenesis (congenital abnormality, developmental disorder, trauma, inflammation, abnormality of other organ systems, neoplasia)

HISTORY

- When the problem start. (e.g., mainly after animal rises from its bed, or when it runs off, or while walking on a gravel path, or when jumping over a fence)

- An accurate description of the progression of the symptoms during the day (lame when getting up from bed, better during walking, worse after excessive exercise—versus an increase in severity of signs during a walk)

HISTORY

- 跛行發生時間
  - DJD (Degenerative joint disease)：休息後的跛行，早上或濕氣重的壞天氣時跛行症狀容易惡化
  - 免疫性關節疾病：中午前跛行症狀較嚴重，中午過後症狀稍微改善

HISTORY: Progression of the problem

- Permanently present (always equally severe)
- Recurrently present (periods of recovery and then recurrence of problems)
- Intermittently present (alternating lameness and normal locomotion)
- Progressive (worsening)
- Migrating (changing from place to place or leg to leg)

HISTORY

- 症狀發生的經過
  - 急性：骨折，韌帶，肌腱的斷裂; 慢性：繼發性DJD
  - 免疫性關節疾病：發燒，對稱性關節腫脹，關節過度彎曲或過度伸展
  - 膝蓋骨脫臼：跛行狀況時好時壞

HISTORY

- Age, breed, inherited disease
- Living condition (guide dog, guard dog, police dog, house pet)
- Walking time and distance
- Drug intake (NSAID, steroid)

TABLE 1-1. Causes of Lameness in the Dog (Excluding Fractures and Minor Soft Tissue Injuries)

Pelvic Limb	Forelimb
GROWING DOG	
1. Hip dysplasia	1. OCD—shoulder
2. Avascular necrosis (Legg-Calvé-Perthes)	2. Luxation/subluxation shoulder—congenital
3. Avulsion of long digital extensor	3. Avulsion supraglenoid tubercle
4. OCD—stifle	4. OCD—elbow
5. OCD—hock	5. UAP
6. Luxating patella complex	6. FCP
7. Genu valgum	7. UME
8. Panosteitis	8. Elbow incongruity
Medium to large breeds = 1, 3-8	a. Congenital
Toy to small breeds =1, 2, 6	b. Physeal injury
Chondrodystrophied breeds = 1, 2, 6, 8	9. Radius curvus
	10. Retained cartilaginous cores (ulna)
	11. Panosteitis
	12. HOD
	Medium to large breeds = 1, 4-7, 8b, 9-12
	Toy to small breeds = 2, 8, 9
	Chondrodystrophied breeds = 5, 8a, 8b, 9, 11, 13
ADULT DOG	
A. Arthritis (or continuum): 1-7	A. Arthritis (or continuum): 1-6, 8, 9
B. Luxating patella complex	B. UME
C. Panosteitis	C. Panosteitis
D. Cruciate/meniscal syndrome	D. Bicipital tenosynovitis/biceps rupture
E. Inflammatory joint disease	E. Calcification of supraspinatus tendon
F. Neoplasia	F. Contracture of infraspinatus or supraspinatus
Medium to large breeds = A1, A 3-7, B, D, F	G. Bone/soft tissue neoplasia
Toy to small breeds = A2, B, D-F	H. Luxation/subluxation—shoulder
Chondrodystrophied breeds = A1, B, D-F	I. Inflammatory joint disease
	J. HO
	Medium to giant breeds = A, 7, 11, I, J
	Toy to small breeds = 2, G, H, I, J
	Chondrodystrophied breeds = A-5, A-8, A-9, C, H, I, J

OCD, Osteochondritis dissecans; *UAP,* ununited anconeal process; *FCP,* fragmented coronoid process; *UME,* ununited medial epicondyle; *HOD,* hypertrophic osteodystrophy; *HO,* hypertrophic osteopathy

Handbook of small animal orthopedics and fracture repair, 2016

FORE LIMB

- 成長期
  - 肩關節分離性骨軟骨炎(OCD)，肘關節發育不良（大型犬），橈尺骨遠端骨折（小型犬），先天性肩關節脫臼，先天性肘關節脫臼，尺骨遠端成長板早期閉鎖
- 中老年期
  - 繼發性DJD，免疫性關節疾病（風濕性關節炎，全身性紅斑狼瘡SLE），腫瘤等

PELVIC LIMB

- 成長期
  - 股關節發育不良(CHD)（大型犬，兩側），股骨頭壞死(LEGG-CALVE-PERTHES)（小型犬，單側），膝蓋骨脫臼（小型犬，內側脫臼；大型犬，外側脫臼），大腿骨遠端成長板骨折
- 中老年期
  - 前十字韌帶斷裂， 免疫性關節疾病，繼發性DJD，腫瘤等

表1-1 シグナルメントと前肢跛行から考えられる鑑別診断リスト





前肢

	若齢犬	成犬
小型犬	橈尺骨遠位骨折 上腕骨遠位骨折 その他の外傷 その他，稀に認められる状態 先天性肘関節脱臼 先天性肩関節脱臼 成長異常	脊髄疾患 外傷 免疫介在性多発性関節炎 上腕骨遠位骨折 その他，稀に認められる状態 肩関節不安定症
中大型犬	肘関節形成不全 肩関節離断性骨軟骨症（OCD） 汎骨炎 手根関節過伸展 前腕成長異常 種子骨損傷 腋窩神経叢損傷 その他の外傷 その他，稀に認められる状態 外傷性肘関節脱臼 肥大性骨異栄養症	肘関節骨関節症 骨腫瘍 二頭筋腱炎 棘上筋腱石灰化症 脊髄疾患 免疫介在性多発性関節炎 種子骨損傷 その他の外傷 その他，稀に認められる状態 外傷性肘関節脱臼 腋窩神経叢腫瘍 肥大性骨症 肩関節不安定症

＊前肢跛行を呈する病態を一般的に多くみられる順に示す。

犬の跛行診断,2016

表1-2 シグナルメントと後肢跛行から考えられる鑑別診断リスト

後肢

小型犬



若齢犬

膝蓋骨脱臼
大腿骨頭壊死
外傷

その他，稀に認められる状態

脊髄疾患（先天的奇形）
成長異常

成犬

膝蓋骨脱臼
前十字靭帯疾患
脊髄疾患
底側靭帯損傷
骨腫瘍
股関節脱臼
股関節骨関節症

その他，稀に認められる状態

浅趾屈筋腱脱臼
免疫介在性多発性関節炎

中大型犬



若齢犬

股関節形成不全
足根関節離断性骨軟骨症（OCD）
汎骨炎
前十字靭帯疾患（裂離骨折）
膝蓋骨脱臼
股関節脱臼
その他の外傷

その他，稀に認められる状態

膝関節離断性骨軟骨症（OCD）
大腿四頭筋拘縮
肥大性骨異栄養症
成長異常

成犬

前十字靭帯疾患
股関節骨関節症
アキレス腱断裂
骨腫瘍
脊髄疾患
膝蓋骨脱臼
免疫介在性多発性関節炎
股関節脱臼
その他の外傷

その他，稀に認められる状態

種子骨損傷
浅趾屈筋腱脱臼
骨以外の腫瘍
長趾伸筋腱断裂
腸腰筋損傷

＊後肢跛行を呈する病態を一般的に多くみられる順に示す。

犬の跛行診断,2016

診察整形外科病症時的問診項目

·品種與年齡　·性別　·過去是否有重大疾病或受傷的病史

·同血脈中是否有同樣的病例　·症狀開始的原因

·目前出現的症狀　·步行時的異狀　·是否有精神與食慾

·目前的飲食內容　·是否有發熱、嘔吐或下痢

·症狀是急性或是慢性 ·散步的時間與距離長短是否有改變

·步行於階梯或坡道時是否有異狀 ·關節是否有腫脹或疼痛現象

·四肢是否有變形 ·症狀是否逐漸惡化

·一天當中哪個時段症狀會惡化　·休息後與起身時是否疼痛

·運動後症狀是否惡化　·天氣是否影響症狀變化

·使用中的藥物與營養品

·NSAID與Steroid當中哪一項會使症狀改善

·服用藥物後症狀是否改變　·家中是否鋪設地板

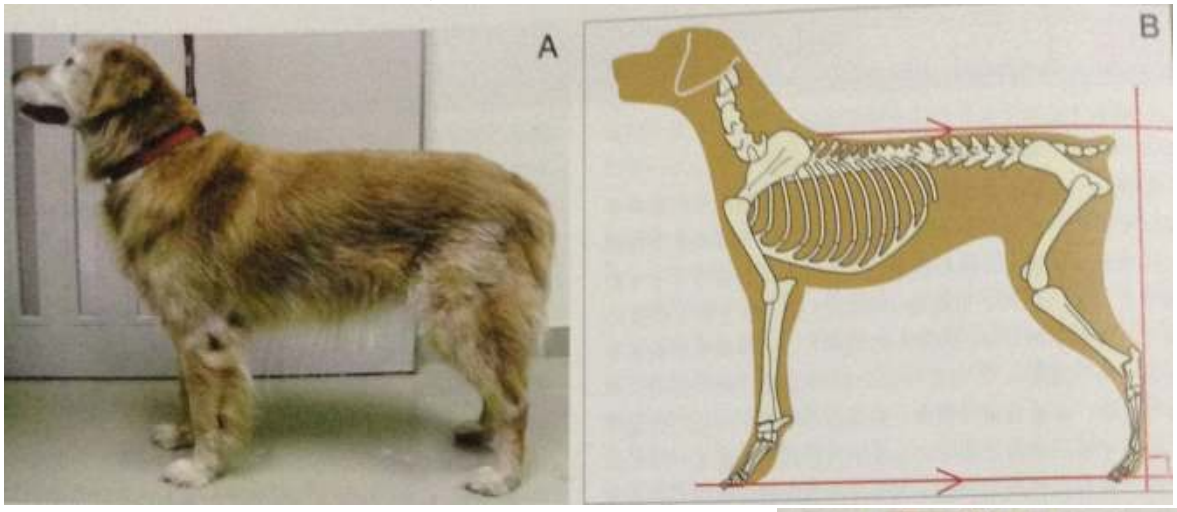
OBSERVATION OF STANCE

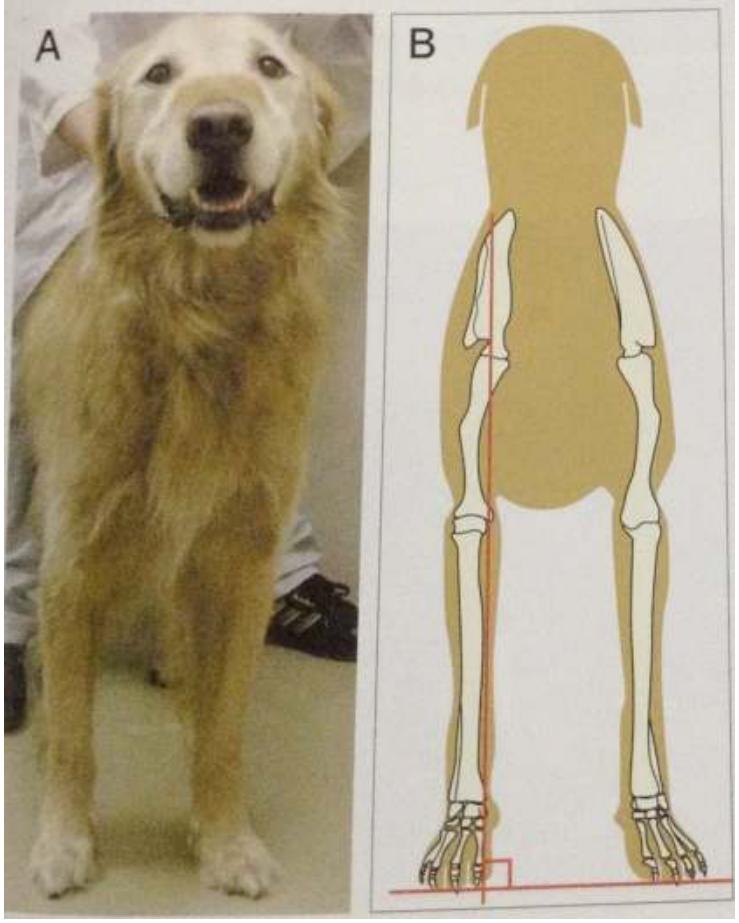
The stance and the weight bearing of the extremities, the posture of the head, neck, and vertebral column

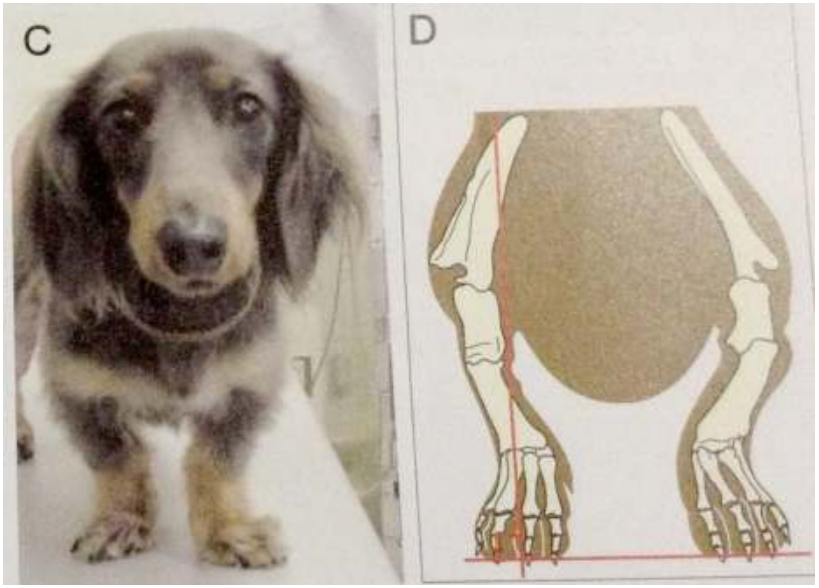
- The stance of the limbs
- The relative positions of the limbs, head, and vertebral column
- The bearing of weight on the limbs

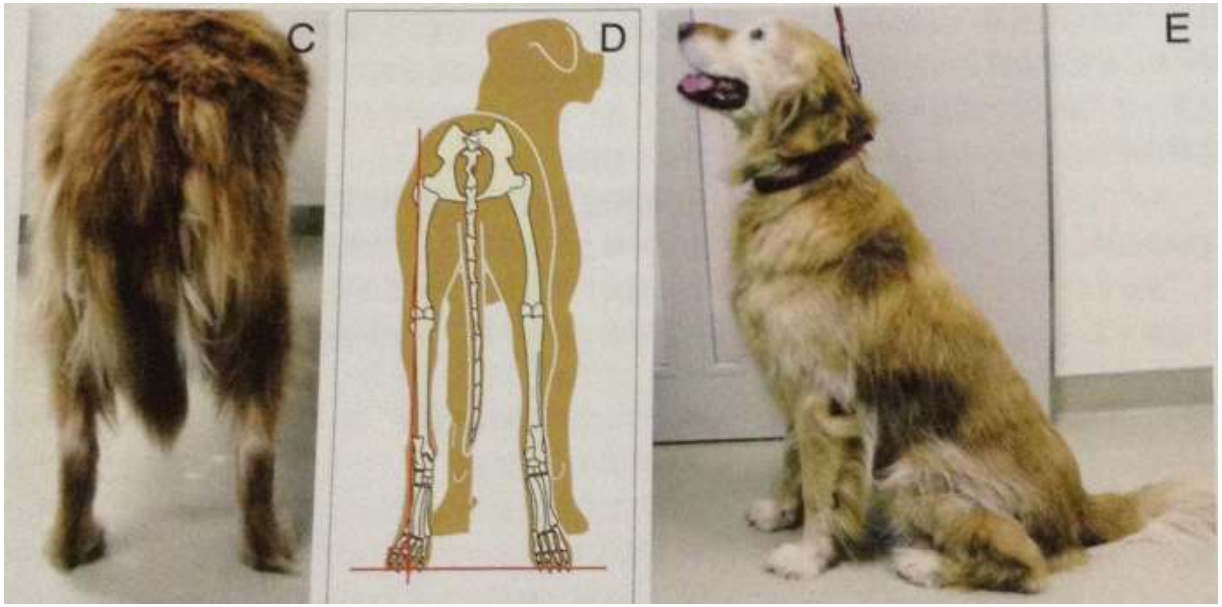
視診

- 姿勢的觀察







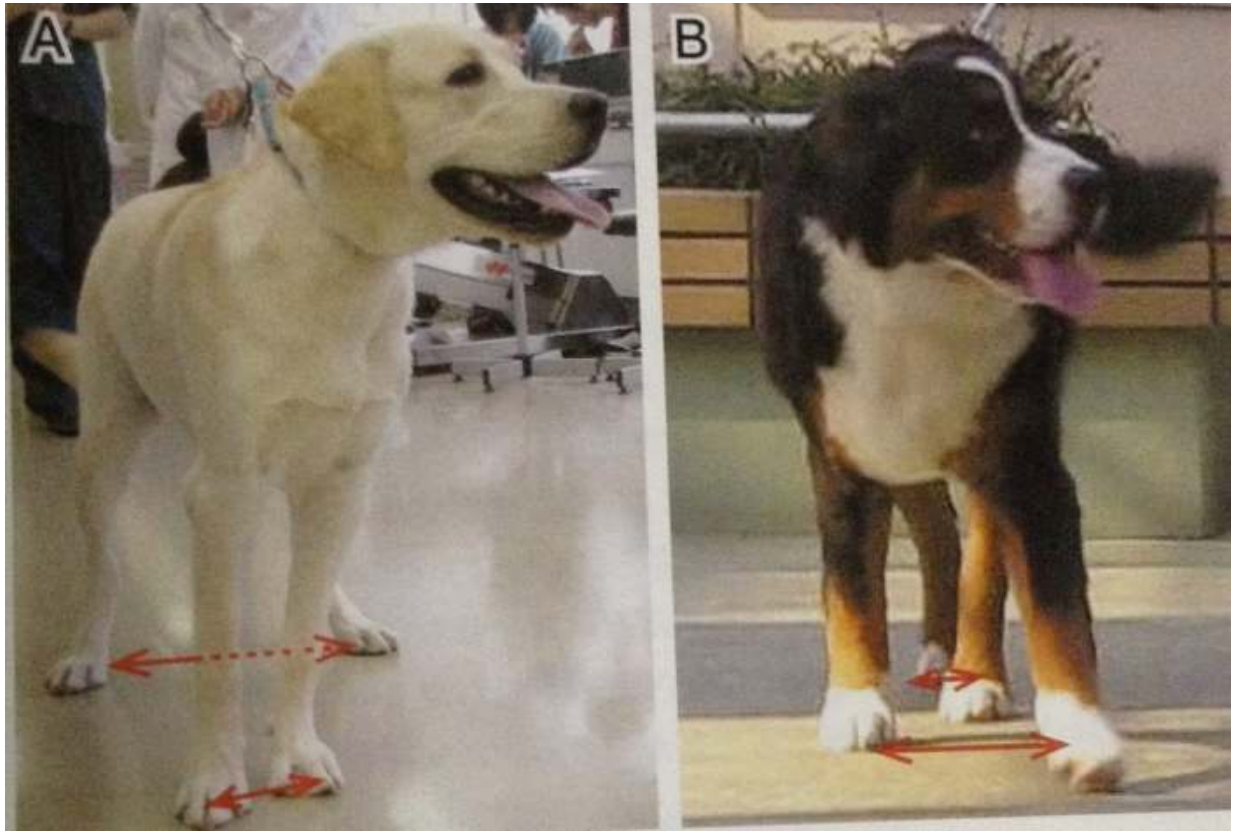


神経学的検査と整形外科学的検査のコツ,2011

視診

- 急性，患肢抬起 ：骨折，脱臼，靭帯斷裂

- 慢性：患肢肌肉量減少（單側），前肢兩側性的成長期疾病或靭帯損傷（前肢的張開幅度小於後肢），後肢兩側性疾病則相反



神経学的検査と整形外科学的検査のコツ,2011

1. 全身状態の評価

患者を注意深く観察し，整形外科学的異常（跛行）なのか，あるいは全身性の虚弱や神経学的異常（運動失調）なのかを鑑別することが重要である。

2. 立位観察

全身状態の評価後，以下の項目に注意して立位の状態を観察し，可能であれば異常肢を特定する。

立位観察のポイント

- 負重の程度
- 奇形（内反，外反など）
- 左右対称性（形態異常や腫脹の有無など）
- 四肢の長さ
- 前肢間，後肢間の幅に違いがないかどうか（図1-1，図1-2）
- 外傷の有無
- ナックリングの有無



図1-1　立位観察（正常例）



図1-2　立位観察（異常例）

犬の跛行診断,2016

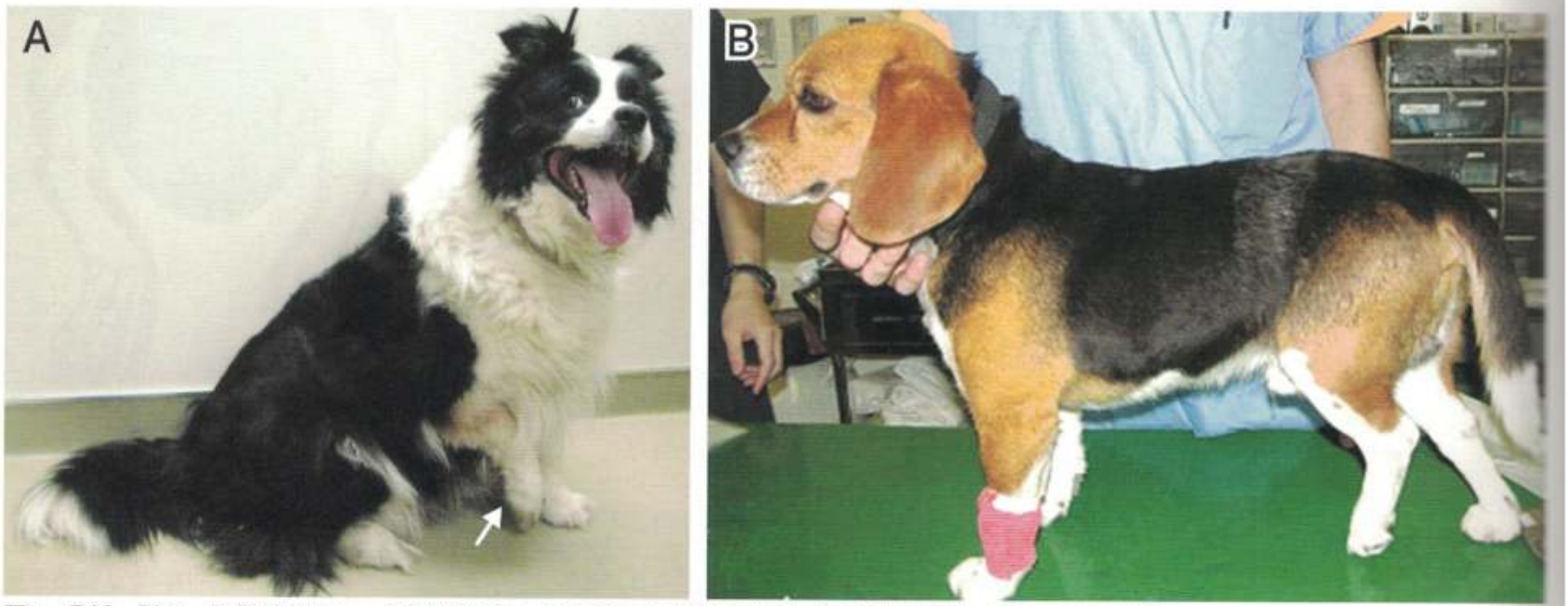


図9　骨折、脱臼、靱帯断裂といった急性例では、起立時および座らせたときに患肢を挙上していることが多い。
A：外傷性靱帯断裂（内側手根側副靱帯断裂）、B：外傷性脱臼（外傷性股関節脱臼）

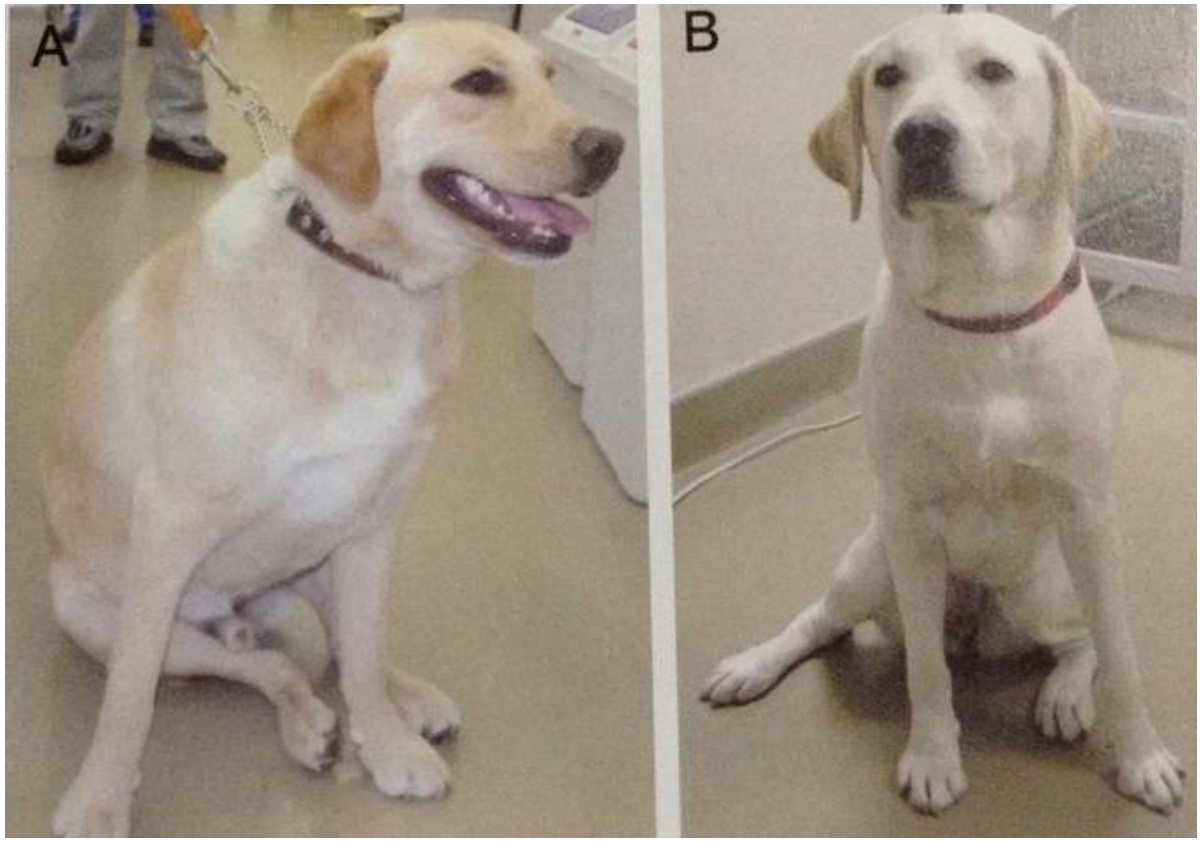


神経学的検査と整形外科学的検査のコツ,2011

視診



- 腕關節過度伸展
- 跗關節過度彎曲
- 股關節發育不全的坐姿
- 膝關節腫脹造成後肢無法彎曲的坐姿



神経学的検査と整形外科学的検査のコツ,2011

CHARACTERISTIC ABNORMALITIES OF THE STANCE

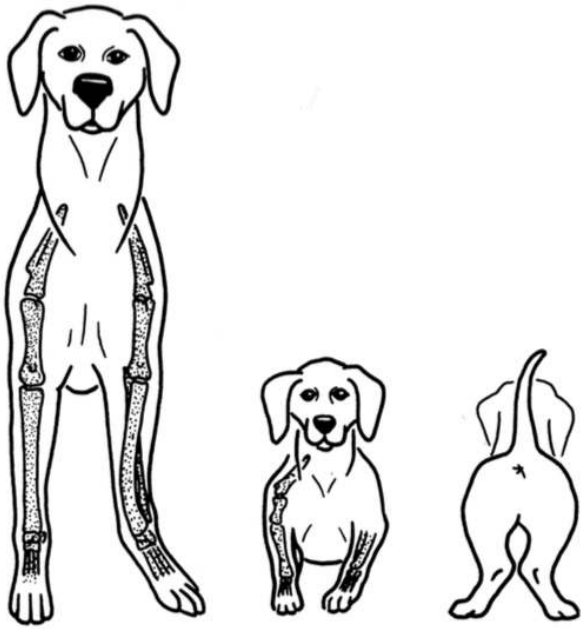


Fig. 17.1 The Labrador and the dachshund illustrate breed differences in the structure and shape of the front leg, as shown by the right front leg in this figure. The stance of the left front leg of each animal is abnormal: the Labrador's leg is in valgus position and the leg of the dachshund is in varus position. Both rear legs of the dachshund are in a French stance due to exorotation.

- **Varus**: an angular deviation of part of the leg toward the median line, in the sagittal plane
- **Valgus**: an angular deviation of part of the leg away from the median line, in the sagittal plane
- The abnormalities are named according to the **origin** of the deviation: in animals whose hind legs form an X there is varus from the hips and valgus from the knees, and in those whose hind legs form an O there is valgus from the hips and varus from the knees.

表3-1　膝蓋骨脱臼のグレーディング・システムの概要

グレード	I	II	III	IV
身体検査	強制的に脱臼可能	間欠的脱臼（自発的）	恒久的脱臼（整復可能）	恒久的脱臼（整復不可能）
臨床症状	正常～最小限の跛行	間欠的跛行	軽度～重度跛行	重度跛行（膝屈曲）
骨格異常	最小限	軽度	中等度	重度

▼重度の膝蓋骨脱臼に特徴的な姿勢



図3-42　重度の膝蓋骨内方脱臼（側方より見たところ）



図3-43　重度の膝蓋骨内方脱臼（尾側より見たところ）



犬の跛行診断,2016

- **Torsion**: twisting of a part of the skeleton or a bone on its long axis
  - **endotorsion**: turning inward (medially) of the distal part of a **long bone**
  - **exotorsion**: turning outward (laterally) of the distal part of a **long bone**

- **Antecurvatum**: anterior bowing
- **Recurvatum**: posterior bowing



獸醫放射診斷學,2014

- There are a few characteristic abnormalities in the **positioning** of the limb relative to the body which affect the way in which the animal stands:
- **Rotation**: turning of a **limb** in a joint on its long axis
    - **exorotation**: lateral turning of a limb or part of a limb
    - **endorotatio**n: medial turning of a limb or part of a limb
- **Abduction**: entire limb moved sideways away from the median line
- **Adduction**: entire limb moved sideways toward the median line

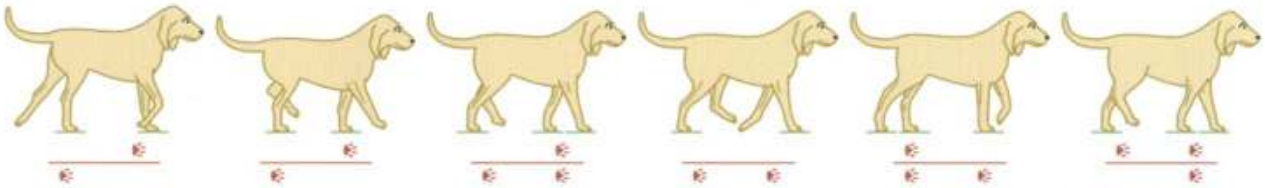
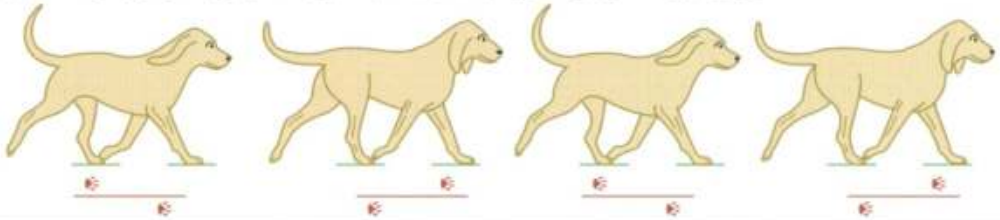
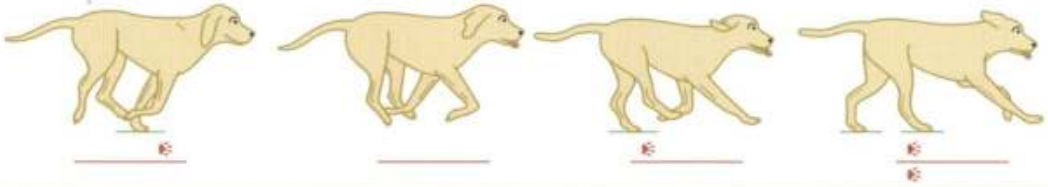
- Some abnormal forms of **weight bearing by limbs** are:
    - **Hyperextension**: overstretching of joint(s)
    - **Hyperflexion**: overflexion (bending) of joint(s)



步行檢查

- 不滑的地板或水泥地
- 可步行10步以上，8~12公尺的距離
- 負重期(STANCE PHASE)，擺動期(SWING PHASE)

表1　犬の歩行様式の分類

種類(英名)	歩行形態
常歩 (Walk)	【通常の歩行】　歩行時に1肢または2肢が挙上する程度の歩行 
速歩 (Trot)	【ジョギング程度の歩行】　歩行時に対角の2肢が挙上する程度の歩行：斜角速歩 
襲歩 (Gallop)	【全力疾走】　すべての肢が宙に浮く瞬間がある走行 

犬の歩様力学（鳳鳴堂書店、2002）の図

神経学的検査と整形外科学的検査のコツ,2011

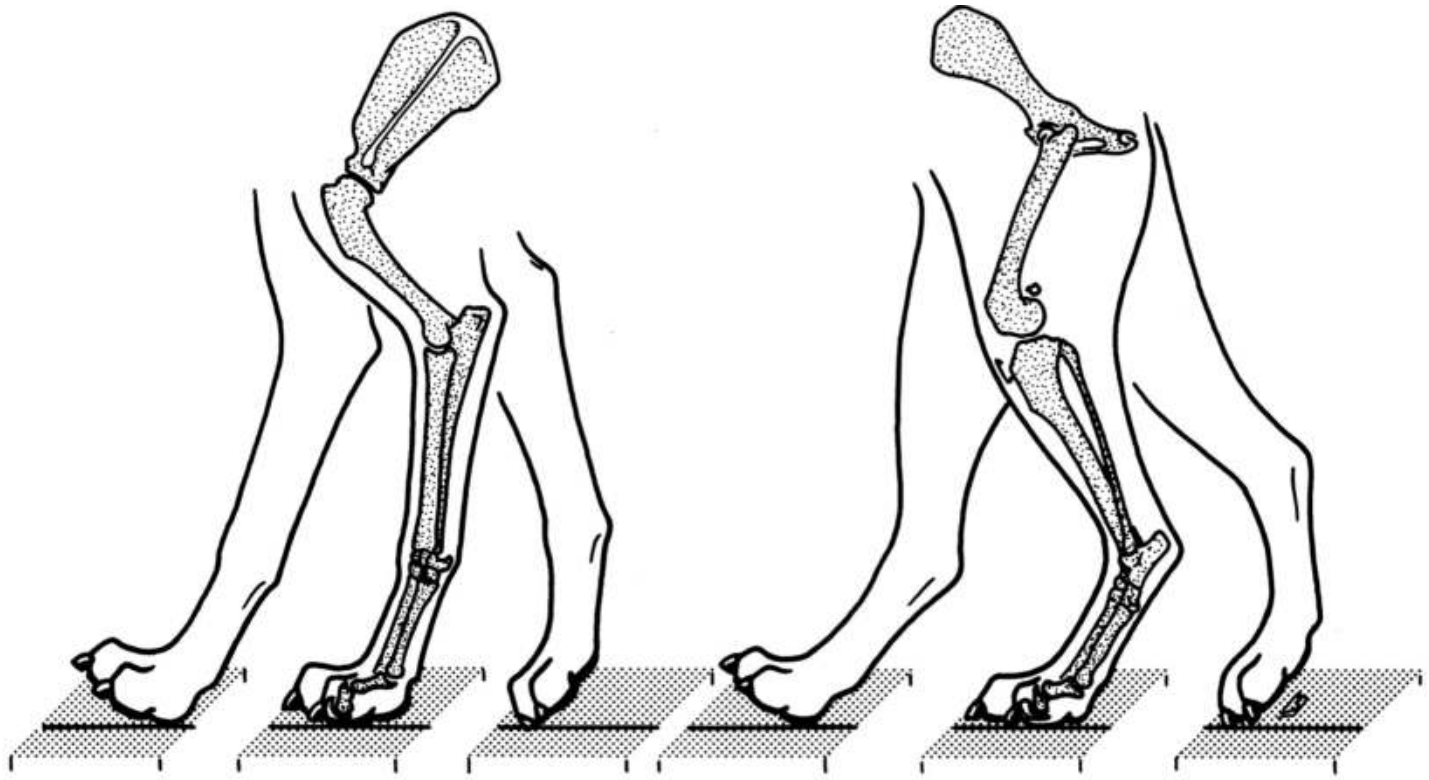


Fig. 17.2 The support phase of both the front leg and the rear leg is from the contact point to the takeoff point.

- Stride：Contact point,　Takeoff point
- Stance phase
- Swing phase

神経学的検査と整形外科学的検査のコツ,2011

LAMENESS SCALE: GRADE THE SEVERITY OF THE LOCOMOTOR DISTURBANCE

- Grade I : barely disturbed locomotion
- Grade II: locomotion disturbed but limb(s) still bearing weight
- Grade III: lameness with limb(s) not always bearing weight
- Grade IV: no weight bearing on limb(s)

步行檢查

- 直線步行：
  - 散步和快步，患肢負重期時間較短，
  - 前肢異常：點頭運動，上下晃動程度由大到小為肩關節>肘關節>腕關節
  - 後肢異常：腰擺動步行和兔子跳步行，出現在成長期動物
- 圓周步行：左右繞圈散步，患肢在圓心處時會比較疼痛
- 特殊步行解析：Force plate gate analysis, Active-ROM

- In order to reduce the load on a **painful front limb**, the animal will **move the center of gravity caudally by raising its head during the weight-bearing phase on the affected limb**. When weight is borne on the unaffected limb, the head will be lowered: ‘falling on the healthy limb’

- To reduce the load on both front limbs, both rear limbs will be brought forward under the body, causing the back to be arched (kyphosis).



神経学的検査と整形外科学的検査のコツ,2011

Physical examination (ROM, muscle volume)

ROM測量

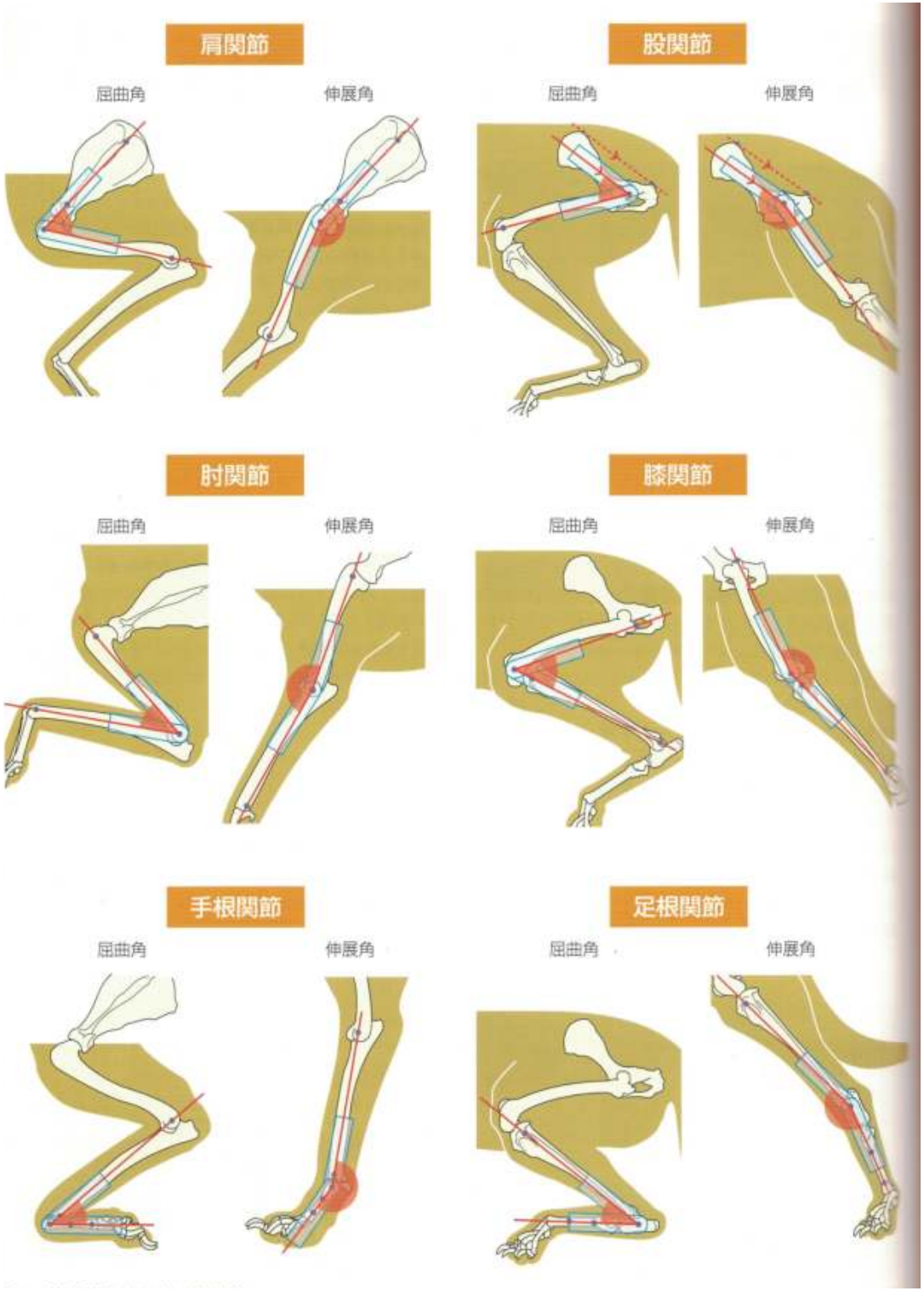
- ROM:兩側互相比較，測3次之平均值
- ROM增大：肌腱或韌帶斷裂
- ROM減少：DJD或關節疾病

表3　ラブラドール・レトリーバーの各関節のROMの平均値

関節	伸展角	屈曲角
手根関節	196 ± 2°	32 ± 2°
肘関節	165 ± 2°	36 ± 2°
肩関節	165 ± 2°	57 ± 2°
足根関節	164 ± 2°	39 ± 2°
膝関節	162 ± 3°	42 ± 2°
股関節	162 ± 3°	50 ± 2°

(Millis, PL. Canine Rehabilitation & Phisical Therapy, 2004)





神経学的検査と整形外科学的検査のコツ,2011

肌肉量測定

- 肌肉量測定：同一人於同一位置測量，可得知時間變化的結果，患肢的肌肉量低於正常肢
- 肢體的外徑





測量肢體部位
肱骨的遠端1/3處
橈骨的近端1/4處
股骨的近端1/4處
脛骨的近端1/4處

Gulick II



神経学的検査と整形外科学的検査のコツ,2011

觸診

- 一般觸診法：前後肢以及脊椎
- 兩側肢體同時進行，左右互相比較，由遠端肢體往體幹部觸診
- 後肢-脊椎-前肢



神経学的検査と整形外科学的検査のコツ,2011

ABNORMALITIES OF PELVIC LIMB

- Hock joint : fracture, luxation, tendon breakage, swelling, fever
- Tibia fracture or tibia varus with lateral patella luxation
- Stifle joint: anterior cruciate ligament rupture, OCD, hind leg form an O (varus from the knee), hind leg form an X (valgus from the knee), swelling
- Femur fracture, hip dysplasia, hip joint luxation

ABNORMALITIES OF FORELIMB

- Carpal joint: swelling, fracture, subluxation, hyperflexion
- Ulnar retained cartilaginous core
- Elbow joint: FMCP, UAP, OCD, fracture, swelling, tumor
- Humerus fracture, osteosarcoma
- Shoulder joint: OCD, luxation, bicipital tenosynovitis

骨科學觸診-前肢

前肢

所見	L	R
肢端的傷口•皮膚炎•化膿	有•無	有•無
腕關節的可動性異常	有•無	有•無
腕關節的可動區域限制	有•無	有•無
前腕的變形	有•無	有•無
前腕的疼痛	有•無	有•無
肘關節內側側副韌帶損傷	有•無	有•無
肘關節外側側副韌帶損傷	有•無	有•無
肘關節伸展時疼痛	有•無	有•無
內側喙狀突起的壓痛	有•無	有•無
肘關節的關節可動區域限制	有•無	有•無
二頭筋腱試驗	＋ • －	＋ • －
肩關節伸展時的疼痛	有•無	有•無
肩關節外轉的角度(＞50°)	＋ • －	＋ • －

CARPAL JOINT



図２　手根関節の解剖
手根関節の周囲には、関節の内・外側に走行している手根側副靭帯や各々の手根骨を連結している短靭帯などがあり、関節を安定化させている。内側手根側副靭帯は手根関節の外反、外側手根側副靭帯は手根関節の内反の動きを制限している

神経学的検査と整形外科学的検査のコツ,2011

- Medial and lateral collateral ligament
- Method: extension of carpal joint with abduction & adduction, ROM (flexion: pad can touch skin. extension: 180+10~20 degree)
- Check: hyperextension, luxation, pain, ligament rupture, reduced ROM

ELBOW JOINT

- ROM (flexion: space for pointer & middle fingers. extension: 170~180 degree)
- Traumatic ligament rupture (medial & lateral collateral ligament), pain, decreased ROM
- Campbell test: both elbow and carpus are flexed 90 degrees externally and internally rotate the carpus



図5　肘関節の屈曲可動域制限の確認方法：正常例
正常な犬や猫の肘関節を最大限に屈曲させたときには、前腕と肩関節の間に人差し指と中指がかろうじて挟まるくらいにまで屈曲させることができる

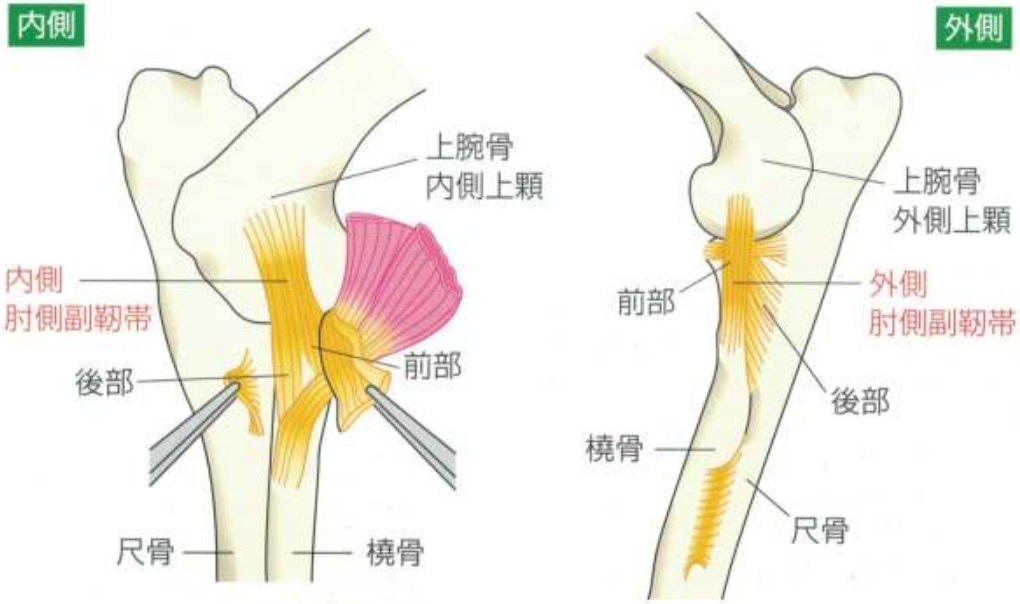


図3　内側肘側副靭帯と外側肘側副靭帯の走行

神経学的検査と整形外科学的検査のコツ,2011

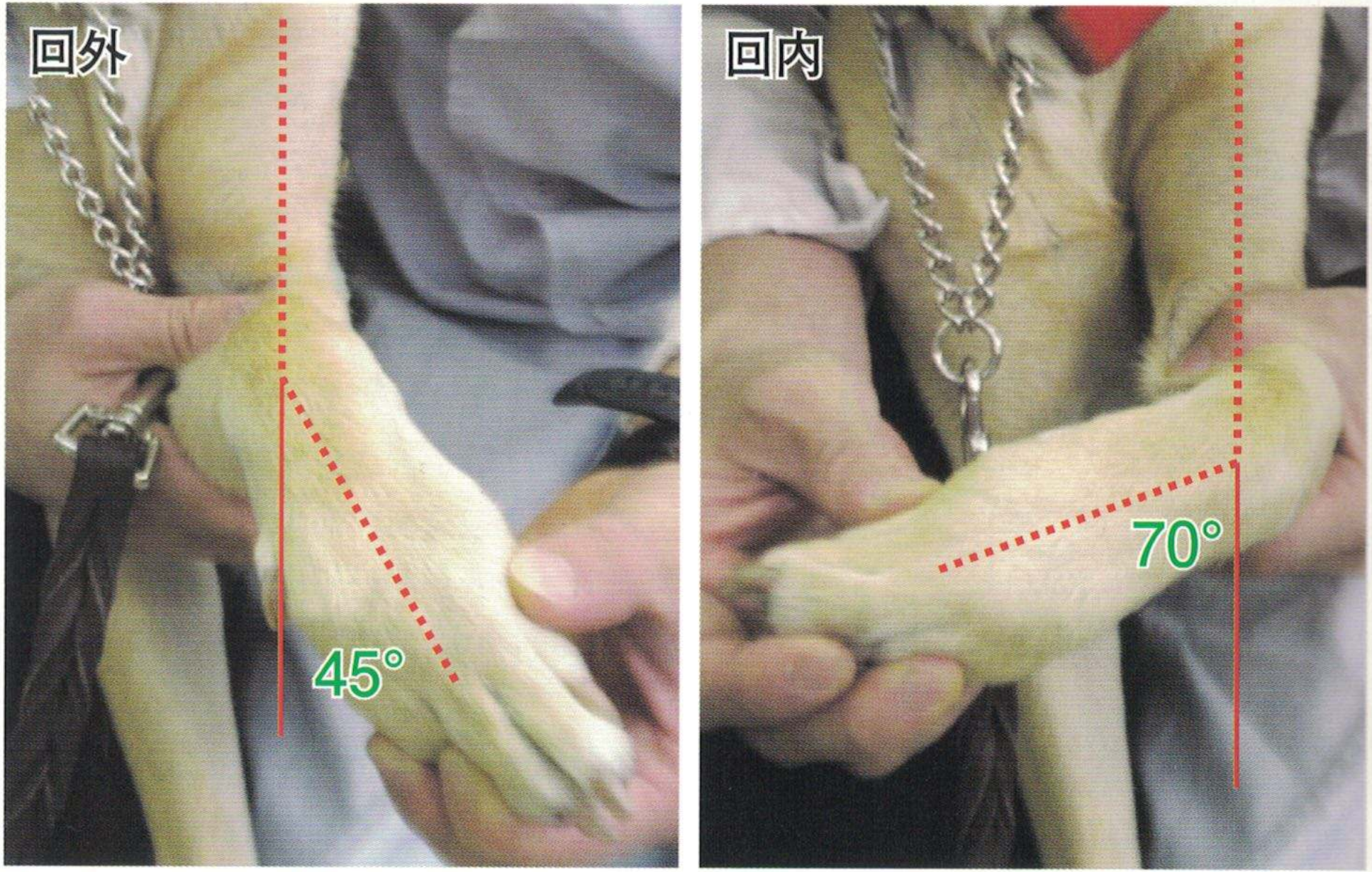


図4　Campbell法の評価方法。正常な犬で前腕を回外させたときには約45°程度回旋するが、明らかにその角度を超えて回外するときには内側肘側副靱帯損傷と診断する。正常な犬で前腕を回内させたときには約70°程度回旋するが、外側肘側副靱帯損傷の場合には約140°回旋することがある

神経学的検査と整形外科学的検査のコツ,2011

FMCP (FRAGMENTED MEDIAL CORONOID PROCESS)



図7　内側鉤状突起の圧痛の確認
肘関節を伸展および90°に屈曲させた状態で、内側鉤状突起の直上を圧迫して圧痛の有無を確認する

- Large breed, growing dog, 4~7 months, fore limb lameness
- Elbow joint with 90 degree, a small concave one finger wide distal to humeral medial condyle, compress this site with exorotation and endorotation
- CT or arthroscopy

神経学的検査と整形外科学的検査のコツ,2011



動画 11-8　内側鉤状突起の位置

肘関節の内側を触診し、最も突出している部位を探す。その部位が上腕骨の内側上顆である。上腕骨の内側上顆よりも一横指遠位に行ったところに小さな凹みを感じる。そこが、内側鉤状突起（赤い部分）の直上である

神経学的検査と整形外科学的検査のコツ,2011

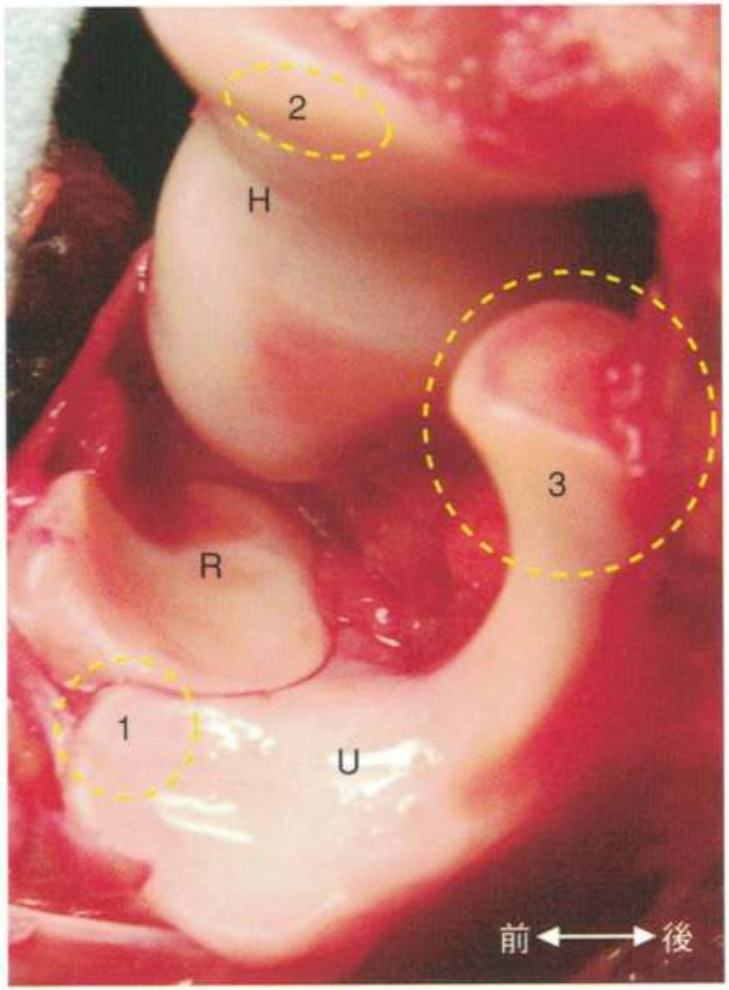


図2-73　肘関節の肉眼所見（内側から観察した関節面）

R：橈骨頭
U：尺骨
H：上腕骨顆
1：FCPの発症部位
2：OCDの発症部位
3：UAPの発症部位

犬の跛行診断,2016

OCD的影像及病灶位置

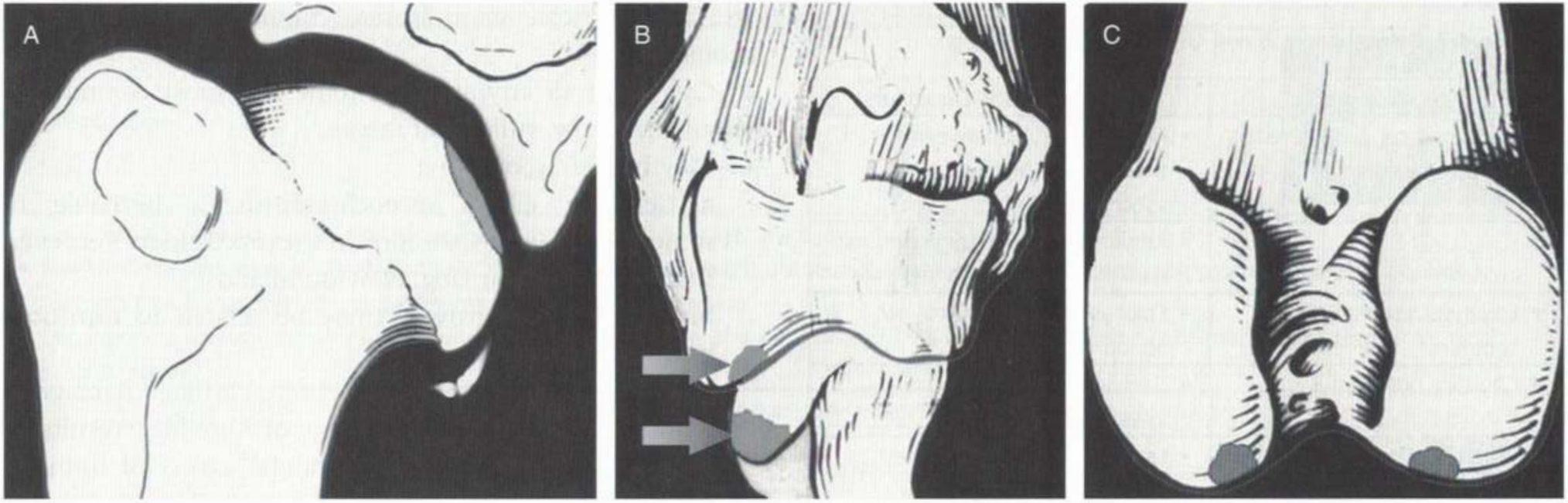


Figure 4.114 Osteochondritis dissecans; gray shaded areas depict common sites of occurrence. **A.** In shoulder on caudal aspect of humeral head (ossified free fragment is present ventral to humeral head). **B.** In elbow on medial aspect of humeral condyle or medial coronoid process (arrows). **C.** In stifle on femoral condyles.

BICIPITAL TENDON TEST

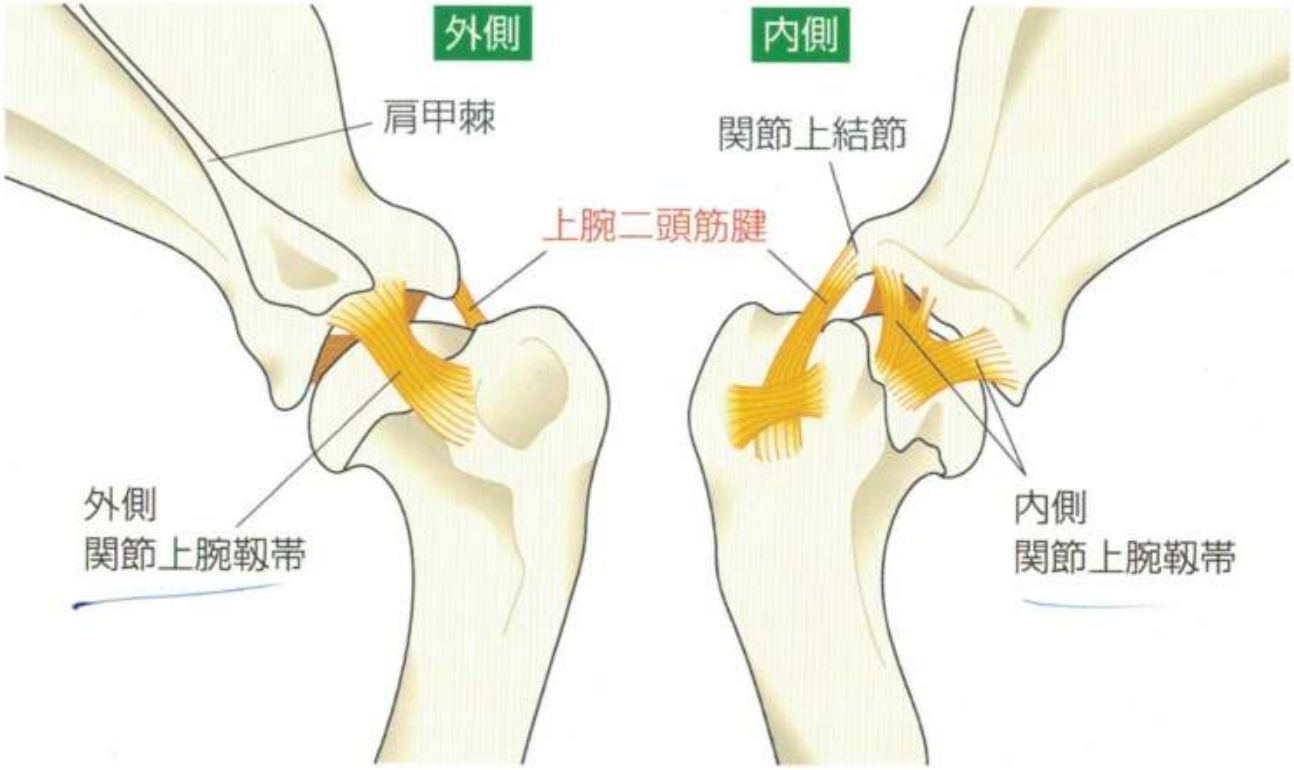
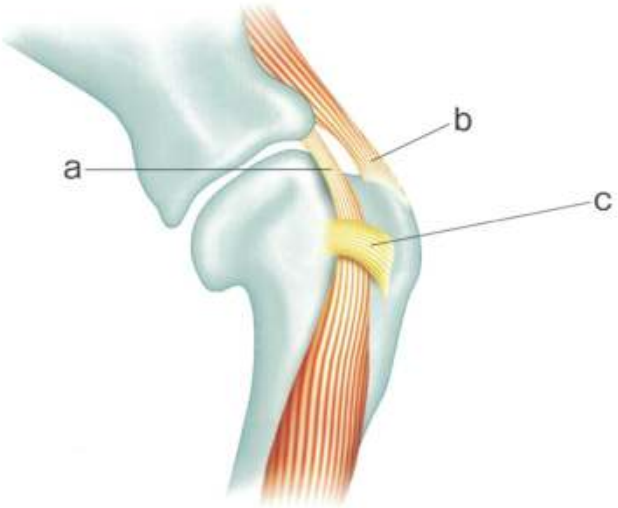


図 9　肩関節周囲の靱帯および上腕二頭筋腱の走行

神経学的検査と整形外科学的検査のコツ,2011



犬の跛行診断,2016

- Lateral and medial glenohumeral ligaments
- Biceps tendon
- Transverse humeral retinaculum
- Bicipital tenosinovitis

BICIPITAL TENDON TEST



動画 11-11 二頭筋腱テスト：正常例

上腕二頭筋腱炎の症例では二頭筋腱テストで判明することがあるため、上腕二頭筋腱炎を疑ったときには必ず本検査を行うことを推奨する。肘関節を伸展させた状態で肩関節をゆっくりと屈曲させていき、最大屈曲位で十分なストレスをかける。次いで、上腕骨の前内側面を走行する上腕二頭筋腱を圧迫して圧痛を誘発させる。肩関節を最大に屈曲させたときや上腕二頭筋腱を圧迫した際に疼痛を認めた場合には、上腕二頭筋腱炎を疑う。これは正常な犬で評価したときの動画である

神経学的検査と整形外科学的検査のコツ,2011

- Middle-old age, medium-large breed
- Too much exercise, progressive inflammation and lameness (worse after exercise)
- Elbow joint extension and shoulder joint flexion with strength
- Compress biceps tendon at medial humerus, both forelimbs perform

SHOULDER JOINT: EXTENSION

- One hand hold shoulder joint, another hand hold humerus
- Slowly extension of shoulder joint
- Pain (OCD, ununited caudal glenoid)

肩関節（上腕骨）OCD

▼身体検査



図2-8　肩関節の伸展

上腕を保持して肩関節を伸展させる（肘関節より遠位を保持しない）。OCDの検査に重要である。疼痛反応に注意する。



図2-9　肩関節の屈曲

上腕を保持して肩関節を屈曲させる（肘関節より遠位を保持しない）。上腕二頭筋腱炎，上腕二頭筋腱起始部を含む裂離骨折，棘上筋腱石灰化症，骨腫瘍，OCDの検査に重要である。疼痛反応に注意する。



神経学的検査と整形外科学的検査のコツ,2011

SHOULDER JOINT: DRAWER TEST



動画 11-14 肩関節の引き出し試験（Drawer test）：正常例

本検査は肩関節の前後方向の安定性を評価するための検査である。片方の手で肩甲棘をつかみ、もう片方の手で上腕骨の近位を保持する。次いで、上腕骨の近位を前後方向に動かし、肩関節の安定性を確認する

- One hand hold spine of the scapula, another hand hold proximal humerus
- Proximal humerus cranial-caudal movement to confirm shoulder joint stability (under anesthesia, sedation)

神経学的検査と整形外科学的検査のコツ,2011

SHOULDER JOINT: ABDUCTION

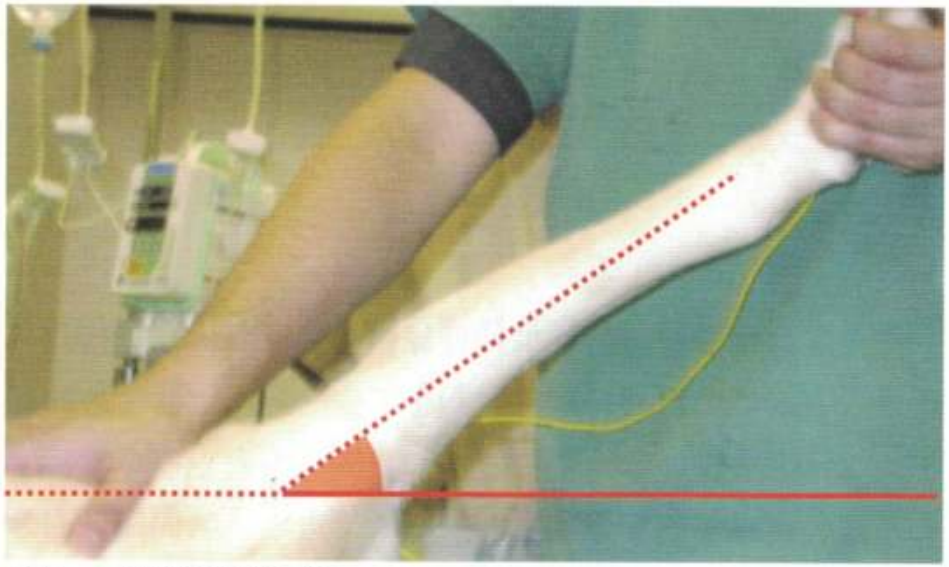


図11　肩関節の外転試験
肩関節の外転試験は、主に肩関節の内側にある靱帯や支帯の緩みを評価するために行う検査である。前肢を完全に伸展させてから、片方の手で肩甲骨を抑える。次いで、もう片方の手で前肢端を保持して、ゆっくりと前肢を外転させていく。肩関節を最大に外転させたときの外転角を計測し、その角度が50°以上のときには肩関節不安定症と診断する

神経学的検査と整形外科学的検査のコツ,2011

- Under anesthesia, one hand compress scapula, another hand hold distal forelimb and abduction performed
- Normal 30 degree, larger than 50 degree is shoulder joint instability
- Both shoulder performed
- Further arthroscopy



神経学的検査と整形外科学的検査のコツ,2011

骨科學觸診-後肢

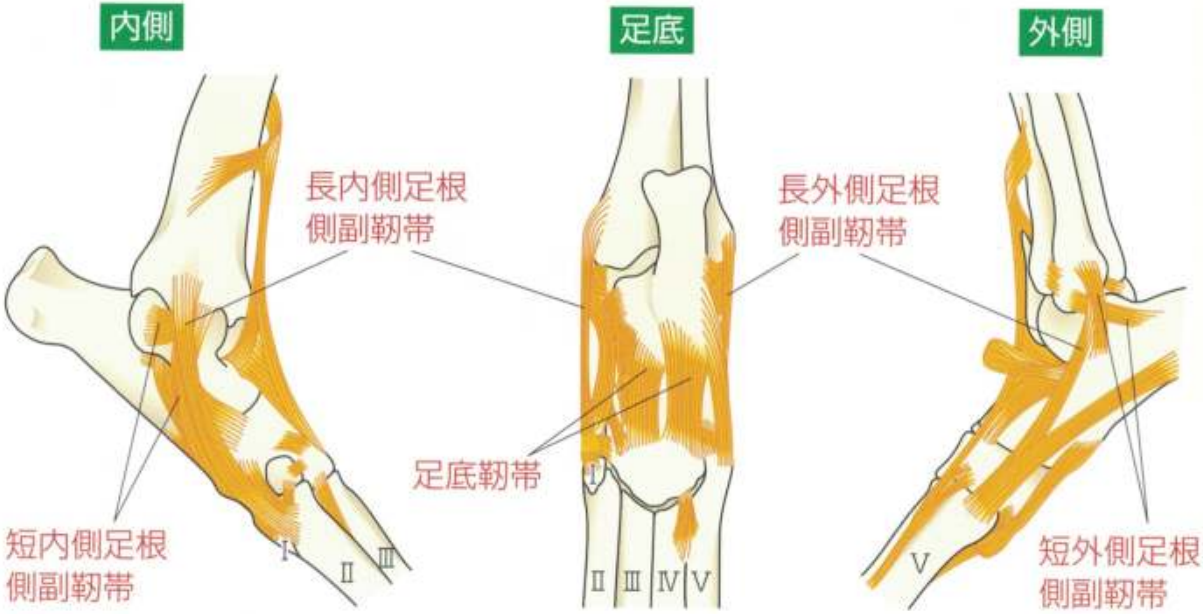
後肢

所見	L	R
肢端的傷口•皮膚炎•化膿	有•無	有•無
跗關節的可動性異常	有•無	有•無
跗關節的可動區域限制	有•無	有•無
阿基里斯腱損傷•移位	有•無	有•無
膝蓋骨脫臼	有•無	有•無
膝蓋骨脫臼的方向	內•外	內•外
膝蓋骨脫臼的Grade	(　)	(　)
膝關節外側側副韌帶損傷	有•無	有•無
膝關節內側側副韌帶損傷	有•無	有•無
脛骨Cranial Drawer Test	＋ • －	＋ • －
脛骨Tibial Compression Test	＋ • －	＋ • －
膝關節屈伸時的click音	＋ • －	＋ • －
Barden Test	＋ • －	＋ • －
Ortolani Test	＋ • －	＋ • －
股關節伸展時的疼痛	有•無	有•無
Thumb Pinch Test •三角試驗	＋ • －	＋ • －

HOCK JOINT



図2　足底靱帯断裂の症例
足根関節を屈曲させたときに足根間関節が異常に屈曲する場合には、足底靱帯断裂の可能性を考慮する



- Medial & lateral collateral ligaments(long & short parts), long plantar ligament
- Maximum extension of hock joint with abduction & adduction
- 90 degree flexion of hock joint with abduction & adduction
- ROM, flexion, extension, pain, crepitus

神経学的検査と整形外科学的検査のコツ,2011

ACHILLES' TENDON IMPAIRMENT



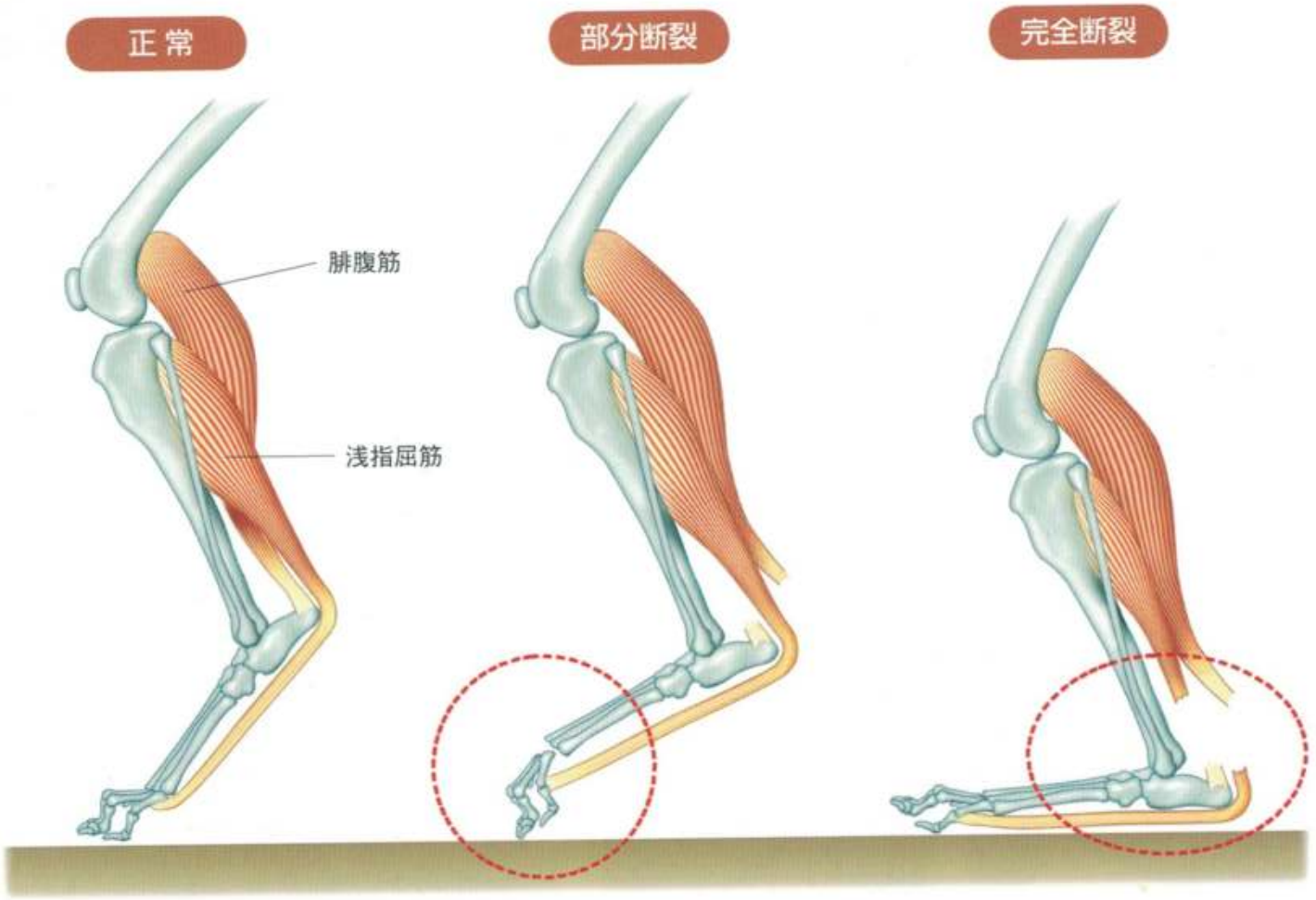


12-2 アキレス腱の損傷と変位の確認：正常例

- Main: Tendon of Gastrocnemius and tendon of superficial digital flexor
- Observe hock joint angle when standing
- Achilles' tendon's motility and compress it to confirm tension

神経学的検査と整形外科学的検査のコツ,2011

図3-121　アキレス腱断裂の模式図



アキレス腱のうち，浅指屈筋腱が断裂せずに残っている場合には，足根関節がある程度沈み，足先が屈曲する（図3-122参照）。

腓腹筋腱，浅指屈筋腱など，アキレス腱がほぼ完全に断裂している場合には，足根関節が完全に沈み，蹠行（plantigrade：足底歩き）を示す（図3-123参照）。

犬の跛行診断,2016





STIFLE: PATELLAR LUXATION

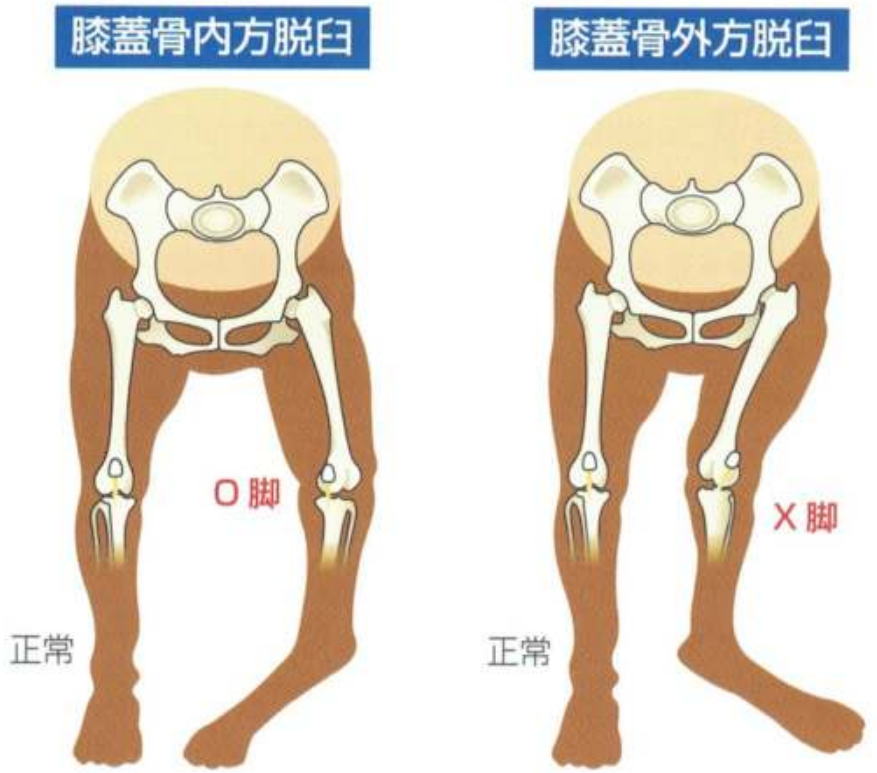


図4　膝蓋骨脱臼の症例の外貌
膝蓋骨内方脱臼の症例では内反膝（O脚）、外方脱臼では外反膝（X脚）といった脚変形を認めることがある

神経学的検査と整形外科学的検査のコツ,2011

Grade I

The patella can be luxated, but spontaneous luxation of the patella during normal joint motion rarely occurs. Manual luxation of the patella may be accomplished during physical examination, but the patella reduces when pressure is released. Flexion and extension of the joint are normal.

Grade II

Angular and torsional deformities of the femur may be present to a mild degree. The patella may be manually displaced with lateral pressure or may luxate with flexion of the stifle joint. The patella remains luxated until the examiner reduces it or it is spontaneously reduced when the animal extends and de-rotates its tibia.

Grade III

The patella remains luxated medially most of the time but may be manually reduced with the stifle in extension. However, after manual reduction, flexion and extension of the stifle result in reluxation of the patella. There is a medial displacement of the quadriceps muscle group. Abnormalities of the supporting soft tissue of the stifle joint and deformities of the femur and tibia may be present.

Grade IV

There may be an 80- to 90-degree medial rotation of the proximal tibial plateau. The patella is permanently luxated and cannot be manually repositioned. The femoral trochlear groove is shallow or absent, and there is medial displacement of the quadriceps muscle group. Abnormalities of the supporting soft tissue of the stifle joint and deformities of the femur and tibia are notable.

膝蓋骨脫臼的分級

- 第1級：用手壓可以脫臼，放開後回到正常位置：幾乎無跛行
- 第2級：有時候會自行脫臼又自行回到正常位置：脫臼時腳會抬起，跳躍樣步行
- 第3級：時常脫臼，可用手移動到正常位置，手放開後又再度脫臼：脛骨內轉，步行困難，可以負重，觸診會輕度疼痛
- 第4級：時常脫臼，無法用手移動到正常位置，步行緩慢，可以負重，輕度疼痛

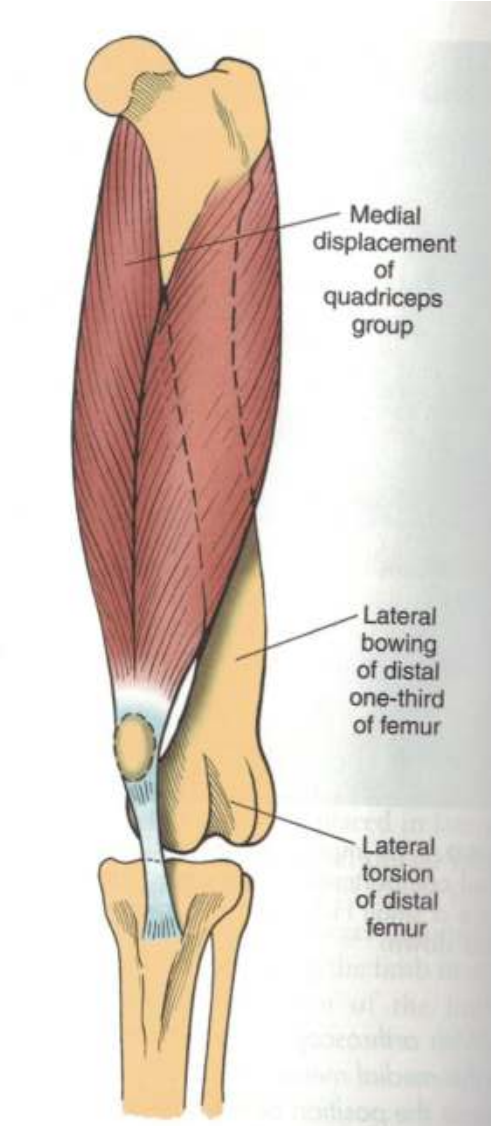
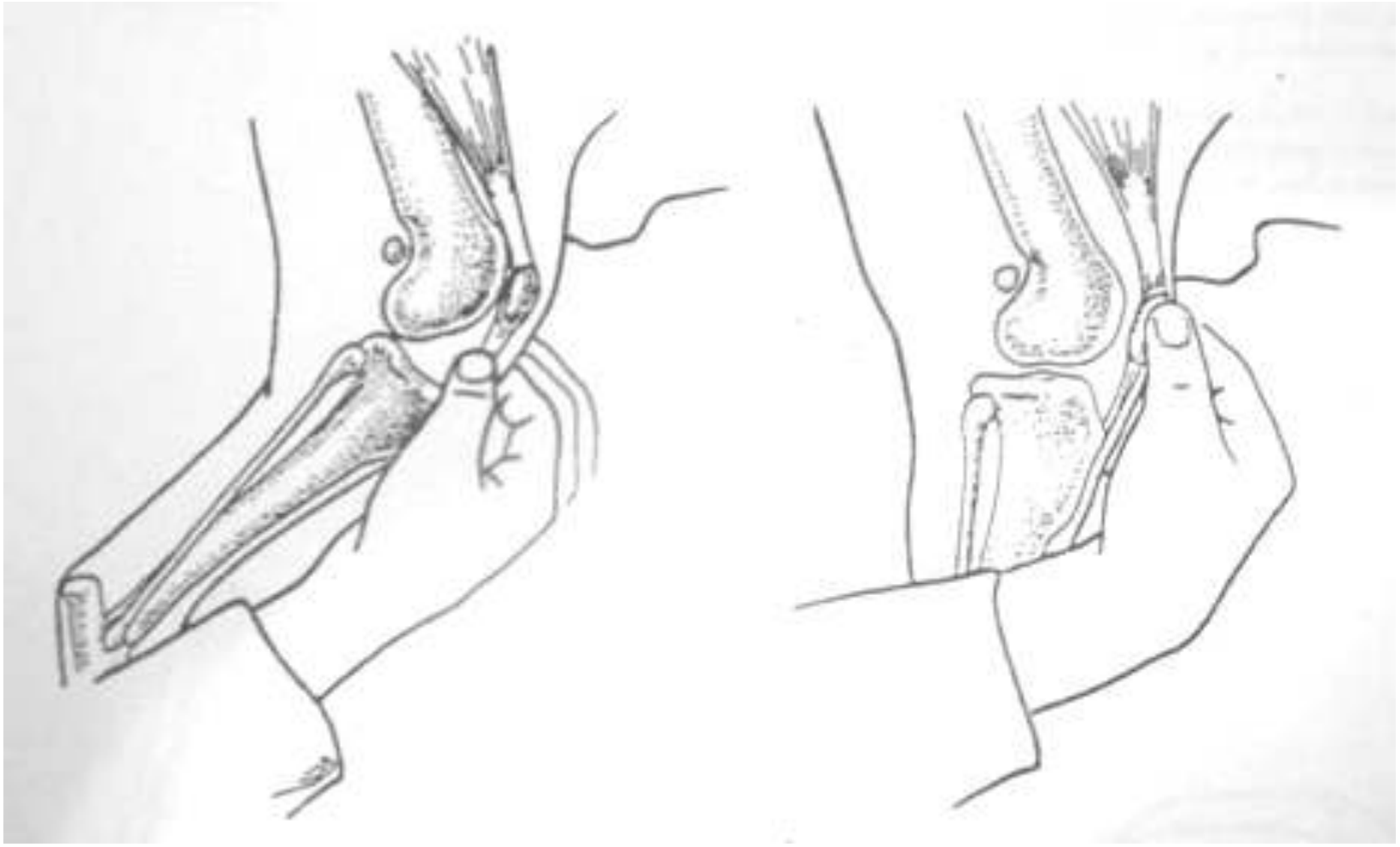


表1　Singleton のグレード分類

グレード	脱臼の程度	脱臼形式	臨床症状
1	完全伸展位や触診時に強制的に脱臼する程度	間欠的脱臼	なし・まれ
2	屈伸時に脱臼し、自然に整復できる	間欠的脱臼	脱臼時のみスキップ歩行・疼痛
3	常に脱臼しているが徒手整復可能	恒久的脱臼	やや屈曲した歩行・疼痛
4	常に脱臼し整復不可能	恒久的脱臼	屈曲した歩行・伸展困難・疼痛

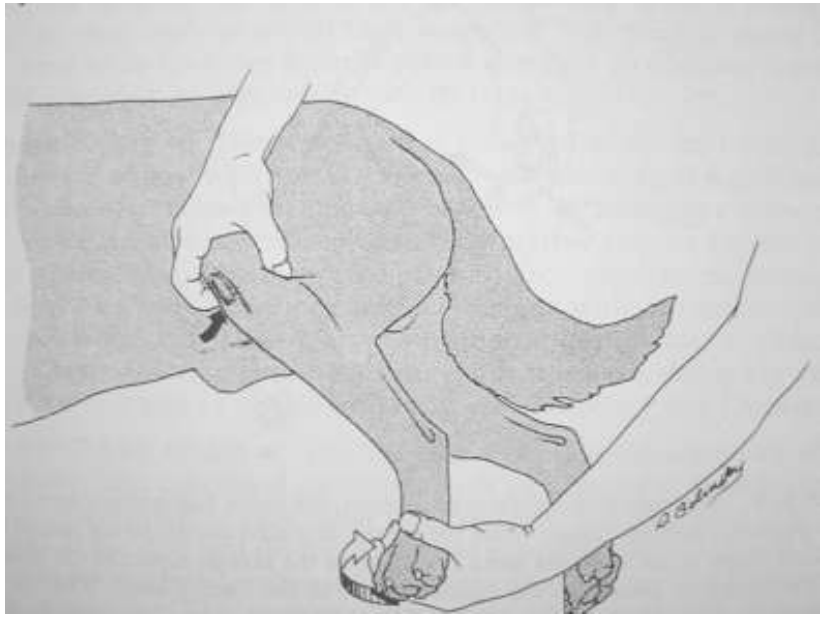
神経学的検査と整形外科学的検査のコツ,2011

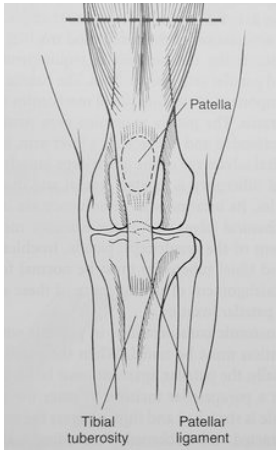
Stifle physical exam – Patella



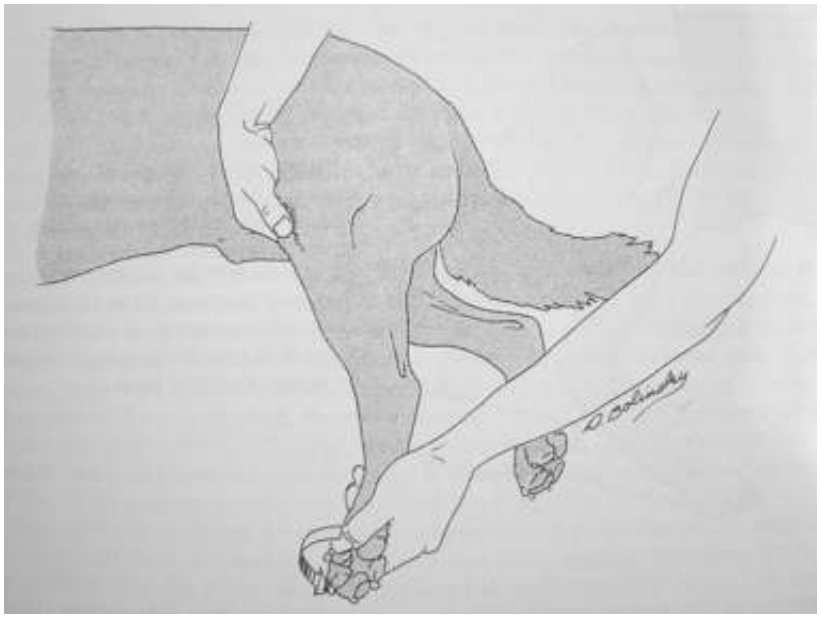
Stifle physical exam – Patella

Detect lateral patella luxation





Detect medial patella luxation



前十字靭帯
ANTERIOR CRUCIATE LIGAMENT

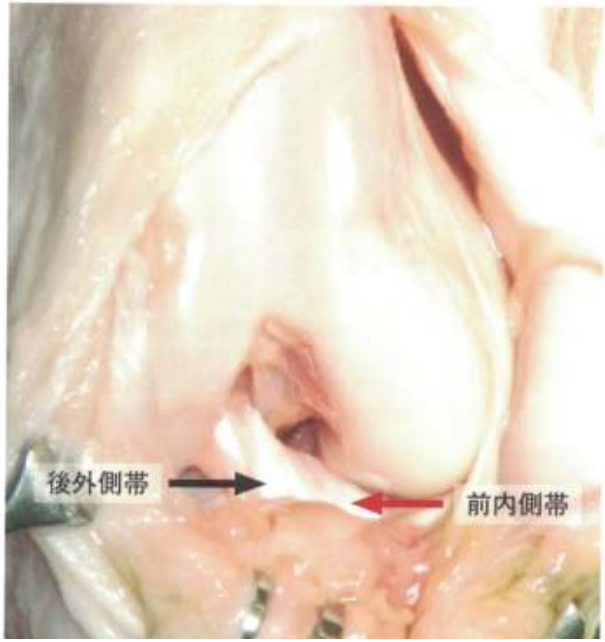


表2　膝関節を屈曲および伸展させたときの前十字靱帯の各帯の安定性

	屈曲位	伸展位
全体	緊張	緊張
前内側帯	緊張	緊張
後外側帯	弛緩	緊張

神経学的検査と整形外科学的検査のコツ,2011

Stifle physical exam － ACL

Cranial drawer test

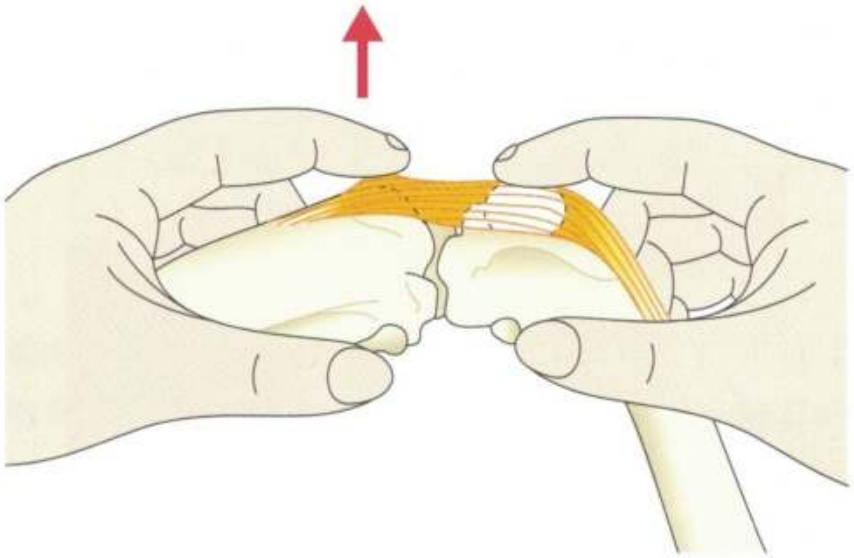
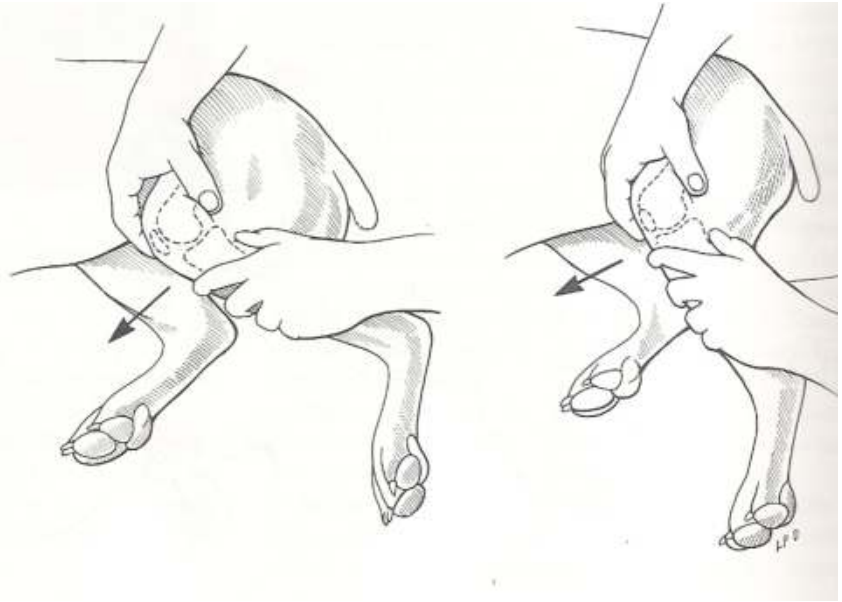


図6　脛骨前方引き出し試験
片方の手で大腿骨遠位を握り、その親指を外側腓腹筋種子骨の上に、そして人差し指を膝蓋骨の上に置く。もう一方の手で脛骨近位を握り、その親指を腓骨頭に、そして人差し指を脛骨粗面に置く。次いで、脛骨を大腿骨に対して頭側に動かし、脛骨の前方への変位の有無を確認する。部分断裂を正確に診断するためには、伸展位と屈曲位で脛骨の前方への変位を試み、その違いを比較する

神経学的検査と整形外科学的検査のコツ,2011



- Flexion and extension
- Patella, Sesamoid, Tibial tuberosity, Fibular head

Stifle physical exam － ACL

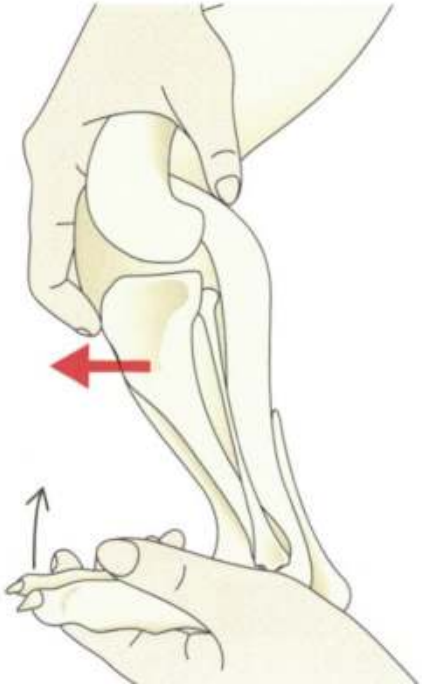
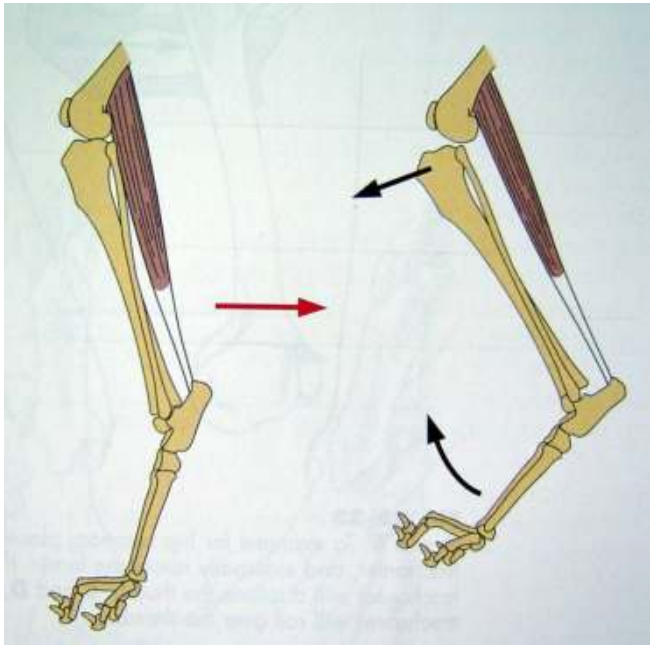


図7　脛骨圧迫試験
片方の手の人差し指を膝蓋骨-膝蓋腱-脛骨粗面に沿うように置いて、膝関節をやや伸展気味に保つ。もう片方の手は中足骨の近位を保持し、膝関節が屈曲しないように足根関節のみを屈曲させる。足根関節を最大に屈曲させたときに脛骨の前方への変位が生じるか否かを確認する

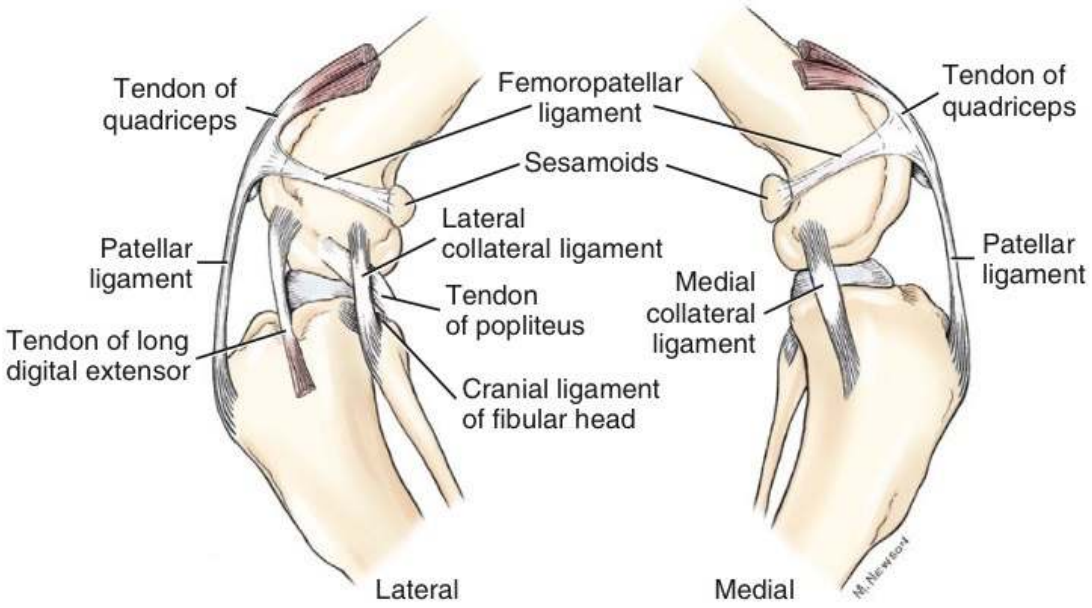
Tibial compression test



- Extension
- Patella, Patellar tendon, Tibial tuberosity
- Tarsal bone, maximum hock joint flexion without stifle flexion

神経学的検査と整形外科学的検査のコツ,2011

Stifle physical exam – Collateral ligaments





Hip physical exam — Hip joint extension

- Animal standing or lateral recumbency
- One hand at dorsal pelvis, the other hand hold femur and caudal extension slowly
- Pain during extension and exorotation

Hip physical exam － Hip joint laxity (Barden test)



動画 12-13 バーデン試験の原理

片方の手の親指を大転子の上に置く。もう片方の手で大腿部をしっかりと握り、外側へと持ち上げる。股関節に緩みがあると大転子が挙上される



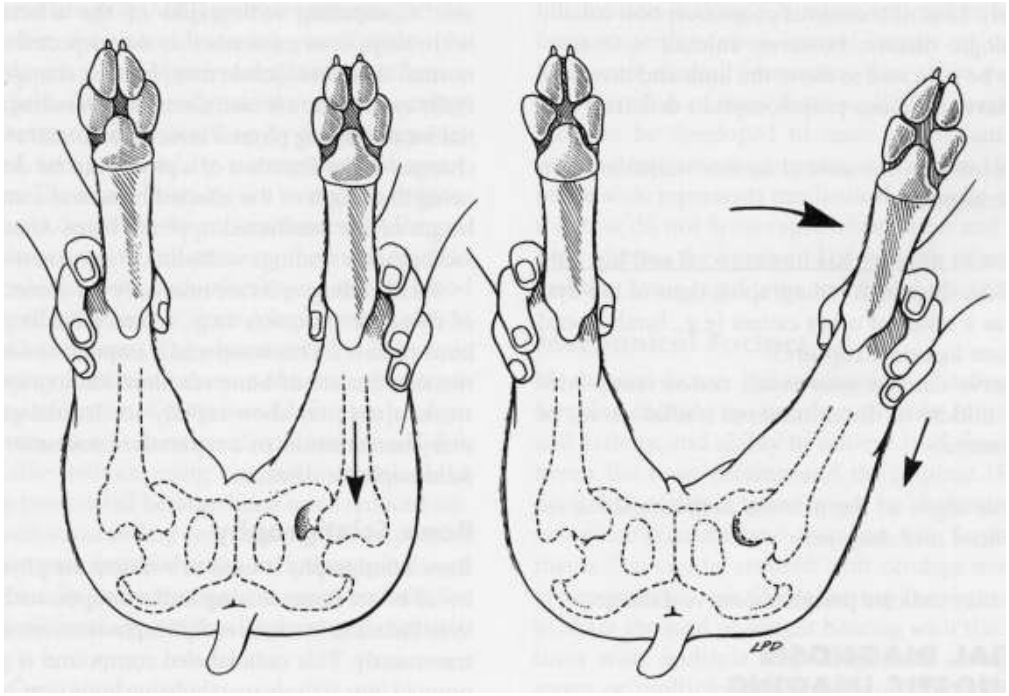
動画 12-15 バーデン試験：異常例

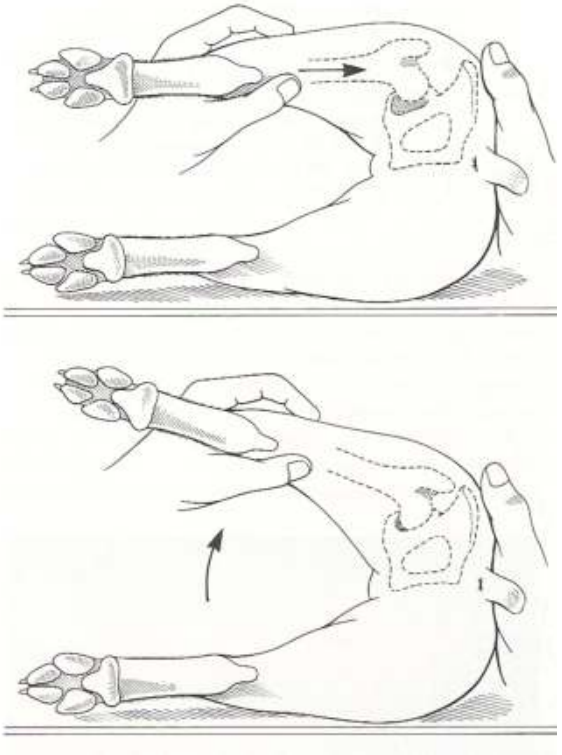
大腿部を外側へ持ち上げた際に、大転子が5～6mm以上挙上された場合にはバーデン試験「陽性」と診断する。本症例は、単純X線検査にて股関節形成不全と診断された

- General anesthesia, lateral recumbency, 6-8 weeks puppy
- Thumb on greater trochanter
- Elevate femur body and check displacement of the great trochanter
- Normal:1-2mm
- Abnormal: 5-6mm or >6mm

神経学的検査と整形外科学的検査のコツ,2011

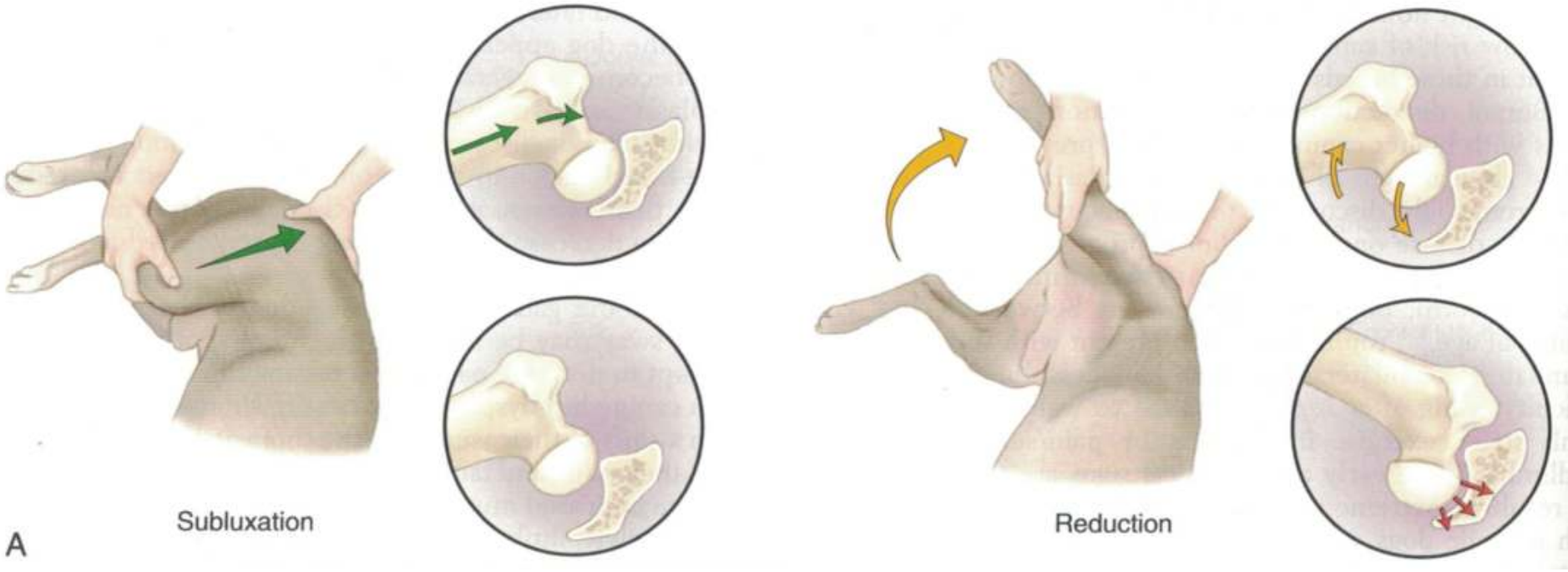
Hip physical exam — Hip joint laxity (Ortolani test)





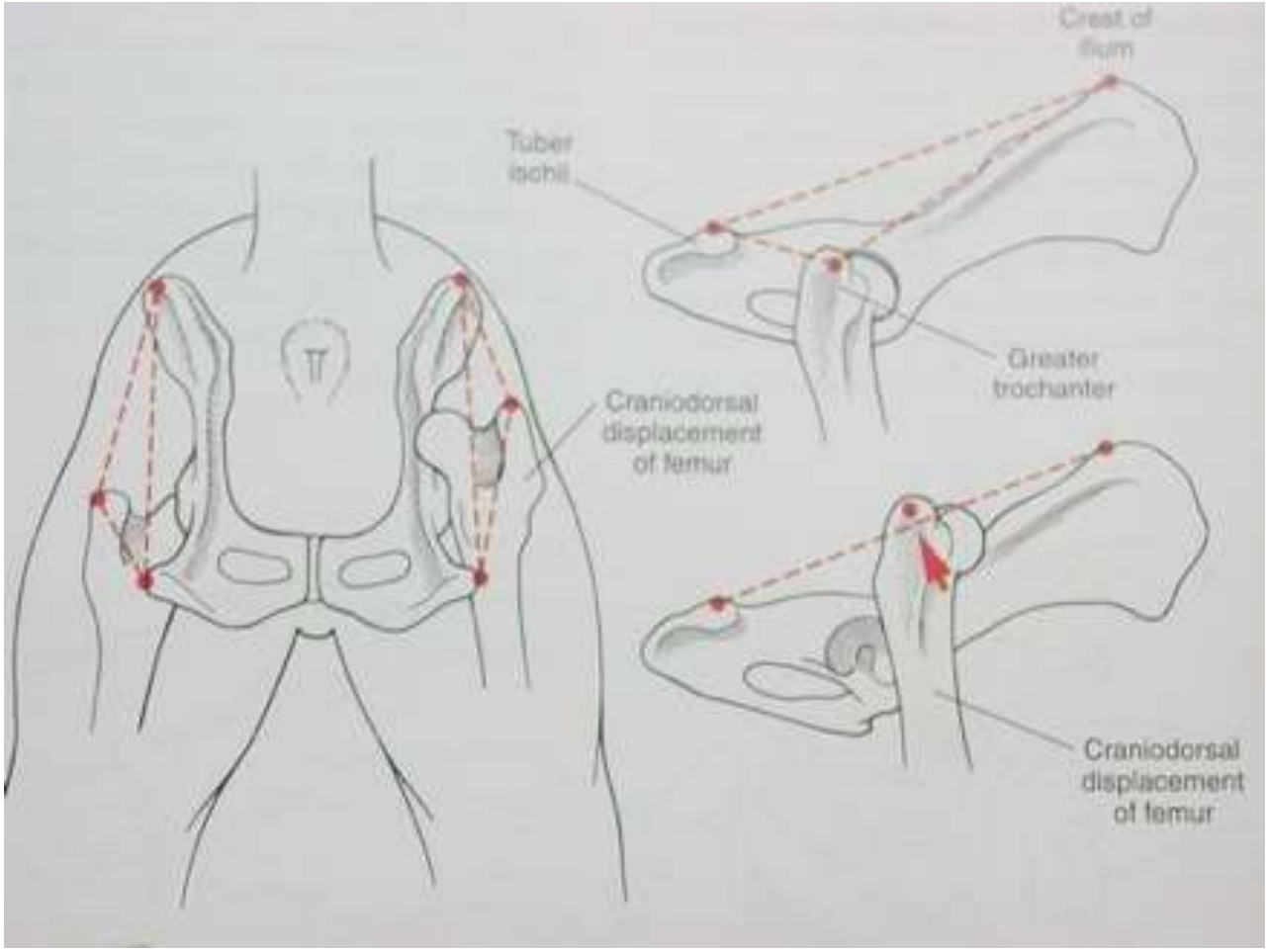
- General anesthesia, lateral recumbency
- 90 degree between femur and pelvis with compression force , and then hip joint abduction

Hip physical exam — Hip joint laxity (Ortolani test)



Hip physical exam － Hip joint luxation

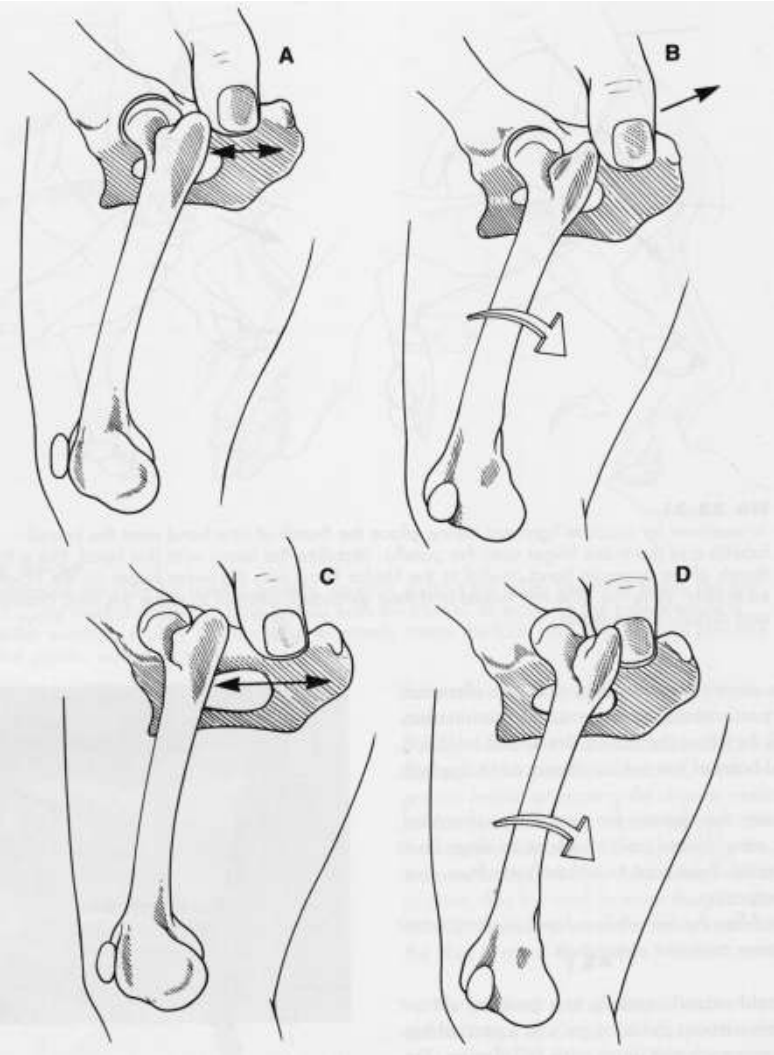
（三角試験）



- Ischiatic tuberosity
- Crest of ilium
- Greater trochanter

神経学的検査と整形外科学的検査のコツ,2011

Hip physical exam — Hip joint luxation (Thumb pinch test)



- Put thumb at the position between greater trochanter and ischiatic tuberosity



神経学的検査と整形外科学的検査のコツ,2011

LOCOMOTOR SYSTEM 2

Liu,ili

BONE RADIOGRAPHY

- Exposure and positioning
  - Sedation or anesthesia required
  - Proper field and centering
    - Poor collimation by include too much soft tissues will increase scattering and decrease definition
- Low kVp and higher mAs to provide better contrast
- **<u>Always making at least 2 views</u>**
  - True A/P, true lateral
  - Optional oblique, stressed, sky line, etc,…..
  - Contralateral limb for reference

POSITIONING FOR SHOULDER JOINT

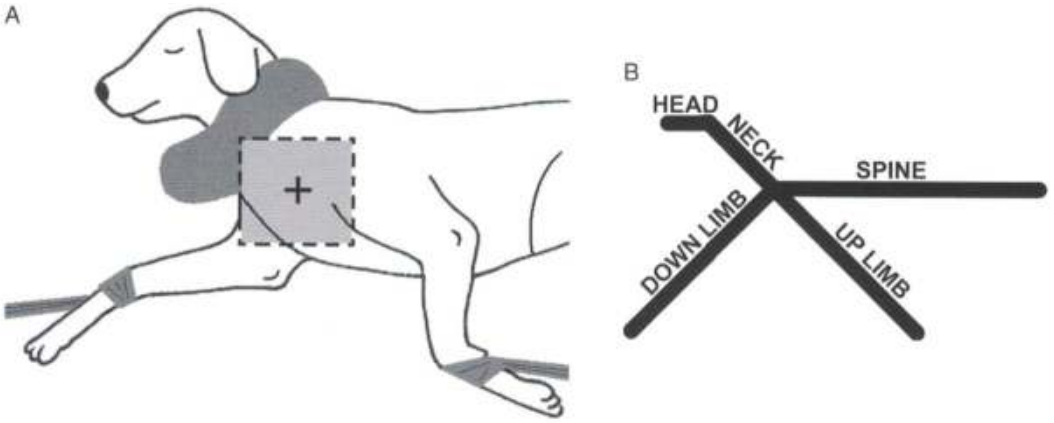


Figure 2.39 Shoulder, lateral radiograph. **A.** Limb is fully extended to avoid superimposition artifacts; opposite limb is pulled caudally until in line with neck; x-ray beam is centered at scapulohumeral joint. **B.** Line drawing to depict correct alignment for lateral radiograph of shoulder.

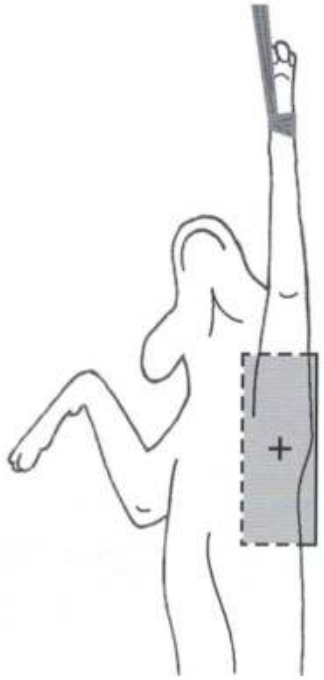


Figure 2.40 Shoulder and scapula, caudocranial radiograph. Limb is fully extended; sternum is rotated 30° toward opposite limb; x-ray beam is centered on scapulohumeral joint.

POSITIONING FOR SHOULDER JOINT

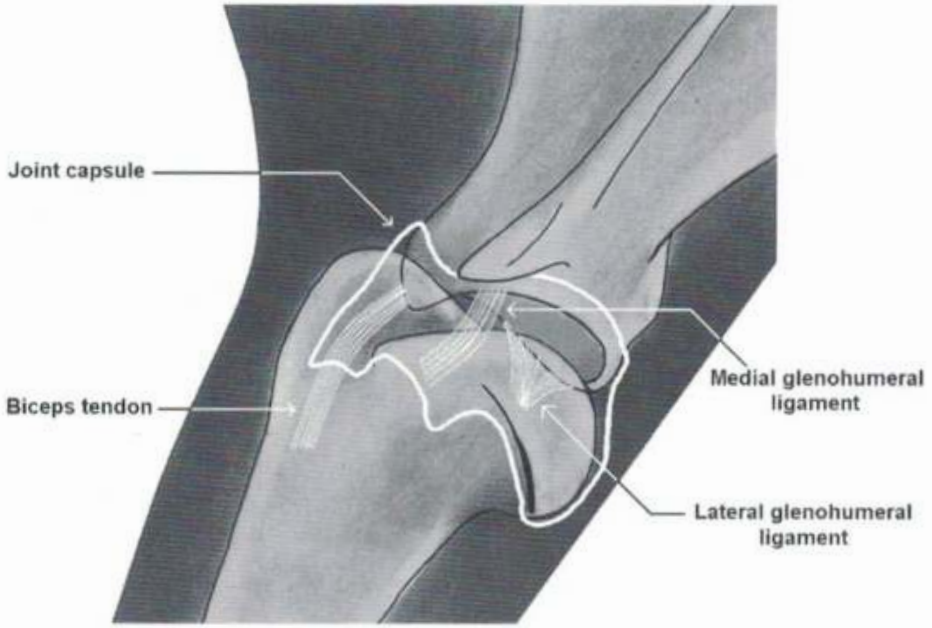


Figure 4.22 Normal canine shoulder (lateral radiograph) with attachment sites for ligaments and joint capsule. (Adapted from Owens and Biery 1999).

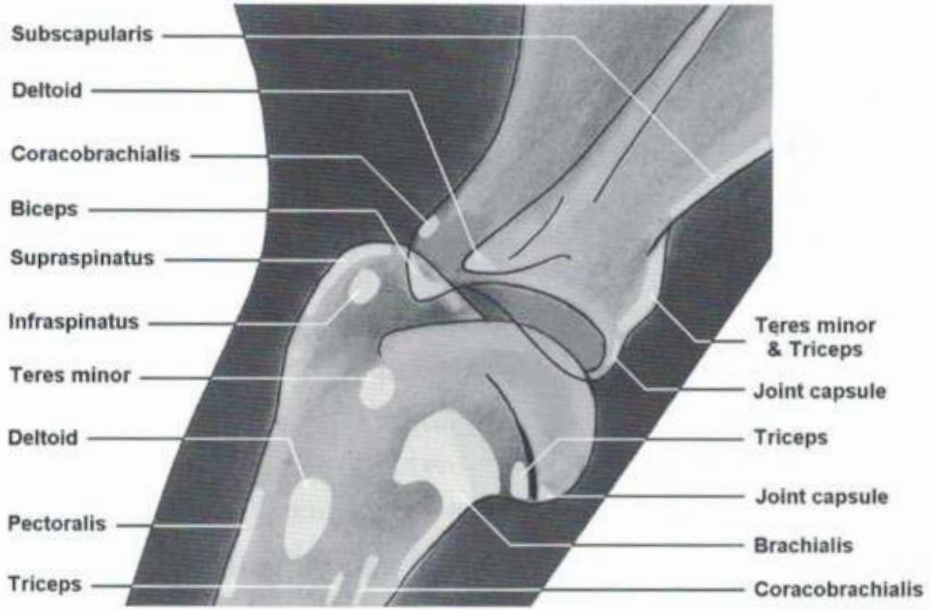


Figure 4.23 Normal canine shoulder (lateral radiograph) with attachment sites for muscle tendons. (Adapted from Owens and Biery 1999).

POSITIONING FOR ELBOW JOINT

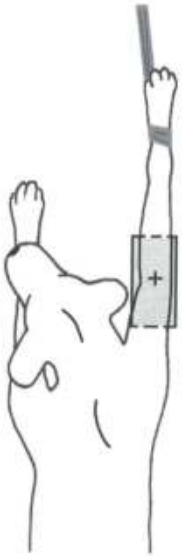


Figure 2.42 Elbow, craniocaudal radiograph.

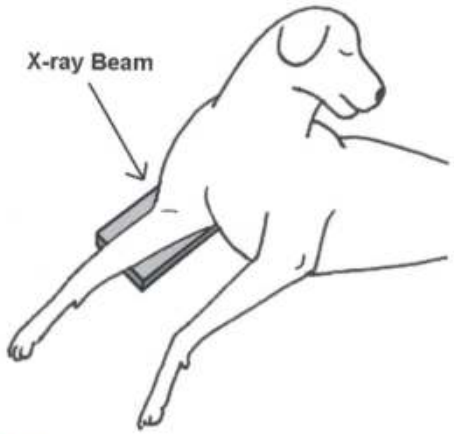


Figure 2.44 Elbow, craniolateral-caudomedial oblique radiograph. X-ray beam is angled 20° craniodistal to caudoproximal (arrow) to enhance visualization of joint space.

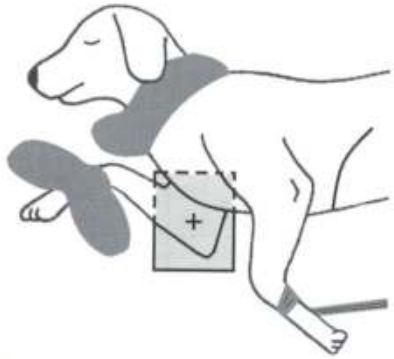


Figure 2.43 Elbow, extreme flexed lateral radiograph.

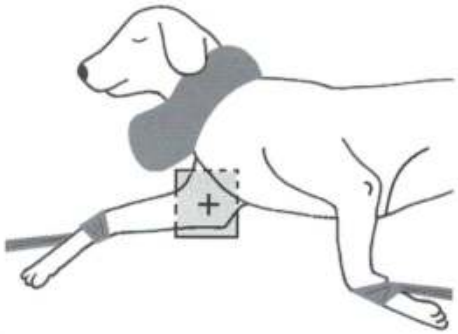


Figure 2.41 Elbow, lateral radiograph.

POSITIONING FOR ELBOW JOINT

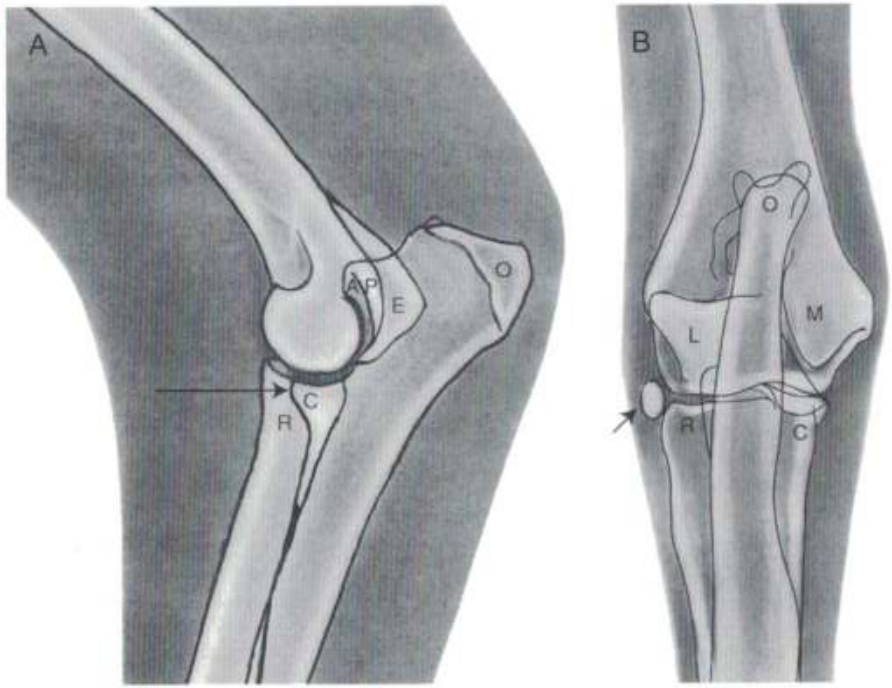


Figure 4.26 Canine elbow with normal anatomic landmarks. **AP** = anconeal process, **E** = medial humeral epicondyle, **O** = olecranon, **R** = radial head, **C** = medial coronoid process (arrow). **A.** Lateral radiograph. **B.** Caudocranial radiograph. (Adapted from Owens and Biery 1999.)

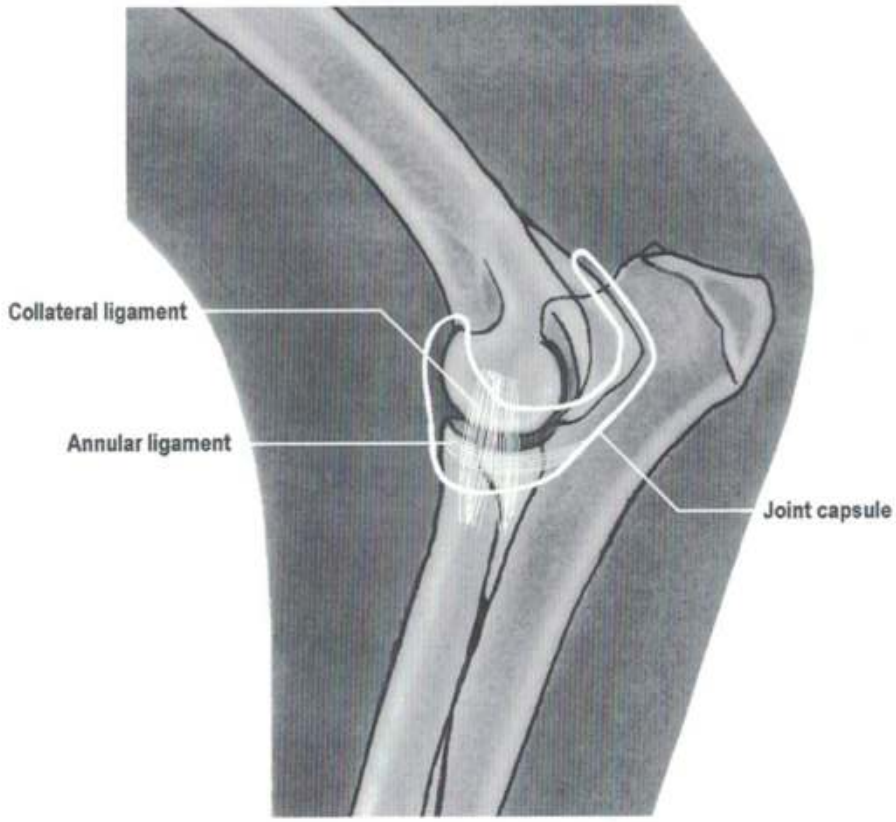


Figure 4.29 Normal canine elbow with attachment sites for ligaments and joint capsule (lateral radiograph). (Adapted from Owens and Biery 1999).

POSITIONING FOR CARPAL JOINT

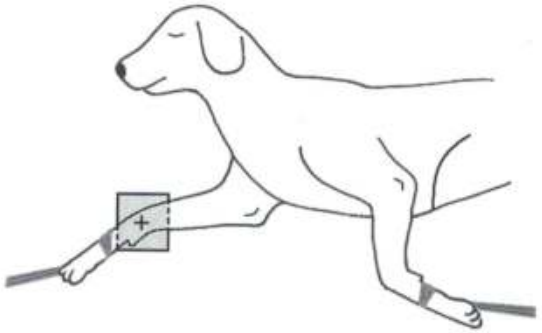


Figure 2.45 Carpus, lateral radiograph.





Figure 2.47 Carpus, stressed dorsopalmar radiograph. Carpus is stabilized and a medial-to-lateral force is being applied to evaluate joint laxity.

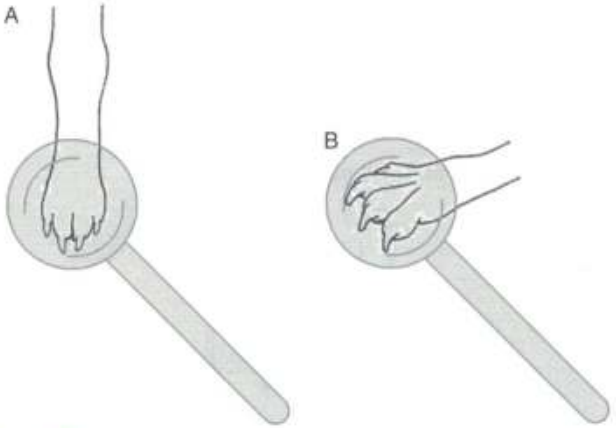


Figure 2.48 Compression radiographs of digits. A low-density plastic or wooden spoon is used to gently compress digits. **A.** Dorsopalmar compression to stabilize foot and digits. **B.** Lateral compression to separate digits.

POSITIONING FOR HIP JOINT

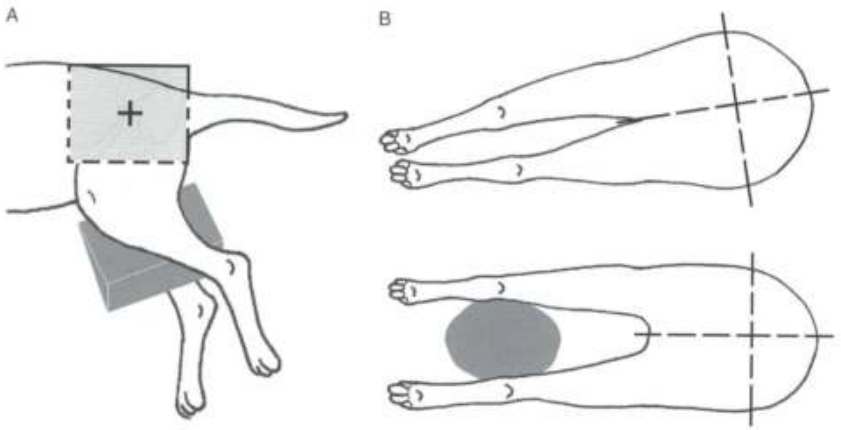


Figure 2.49 Pelvis, lateral radiograph. A foam wedge or towel is placed between stifles to correct rotation of pelvis. **A.** Lateral view. **B.** Cau cranial view.

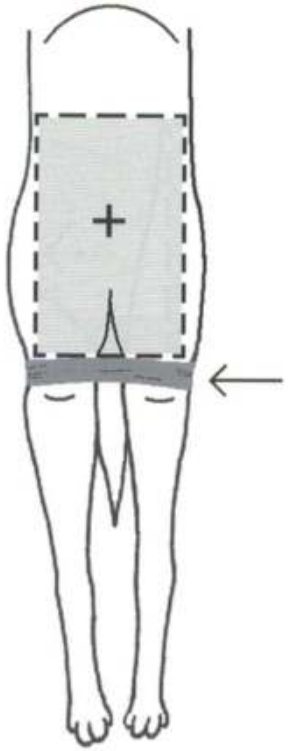
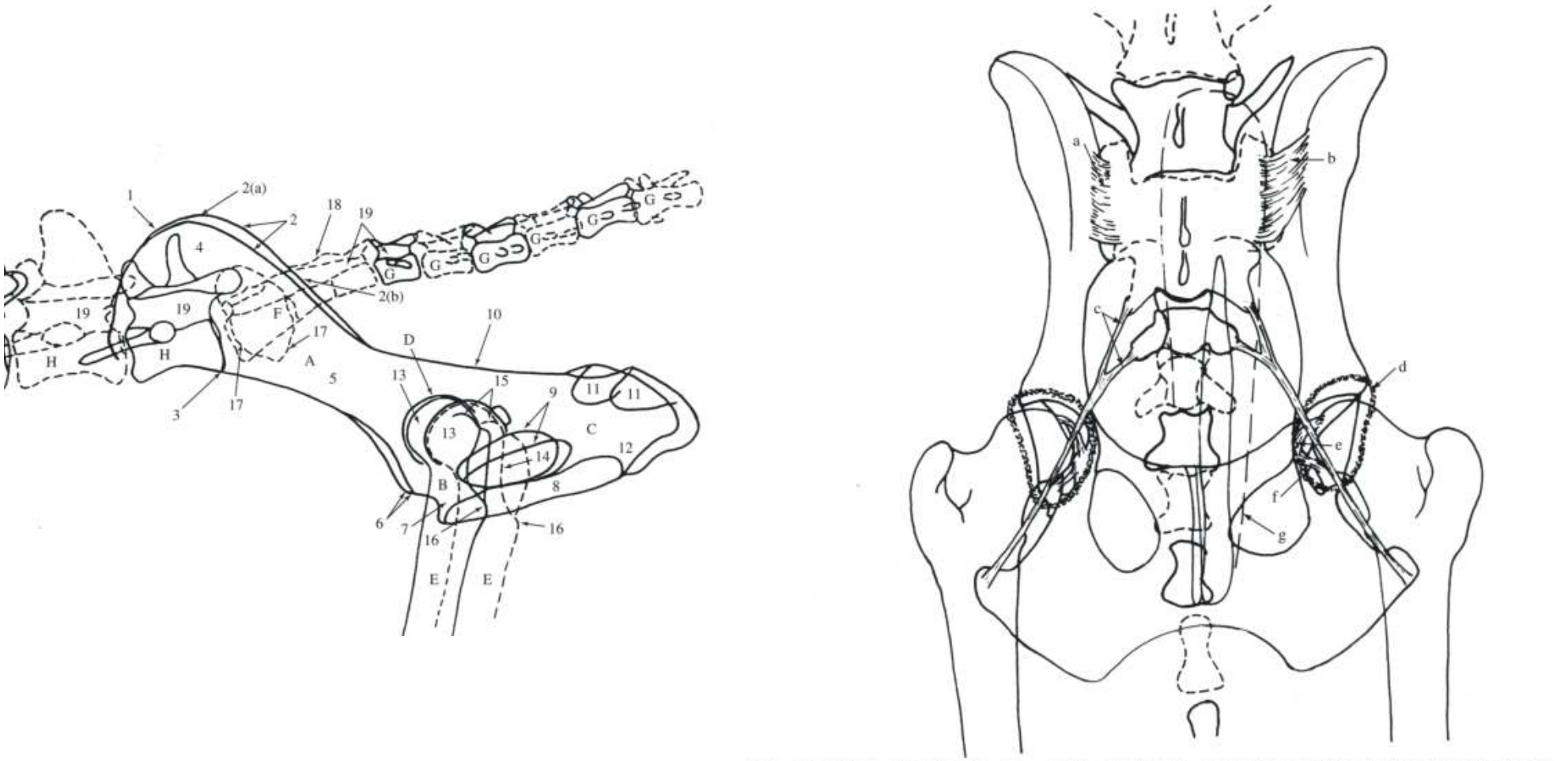


Figure 2.50 Pelvis, VD extended radiograph. Stifles are temporarily bound together to keep femurs parallel (arrow).

POSITIONING FOR HIP JOINT



114 関節と靱帯の範囲を示すために、大腿骨を十分伸展させた股関節および骨盤の腹背方向撮影像の模式図。

POSITIONING FOR HIP JOINT



チェックポイント

1. 背中をテーブルにつける
2. 第７腰椎と膝関節を含む
3. 閉鎖孔、骨盤が左右対称
4. 両側の大腿骨が互いに平行
5. 膝蓋骨が大腿骨の中心
6. 棘突起が第７腰椎の中心

POSITIONING FOR STIFLE JOINT

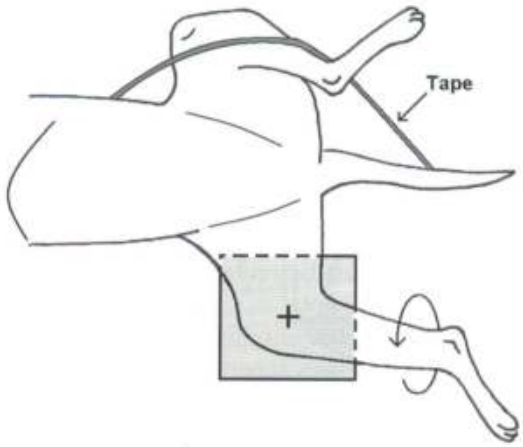


Figure 2.54 Stifle, lateral radiograph. Limb of interest is down and slightly rotated (curved arrow) to move patella closer to detector and avoid rotation. Opposite limb is abducted and positioned out of field of view using tape.

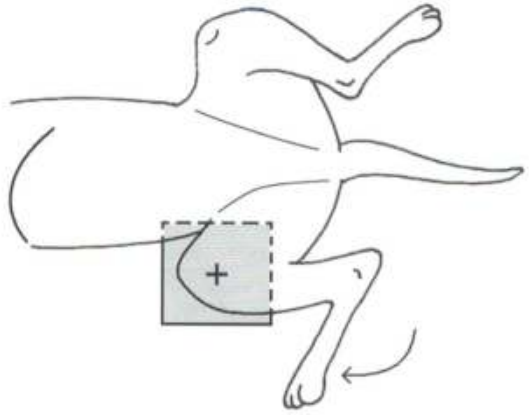


Figure 2.56 Stifle, flexed lateral radiograph with tibial compression. Stifle is flexed 90° and tarsus is maximally flexed (curved arrow), which creates cranial displacement of proximal tibia.

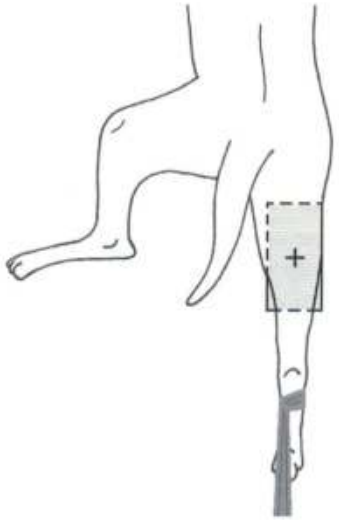


Figure 2.55 Stifle, caudocranial radiograph. Animal in ventral recumbency with limb of interest extended caudally and x-ray beam centered on stifle. Alternatively, a craniocaudal radiograph can be made with animal in dorsal recumbency and limb of interest extended caudally.

POSITIONING FOR STIFLE JOINT

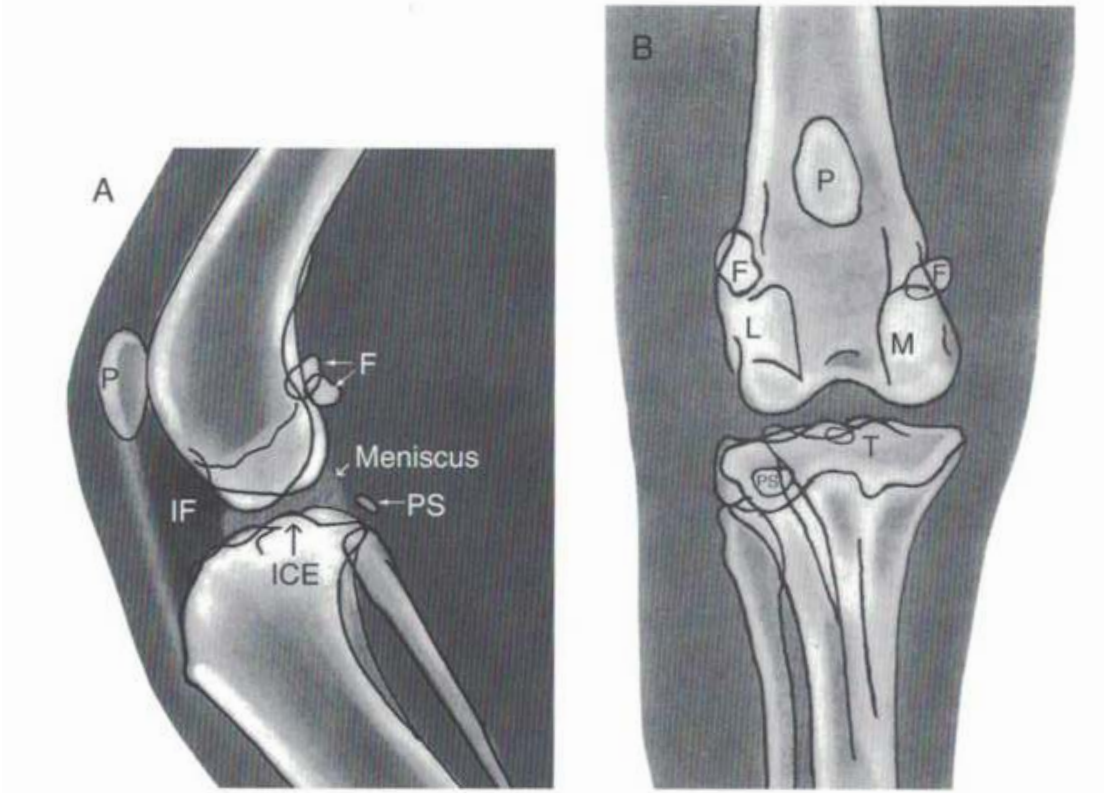


Figure 4.41 Normal canine stifle. **P** = patella, **F** = fabellae, **PS** = popliteal sesamoid, **IF** = infrapatellar fat pad, **ICE** = intercondylar eminence, **M** = medial femoral condyle, **L** = lateral femoral condyle, **T** = tibial plateau. **A.** Lateral radiograph. **B.** Craniocaudal radiograph. (Adapted from Owens and Biery 1999.)

POSITIONING FOR STIFLE JOINT

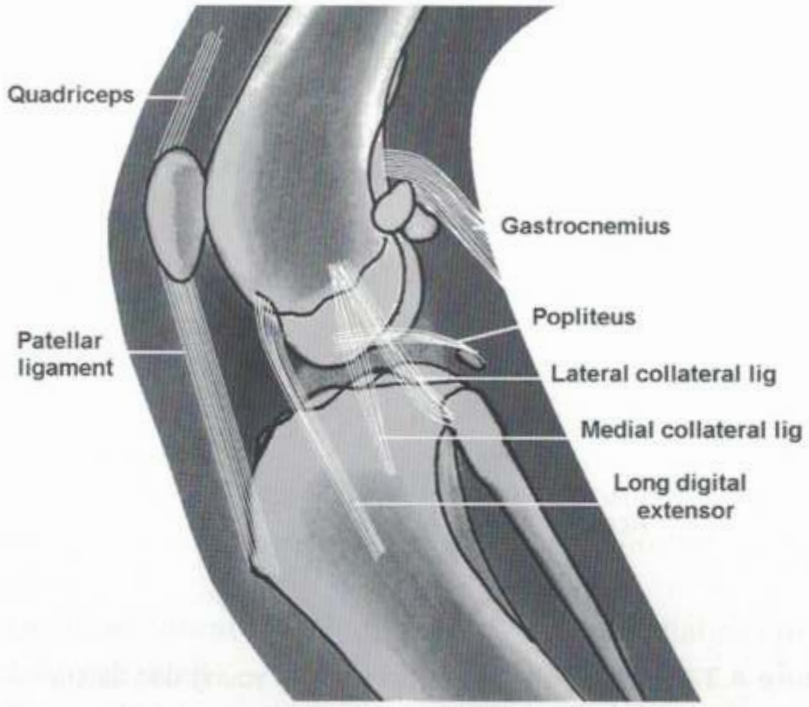


Figure 4.42 Normal canine stifle with ligaments (lateral radiograph).

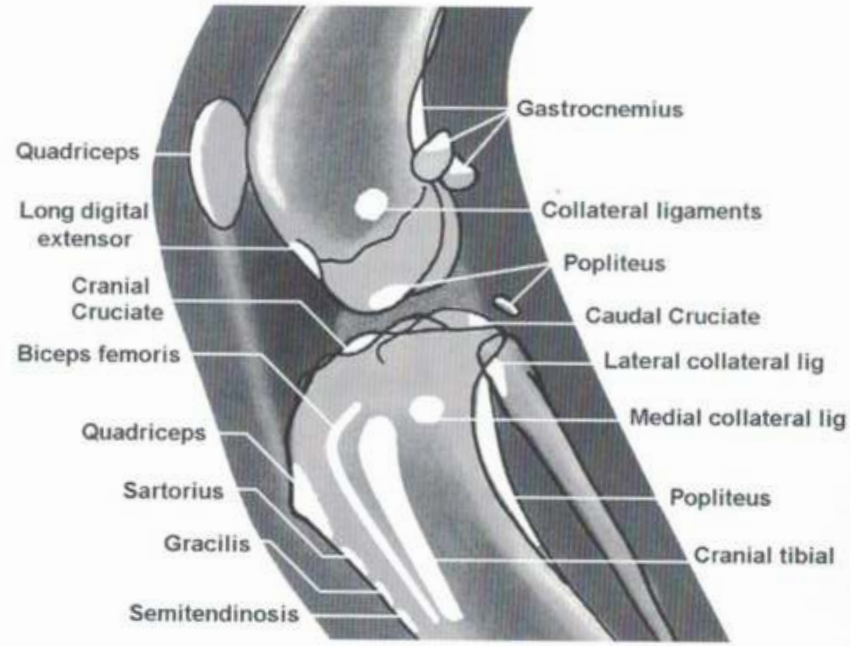


Figure 4.43 Normal canine stifle with attachment sites for tendons (lateral radiograph).

POSITIONING FOR STIFLE JOINT

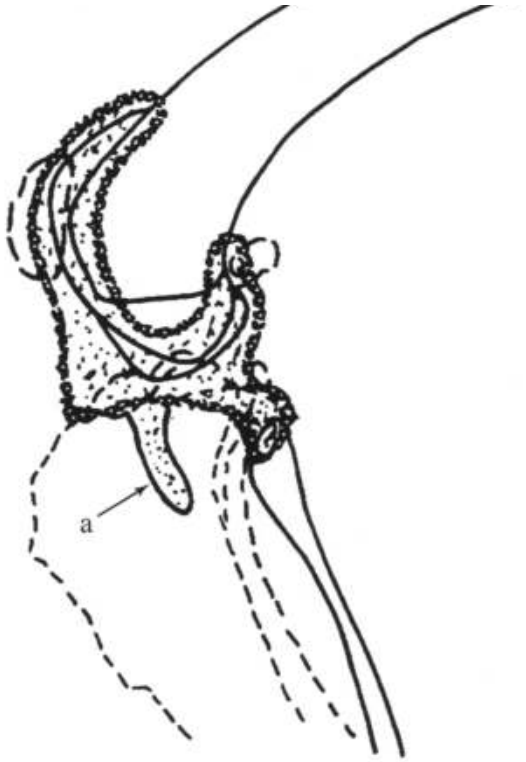
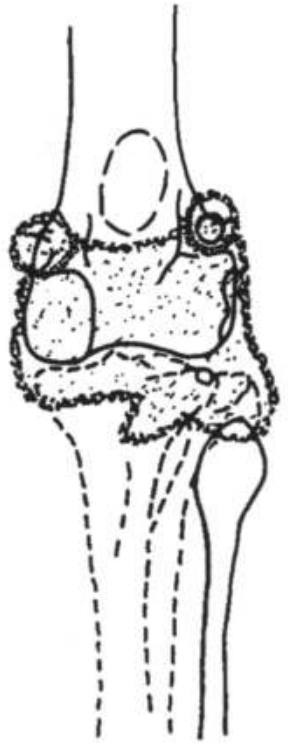


図134　関節包の拡がりを示す膝関節における内外方向撮影像の模式図。

= 関節包
= 滑膜腔



膝関節の尾頭方向撮影像の模式図。



膝関節の尾頭方向撮影像の模式図。

POSITIONING FOR HOCK JOINT

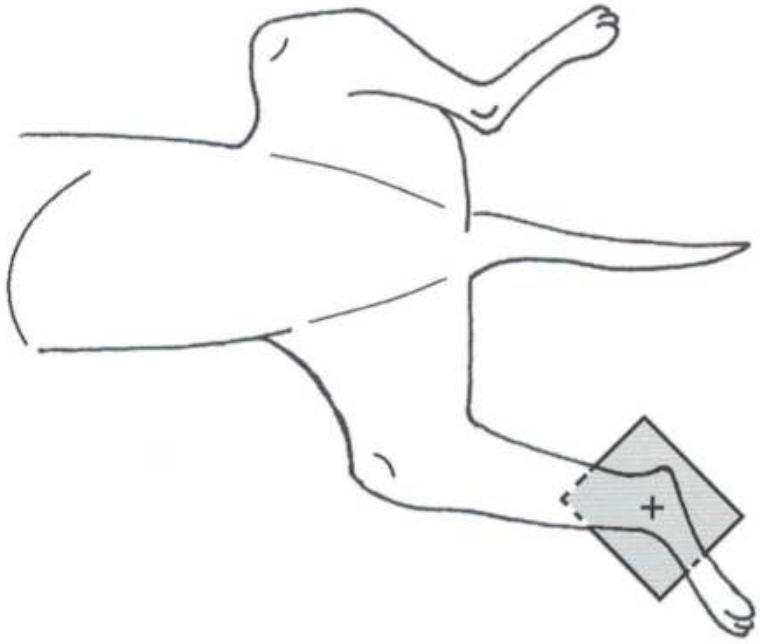


Figure 2.57 Tarsus, lateral radiograph. Limb of interest is down and extended caudally; opposite limb is abducted out of the field of view.

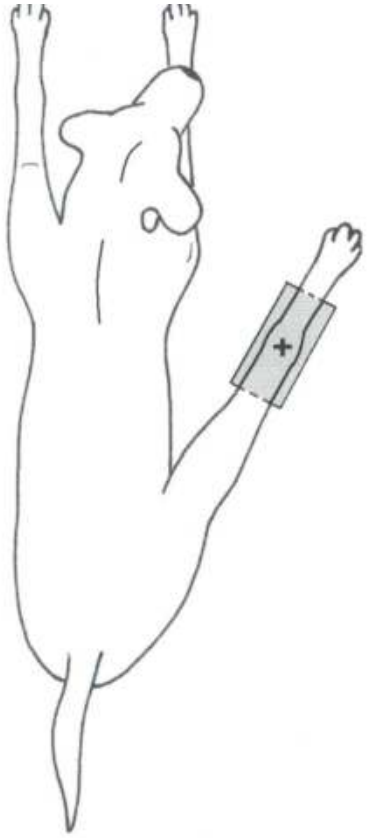
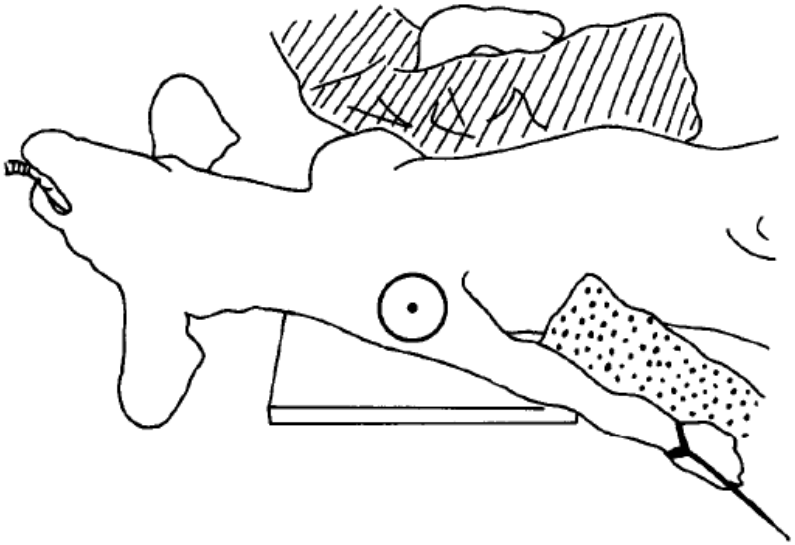


Figure 2.58 Tarsus, dorsoplantar radiograph. Animal is in ventral recumbency with limb of interest extended cranially and slightly laterally.

FRONTAL PLANE: HUMERUS

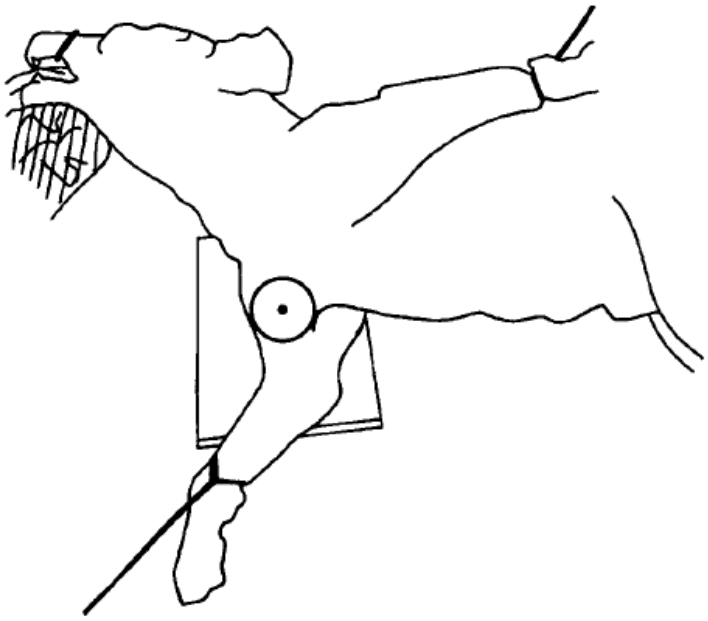




- Lateral and medial side of the anconeal process should not appear

SAGITTAL PLANE: HUMERUS

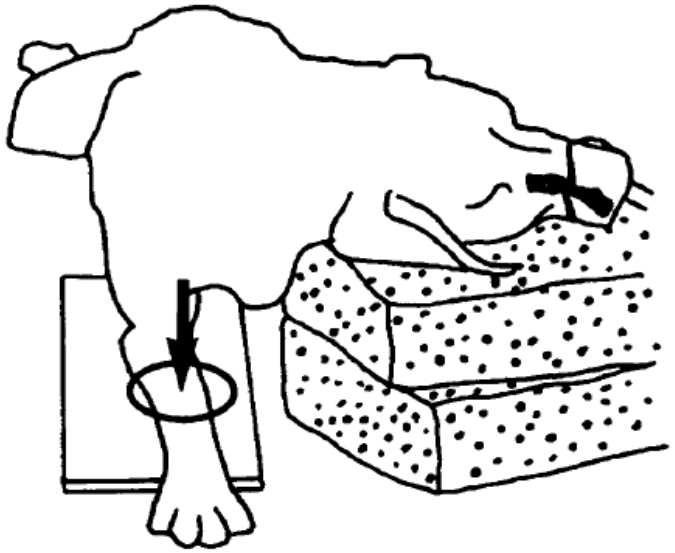




- Well defined trochlear notch at distal side
- Over lapping of two humeral condyles

FRONTAL PLANE: RADIUS

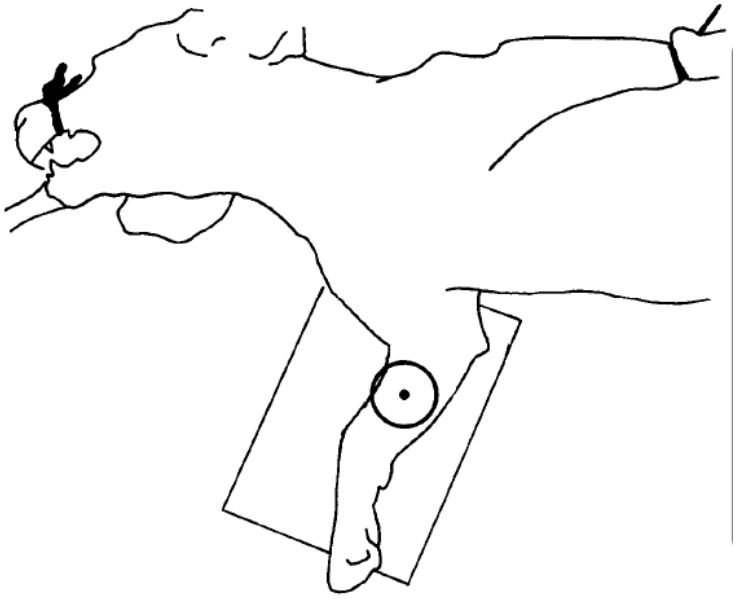




- Lateral and medial side of the anconeal process should not appear
- Well defined distal radial articular surface

SAGITTAL PLANE: RADIUS

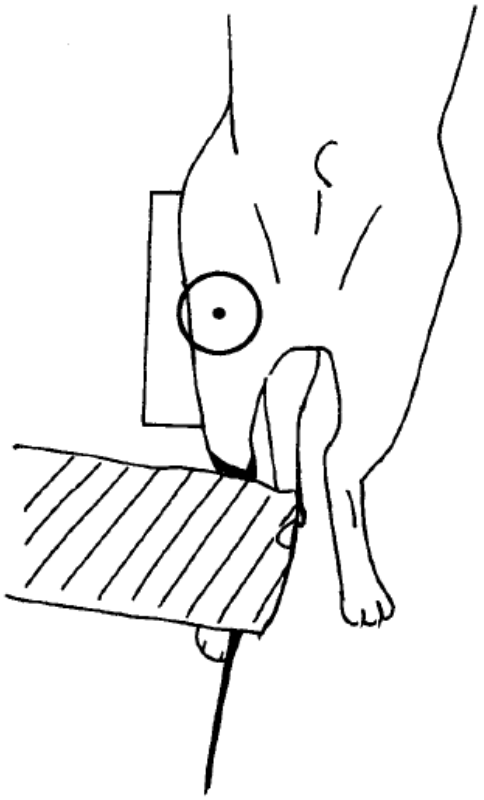




- Well defined trochlear notch
- Over lapping of two humeral condyles
- Well defined joint space between accessory carpal bone and ulnar carpal bone

FRONTAL PLANE: FEMUR

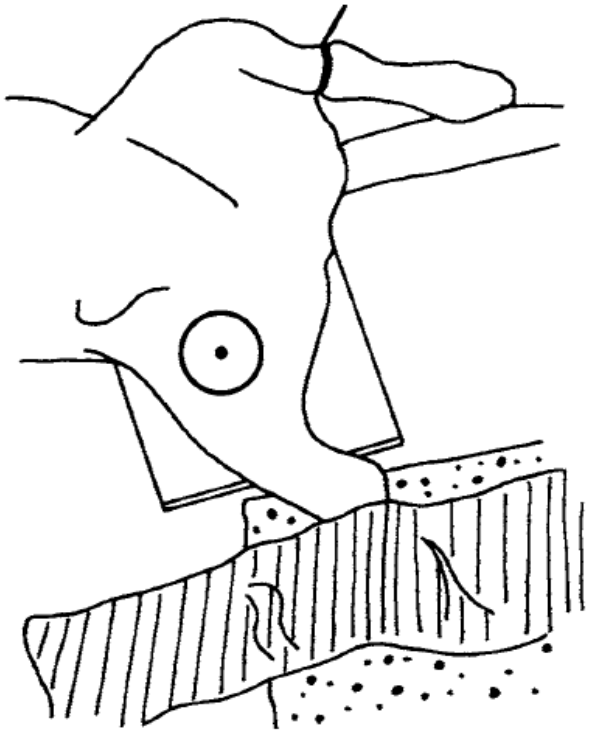




- Patella is in the middle of two femoral condyles
- Well defined femoral condyle margin
- Lesser trochanter

SAGITTAL PLANE: FEMUR

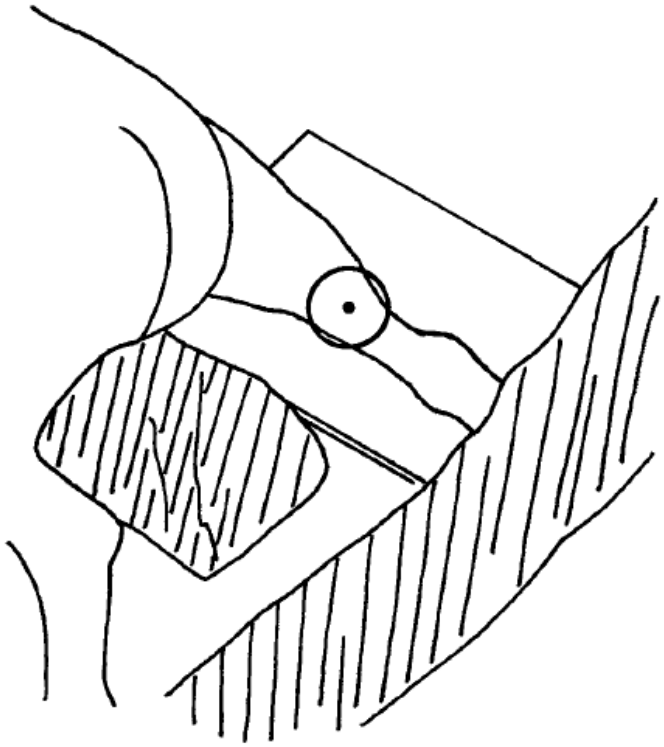




- Femoral head is divided by cortex
- Over lapping of two femoral condyles

FRONTAL PLANE: TIBIA

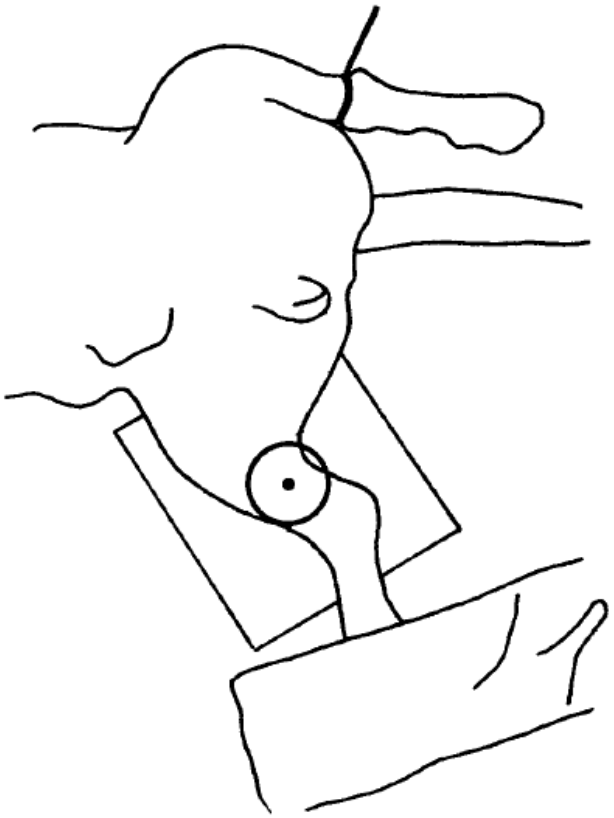




- Femoral condyles
- Well defined arciform groove

SAGITTAL PLANE: TIBIA





- Over lapping of two lateral and medial condyles at proximal site

READING OF RADIOGRAPH

- Under/Over exposure
- (VD, DV view) Animal's Right on your Left (animal facing you)
- (VD, DV view)Head always Up
- (Lateral view) Head always on your Left
- Anatomy: regions of interest (centering)
- Bone anatomy
    - Epiphysis (articular cartilage, cancellous bone and subchondral bone)
    - Physis (growth plate) present only in immature animal, composed of hyaline cartilage
    - Metaphysis (cut bone zone)
    - Diaphysis
    - Joint space; Periosteum; Soft tissues

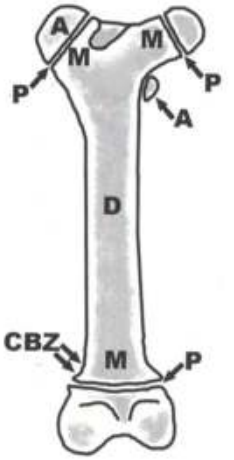


Figure 4.1 Normal immature canine femur. **A** = apophysis, **E** = epiphysis, **P** = physis, **M** = metaphysic, **D** = diaphysis, **CBZ** = cut back zone.

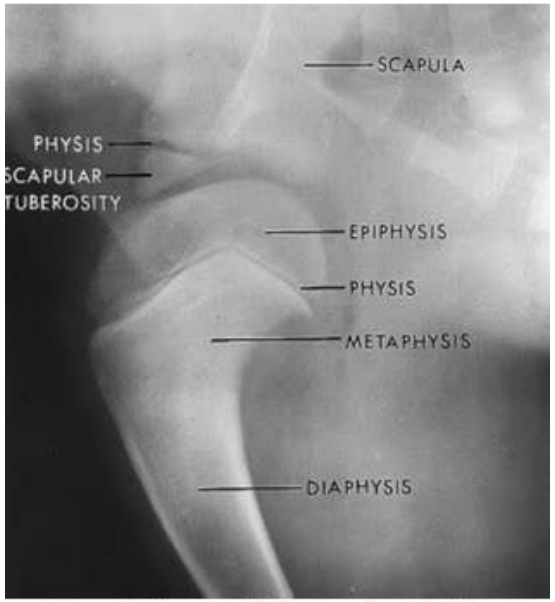
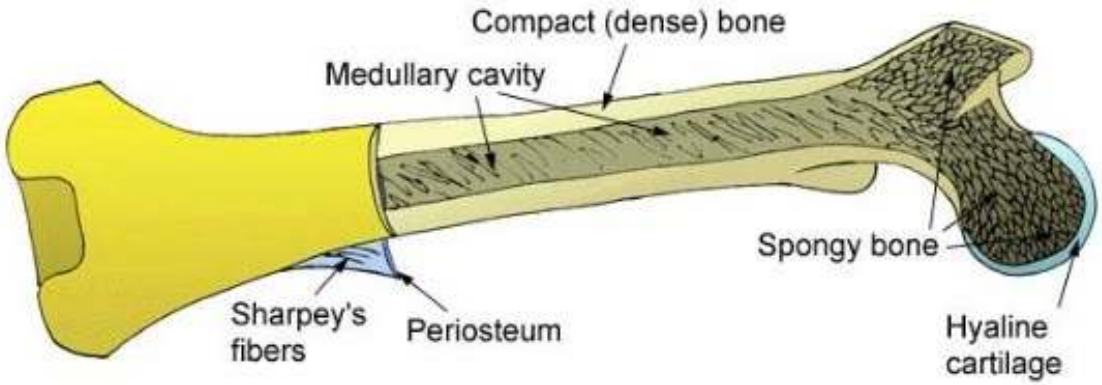


Figure 4-2 A normal immature shoulder joint.

BONE STRUCTURE





- Cancellous (spongy) bone : made up of tiny flakes of bone called trabeculae（小梁）

- Compact, Cortical bone, Cortex: dense, smooth appearance

Box 1.1 Checklist for interpreting orthopedic radiographs.

- ✓ At least two orthogonal radiographs
- ✓ Radiographs are those of the animal being evaluated
- ✓ Technique:
  - ✓ Exposure
  - ✓ Positioning
- ✓ Animal's age, body condition, and conformation
- ✓ Bones:
  - ✓ Margination
  - ✓ Opacity
  - ✓ Geometry
  - ✓ Lesion distribution
- ✓ Joints:
  - ✓ Alignment
  - ✓ Width of joint space
  - ✓ Periarticular findings
  - ✓ Lesion distribution
- ✓ Soft tissues:
  - ✓ Thickness
  - ✓ Opacity
- ✓ Reassess based on clinical, physical, and laboratory findings
- ✓ Prioritize list of differential diagnoses
- ✓ Make follow-up radiographs (assess rate of change)

GENERAL DESCRIPTION OF ABNORMALITIES

A. Shape and contour:

1. Deformed 2. Valgus, varus, antecurvatum, recurvatum, torsion

3. Fracture a. Open / close b. Type: simple / comminuted, Transverse, oblique, spiral c. Other description: dislocation, fragment

B. Description of density (calcification)

1. Radio lucent (透空的) 2. Osteolytic (骨溶解的) 3. Sclerotic (硬化的) 4. Granular (粒狀的) 5. Patchy (小片狀的) 6. Mottled (斑駁的)

C. Cortex :

1. Thicker, thinner 2. Contrast between cortex and medullary canals : diminished

D. Trabecular structure of cancellous bone

1. Blur, hazy (模糊、朦朧的) 2. Accentuation (加強的)

E. Epiphysis and joint

1. Open, close, retained epiphyseal line 2. Increased, decreased space

F. Lesion distribution

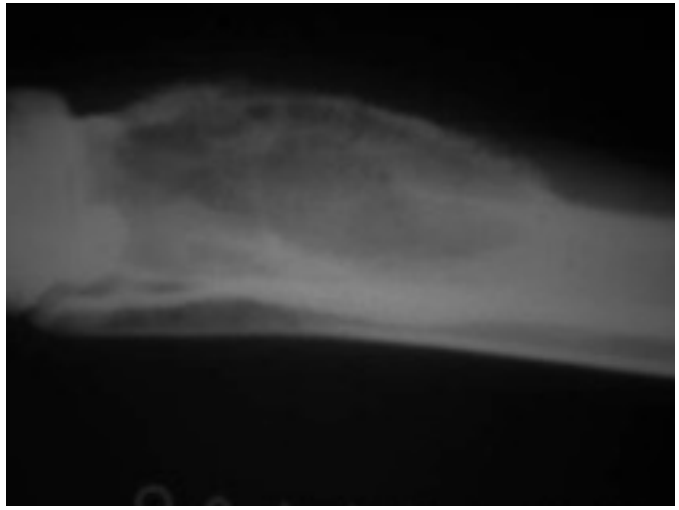
1. Monostotic, polystotic 2. Unilateral or bilateral 3. Symmetrical or asymmetrical

- **<u>Increased opacity</u>**: increased mineralization or production of new bone. It may result from disease process within the bone or response to trauma or stress.
  - **Sclerosis** 硬化: increased radiographic opacity in bone.
    - 在X光下骨質較密的部位
    - 可能因為鈣化較多或骨密度較高
    - 為骨組織對各種傷害的反應，也可以說是骨的疤痕
    - 可以在關節軟骨erosion之下方觀察到，可在骨髓炎周圍看到，也可在較早發生、已癒合的hair-line fracture處看到。
  - **Sequestrum** 死骨：dead bone fragments without cellularity, pure calcium deposit.





- **<u>Decreased opacity</u>**: structure destroyed or resorbed due to trauma, disease, infection, neoplasia.
  - **Change in trabecular pattern**：may become hazy or coarse or disappear. It may be the first indication of a disease process.
  - **Decalcification (generalized osteoporosis)**
    - Deficiency of mineralization and matrix
    - 骨骼全面失去密度及細節。
      - 幼年動物營養不足或不平衡Rickets。
      - Osteomalacia
        - 成年動物的骨鈣流失
        - sufficiency of matrix but inadequate mineralization.
        - e.g., renal rickets, disuse atrophy
  - **Osteolysis: (localized osteoporosis)**
    - 骨骼局部失去應有的密度
    - 大多數因為某部份骨頭死亡，並被溶解
      - 骨腫瘤、骨髓炎，界限不規則
      - 死骨(sequestra)周圍，界限整齊
  - **Bone cyst**:因為局部未骨化而形成



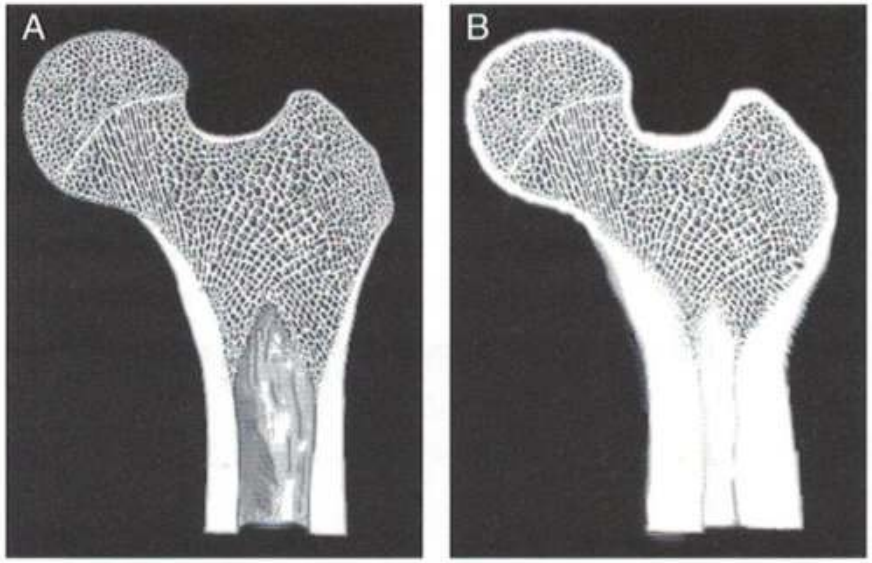


Figure 4.78 Osteopetrosis. **A.** Normal bone. **B.** Thickened, dense bone due to abnormal osteoclastic activity in growing animal.

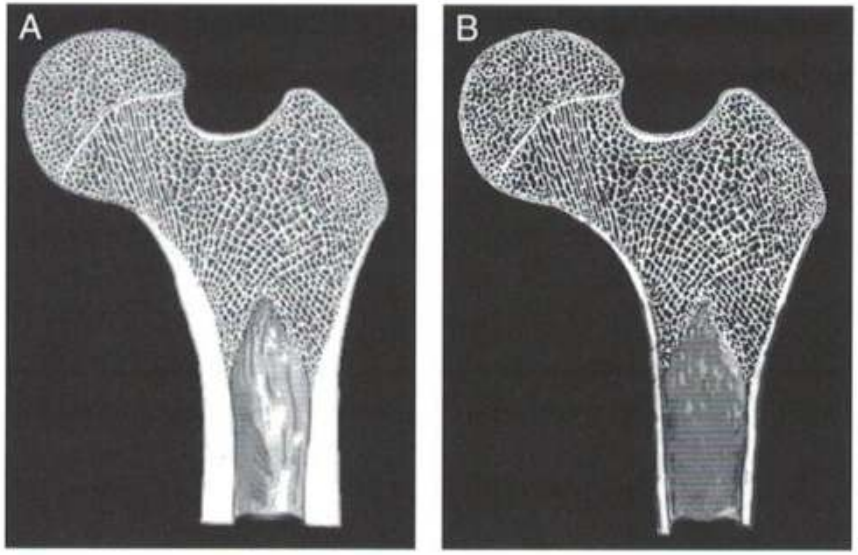


Figure 4.80 Osteopenia. **A.** Normal bone. **B.** Decreased bone mass and thin cortices due to an imbalance between bone formation and bone resorption.

OSTEOLYSIS

Table 1.1 Patterns of osteolysis

Pattern	Geographic	Moth-eaten	Permeative
Radiographic appearance			
Size, number of lesions	Large, solitary (over 10 mm)	Small, multiple (3–10 mm)	Tiny, numerous (1–2 mm)
Margination	Well-defined	Poorly-defined	Ill-defined
Zone of transition	Short, well-defined	Long, poorly-defined	Diffuse, nondefined
Cortical remodeling	Normal or thinned	Irregular thinning, sometimes disrupted	Thin, indistinct, often disrupted
Osteosclerosis	May surround lesion	Absent	Absent
Type of lesion	Benign, nonaggressive	Aggressive	Very aggressive
Examples	Bone cyst, pressure remodeling	Tumor, osteomyelitis	Malignant tumor, severe osteomyelitis

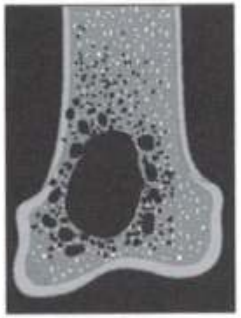


Figure 1.1 Mixed pattern of osteolysis. An aggressive disease process causing peripheral permeative osteolysis with an inner moth-eaten pattern and central geographic pattern (latter due to coalescence of moth-eaten osteolysis).

PERIOSTEAL RESPONSE

Table 4.3 Stages of periosteal response

Stage	Appearance	Description
I		Bone injury or disease causes separation of periosteum from cortex.
II		Separation may result from subperiosteal accumulation of blood, edema, purulent material, or tumor cells.
III		Blood vessels and Sharpey's fibers extend from cortex perpendicular to periosteum.
IV		Mineral deposits are laid down along these fibers and blood vessels, perpendicular to periosteum.
V		Mineral deposits continue until subperiosteal space is filled.
VI		When disease process stops, mineral deposits can mature into more organized and homogeneous bone.
VII		New bone continues to remodel and develop more well-defined margins.
VIII		Remodeling gradually creates trabeculation and new bone begins to resemble parent bone.
IX		Normal stresses on bone gradually return it to normal shape.

PERIOSTEAL RESPONSE

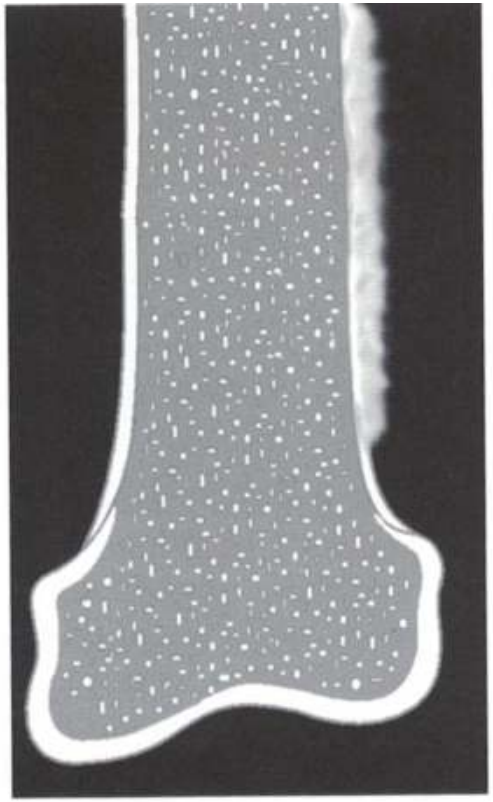


Figure 1.2 Active mild or early periosteal response. Smooth but ill-defined new bone production is visible along right cortex of long bone diaphysis; margins of new bone are hazy and ill-defined (difficult to trace with a pencil).



Figure 1.3 Inactive periosteal response due to benign disease. Smooth new bone production is visible along right cortex of long bone diaphysis; margins of new bone are well-defined (easy to trace with a pencil).

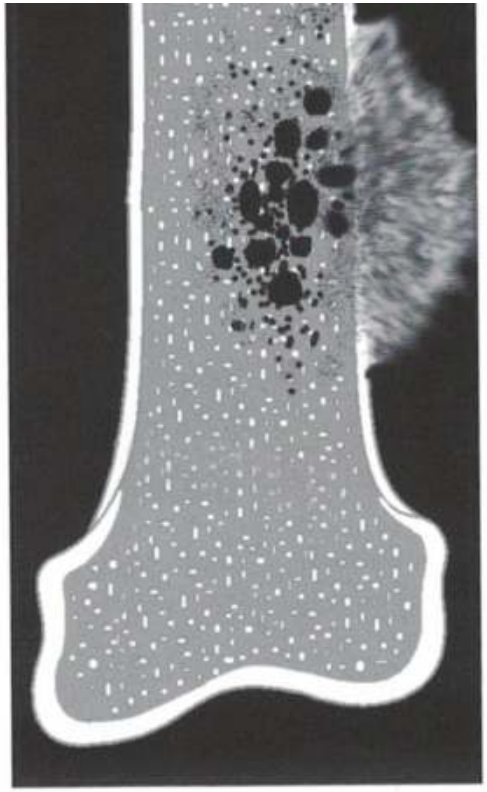
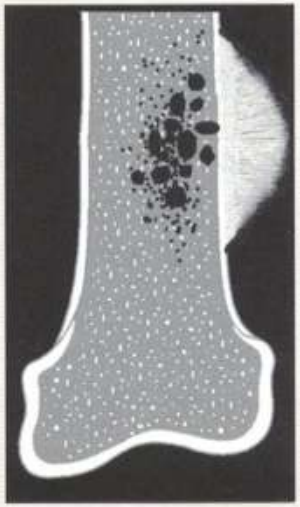


Figure 1.4 Active periosteal response with aggressive disease. Irregular new bone is visible along right cortex of long bone diaphysis; margins are ill-defined. Mixed pattern of osteolysis is present in adjacent bone.

Table 1.2 Activity and aggressiveness of bone disease

Lesion	Active and Aggressive	Active and Nonaggressive	Active and Chronic	Inactive and Nonaggressive
Radiographic Appearance				
Margination	Ill-defined	Ill-defined	Ill-defined	Well-defined
Size and shape of new bone*	Large; irregular	Medium; smooth or irregular	Organized (maturing)	Small; smooth
Osteolysis	Often severe; ill-defined	Well-defined or absent	Well-defined or absent	Usually absent
Appearance of cortex	Destructive remodeling	Displaced, thinned, or normal	Normal, displaced, or thinned	Normal
Soft tissue swelling	Moderate to Severe	Mild	Mild or absent	Absent
Rate of change	Rapid	Slow	Minimal	None
Zone of transition	Long and poorly demarcated	Indistinct but visible	Well-defined	Well-defined
Osteosclerosis	Absent	Ill-defined	Well-defined	Well-defined
Examples	Malignant neoplasia, severe osteomyelitis	Healing fracture, low grade osteomyelitis	Healing fracture, low grade osteomyelitis	Healed fracture

*NOTE: All lesions start small, but active and aggressive lesions tend to grow quickly, while active and chronic lesions tend to enlarge slowly over a long period of time.

- Codman's triangle:
  - At periphery of rapidly growing disease process
  - Result from lifting of periosteal new bone
  - A small, solid, wedge-shaped area of mineralization at periphery of lesion
  - Well defined margins and cortex underlying Codman's triangle is intact



Figure 4.76 Irregular-shaped periosteal response with ill-defined margins, cortical destruction, and soft tissue swelling (active and aggressive disease process).

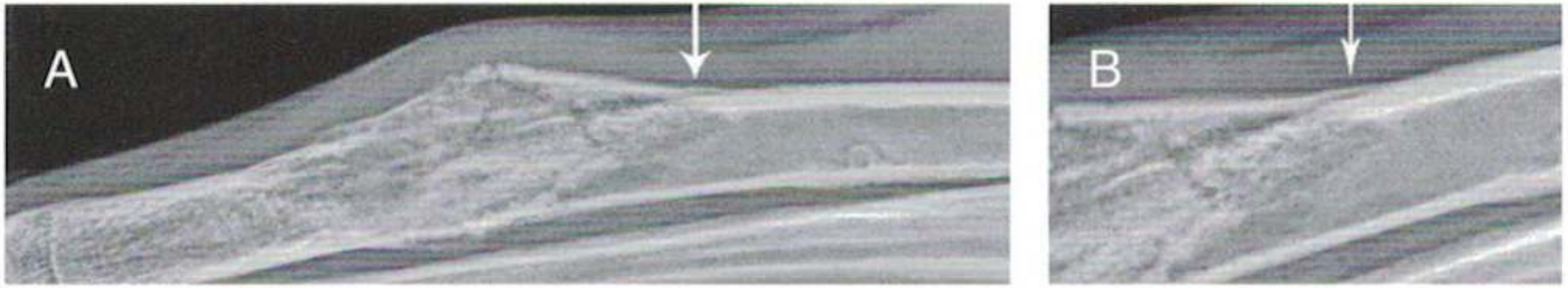
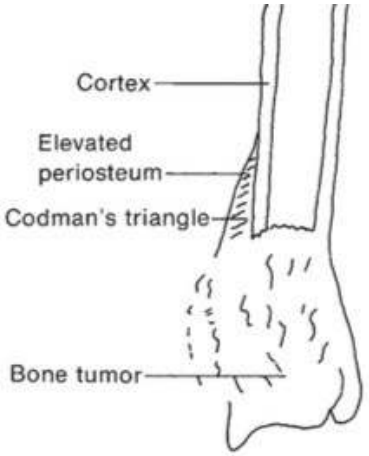
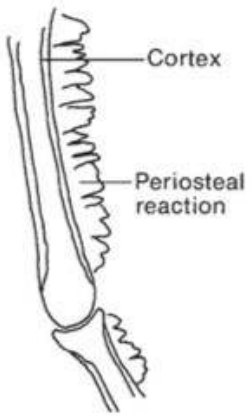


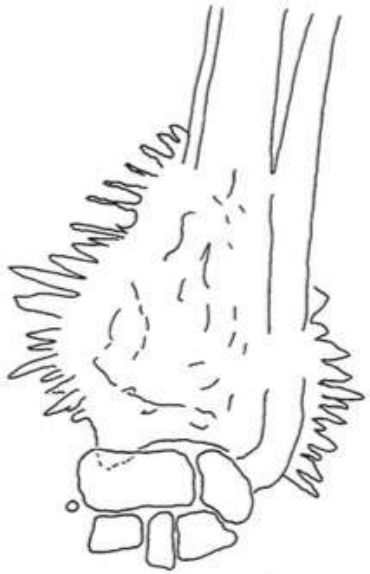
Figure 4.77 Codman's triangle (arrow), a well-defined wedge-shaped area of new bone at periphery of an active, aggressive lesion. Underlying cortex is lytic but visible. **A.** Wide view. **B.** Close-up view.



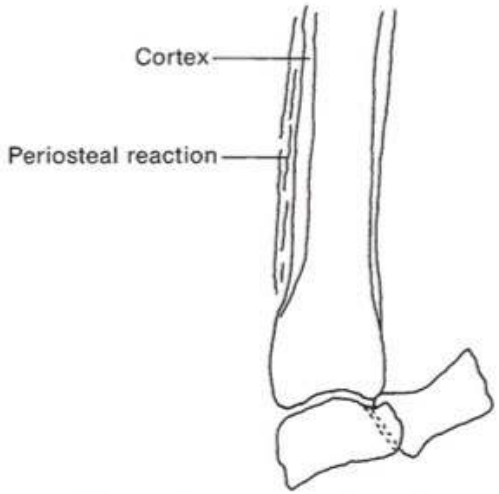
The tumor elevates the periosteum, resulting in subperiosteal new bone formation — often a feature of malignancy, but also seen in some cases of osteomyelitis.



This palisade-like periosteal reaction is seen in hypertrophic osteopathy and sometimes in osteomyelitis.



A "sunburst" type of periosteal reaction is sometimes, but not always, seen associated with osteosarcoma or chondrosarcoma.



A linear (lamellar or laminated) type of periosteal reaction may be seen with callus formation, or it may result from metabolic disease or irritation of the bone from a variety of causes.

Figure 4-4 Four kinds of periosteal reaction. The tumor elevates the periosteum, resulting in subperiosteal new bone formation, often a feature of malignancy but also seen in some cases of osteomyelitis. This palisade-like periosteal reaction is seen in hypertrophic osteopathy and sometimes in osteomyelitis. A "sunburst" type of periosteal reaction is sometimes seen associated with osteosarcoma or chondrosarcoma. A linear (lamellar or laminated) type of periosteal reaction may be seen with callus formation, or it may result from metabolic disease or irritation of the bone from a variety of causes.

SIGNS OF JOINT DISEASE

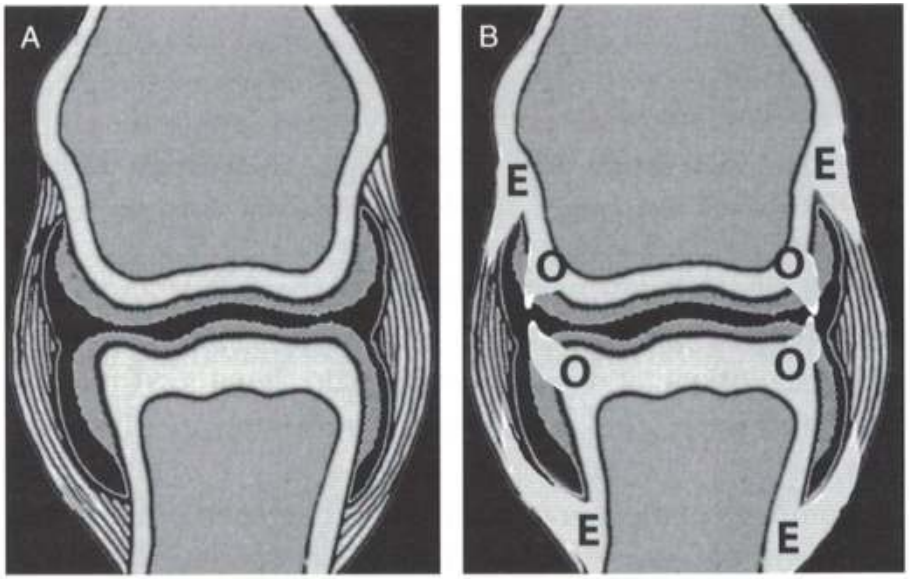


Figure 1.5 Osteophytes and enthesophytes. **A.** Normal joint. **B.** Degenerative joint disease with osteophytes **(O)** forming at periphery of articular cartilage and enthesophytes **(E)** forming at attachments of joint

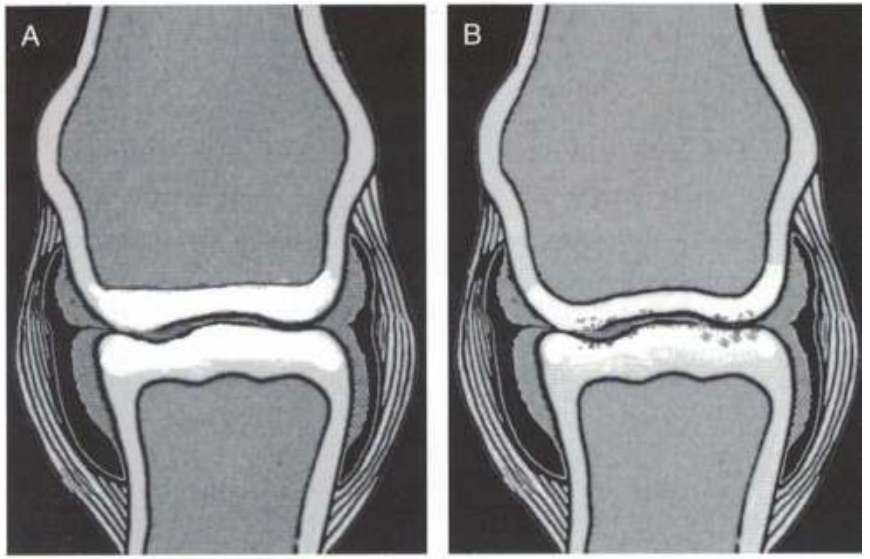


Figure 1.6 Degenerative joint disease. Joint space is narrowed due to loss of articular cartilage. **A.** Subchondral bone is increased in opacity (osteosclerosis). **B.** Focal osteolysis (cystlike lesions) in subchondral bone due to erosion. Normal joint is depicted in Figure 1.5A.

Osteophytes:

- Develop at peripheral margins of articular cartilage in non weight bearing area.
- Caused by damage to cartilage which leads to mineralization of cartilage and development of a periarticular bony spur.

Enthesophytes:

- Occur at attachment sites of tendons, ligaments, joint capsules, fascia
- Caused by chronic abnormal tension or trauma to the attachment, leading to mineralization of the fibrous tissue and formation of bony spur.

SIGNS OF JOINT DISEASE

Box 1.2 Radiographic signs of joint disease.

- Malalignment of bones (may be mimicked by poor positioning)
- Joint capsule distension (e.g., joint effusion, synovial thickening)
- Altered width of joint space (widened or narrowed)
- Osteophytes (form at edges of articular margins)
- Enthesophytes (form at attachment sites)
- Subchondral osteosclerosis or osteolysis
- Soft tissue mineralization (periarticular or intra-articular)
- Intra-articular gas (vacuum phenomenon)

Fracture healing

- Hematoma initially forms at fracture site
- Increased blood flow leads to active remodeling of fracture margins
- Granulation tissue develops and begins to mineralize, forming a callus between fragments
- Callus matures to bone and gradually remodels over time

Table 4.4 Fracture healing

Time	Appearance	Description
0 days		• Initial fracture • Soft tissue swelling
5–10 days		• Active bone remodeling results in widening of fracture line and fracture • Soft tissue swelling decreases. • Subcutaneous emphysema (postoperative) resolves.
10–20 days		• Endosteal and periosteal callus becomes visible. • Gap between fracture fragments narrows. • Fracture ends become less opaque due to bone resorption.
30 days		• Fracture lines gradually disappear. • Callus increases in opacity and becomes smoothly marginated.
90 days		• Callus continues to mature and develops trabeculation. • Medullary cavity gradually is reestablished. • Cortical margins become more distinct.

犬常見骨科疾病之分類

- 發育期骨關節疾病 ＝發育期
- 外力造成的疾病一挫傷，脫臼，外傷性發炎，韌帶斷裂等＝任何年齡
- 炎症性關節炎（感染，免疫性：風濕性關節炎等）＝感染好發於幼犬，免疫性好發於中高齡犬
- 退化性疾病（軟骨的退化）＝多發於老犬，但幼犬也可能因為其他疾病而造成此症狀
- 腫瘤＝多發生在成年犬及老犬
- 其他關節疾患

發育期骨科疾病

- 主要:肩關節分離性骨軟骨炎，肘發育不良（肘突起癒合不良，內側勾狀突起斷裂，分離性骨軟骨炎），髖關節發育不良，Legg-Calve-Perthes’疾病，留存性軟骨核
- 常發生在大型犬
- 遺傳有關
- 因軟骨受損而導致DJD

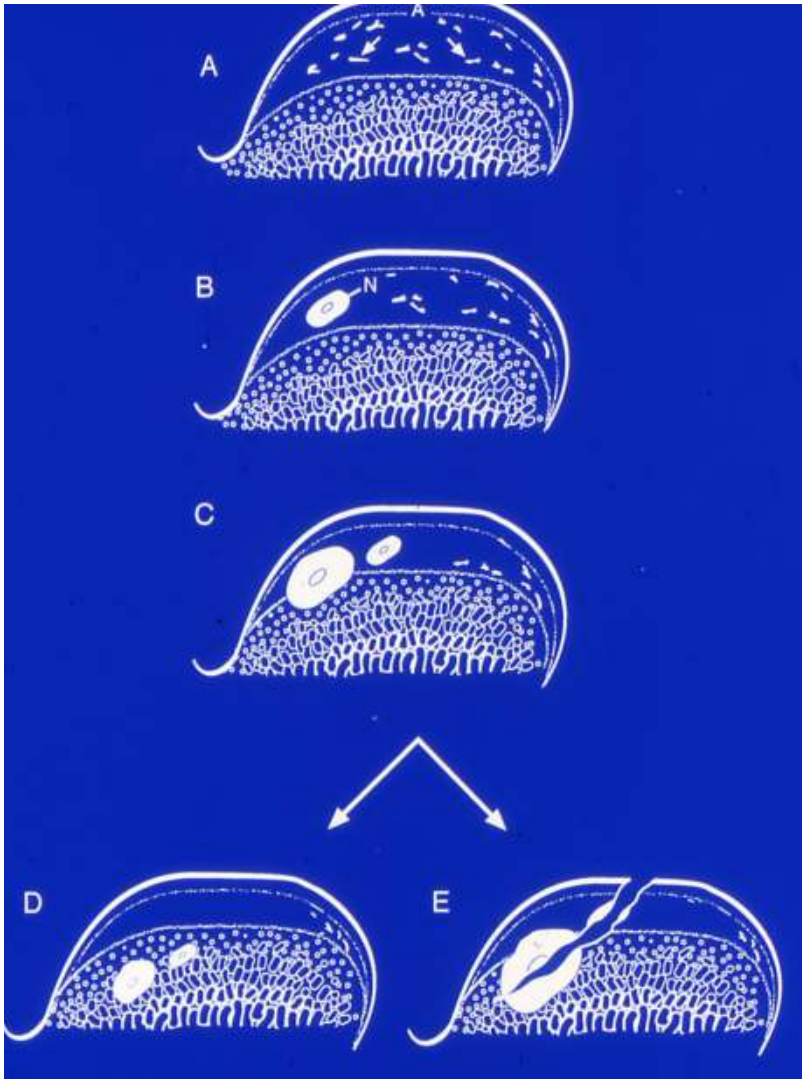
分離性骨軟骨炎
(Osteochondritis dissecans: OCD)

- 大型狗
- 好發關節：肩關節、肘關節、膝關節、跗關節
- 公 vs 母＝３：１
- 約有50％的病例是雙腳皆有



OCD的發病機制以及病理生理學

- 軟骨內骨化異常
- 關節軟骨缺血
- 軟骨層肥厚及易脆
- 軟骨層剝離（軟骨瓣＝關節鼠）
- 滑膜炎以及軟骨下骨的腐蝕，發炎引起的疼痛



OCD的影像及病灶位置

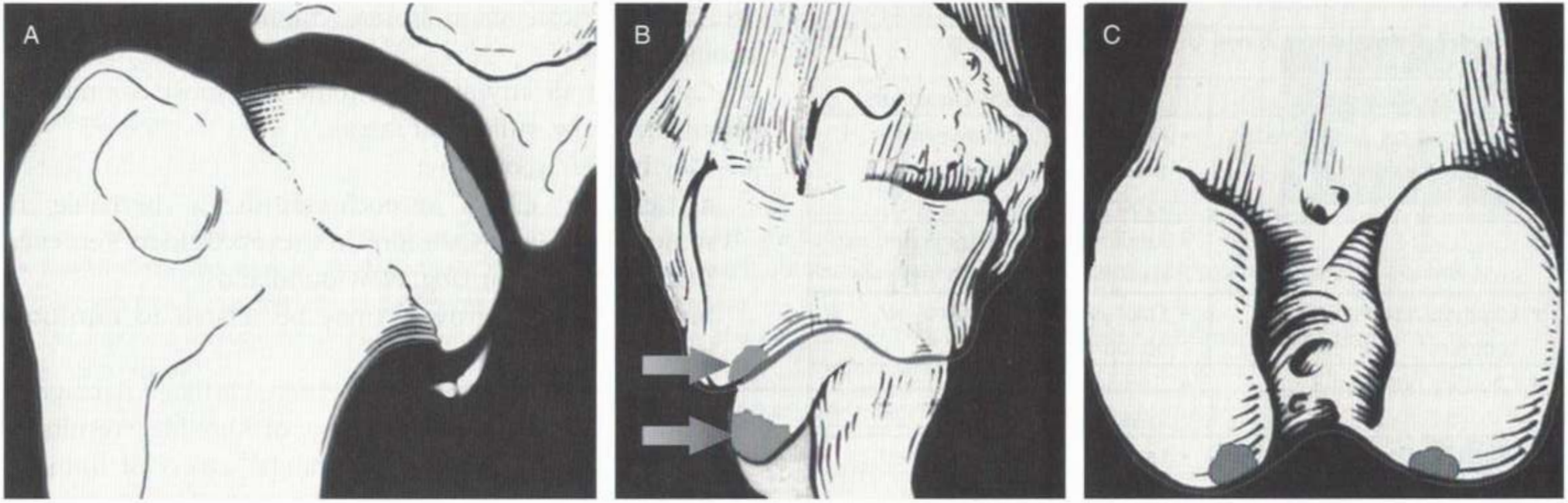


Figure 4.114 Osteochondritis dissecans; gray shaded areas depict common sites of occurrence. **A.** In shoulder on caudal aspect of humeral head (ossified free fragment is present ventral to humeral head). **B.** In elbow on medial aspect of humeral condyle or medial coronoid process (arrows). **C.** In stifle on femoral condyles.

肘關節發育不良

- 肘突起癒合不良 (ununited anconeal process:UAP)
  - 多發生於德國狼犬，遺傳的背景？
- 內側勾狀突起斷裂（fragmented medial coronoid process:FMCP)
- 分離性骨軟骨炎(OCD)

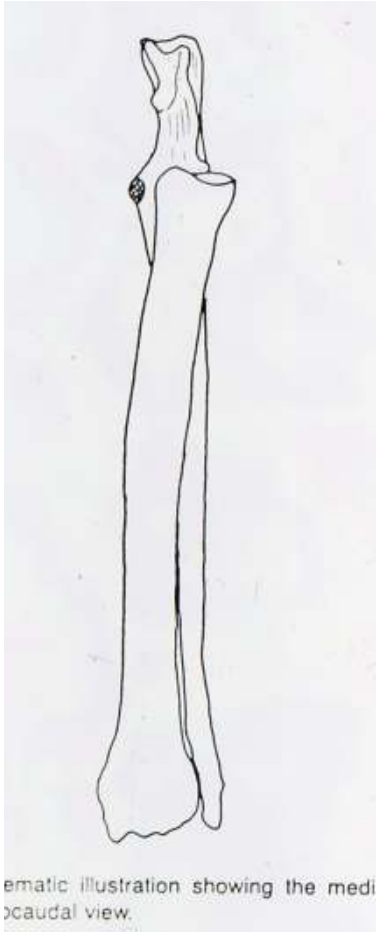
UAP



內側勾狀突起斷裂（FMCP)

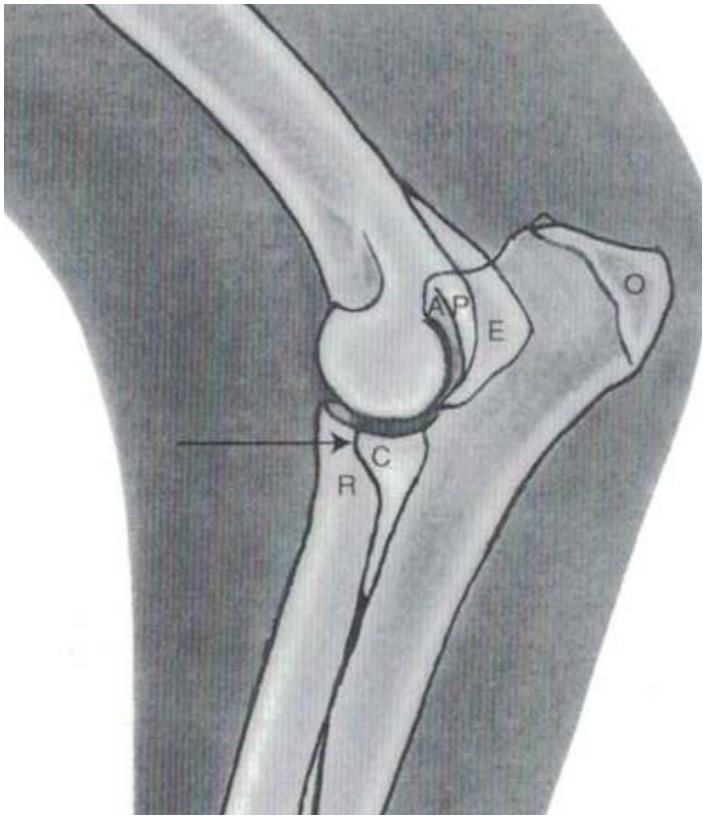
- 因為骨軟骨病造成勾狀突起斷裂？
  - 肱骨-橈骨之間的關節面不一致，或是橈骨尺骨間的生長不一致？
  - 此結果造成嚴重DJD以及慢性疼痛
- 7～10個月齡發病
- 通常是兩側
- 好發品種：
  - 伯恩山犬，挪威那，黃金獵犬，拉不拉多犬，德國狼犬（較少見於喜樂蒂牧羊犬，博美狗以及米格魯）

FMCP的X光攝影

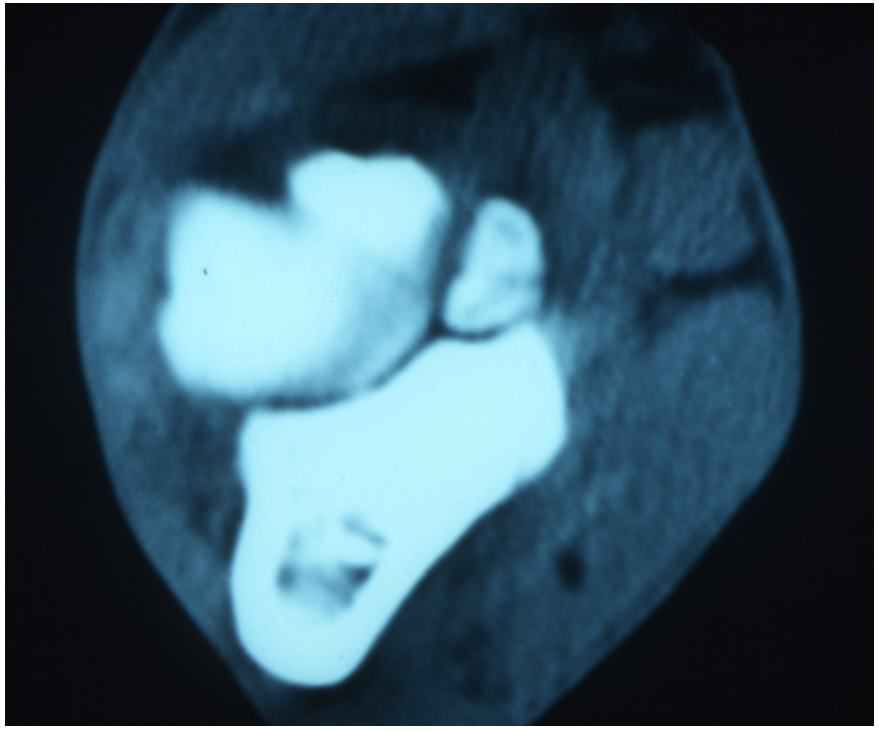


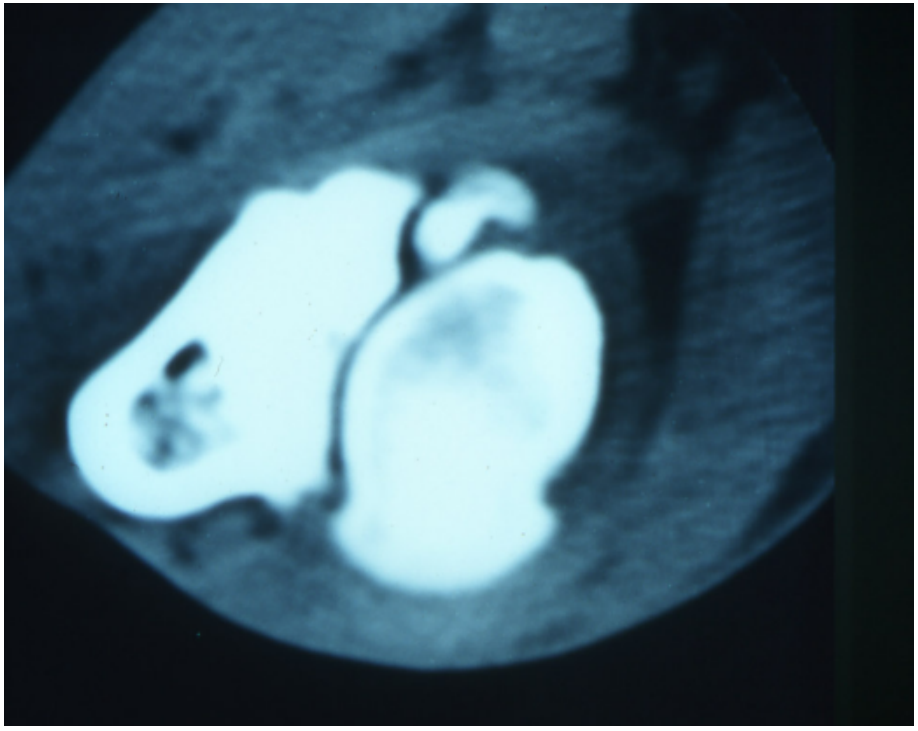
ematic illustration showing the medi
caudal view.





Computed Tomography (CT)



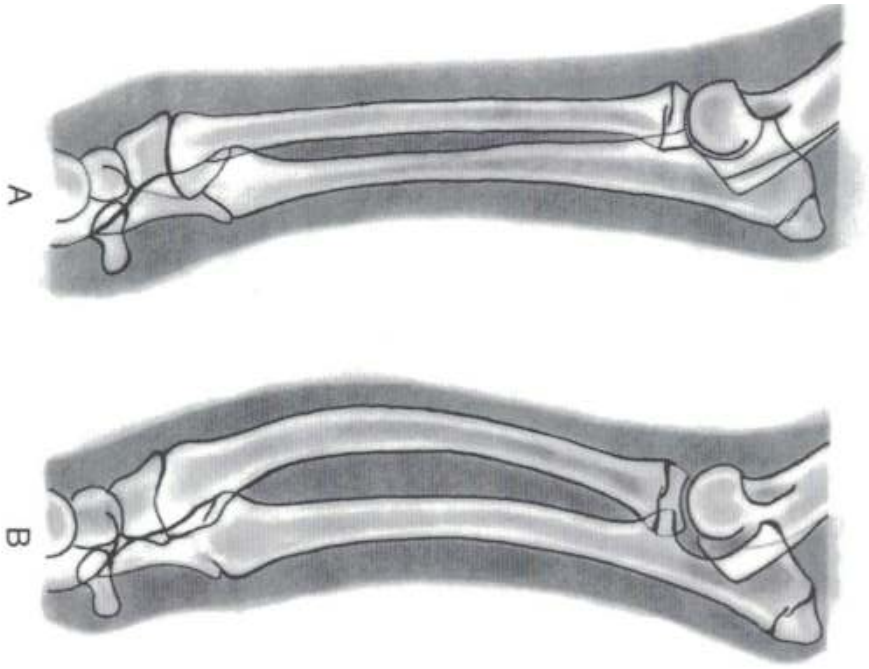


留存性軟骨核
(RETAINED CARTILAGE CORE)



圖 15-18　遠端尺骨內的存留性軟骨核。尺骨遠端的幹骺端中可見一個三角形透明區（白色箭號）。鄰近放射線透明區處則可見骨質硬化的薄緣。

1. Large breed dogs
2. Distal ulna physis
3. Disruption of endochondral ossification
4. Retention of hypertrophic chondrocytes
5. Angular limb deformity



髖關節發育不良(Hip Dysplasia)



正常髖關節





髖關節發育不良(Hip Dysplasia)

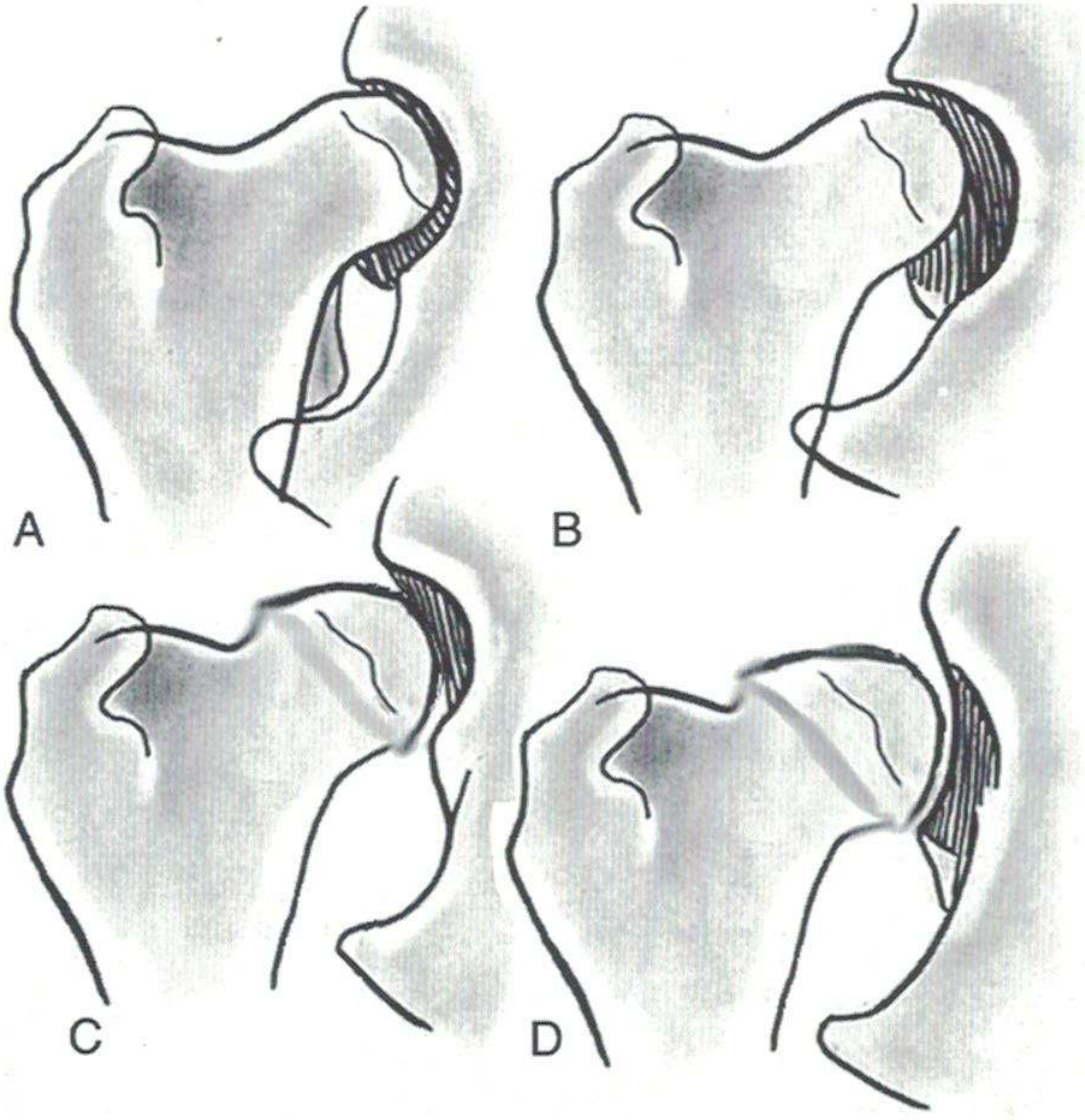


Figure 4.146 Grades of canine hip dysplasia. **A.** Normal hip. **B.** Mild, subluxation. **C.** Moderate subluxation, shallow acetabulum, and thickened femoral neck. **D.** Severe luxation, shallow acetabulum, secondary degenerative joint disease. (Adapted from Owens and Biery 1999.)

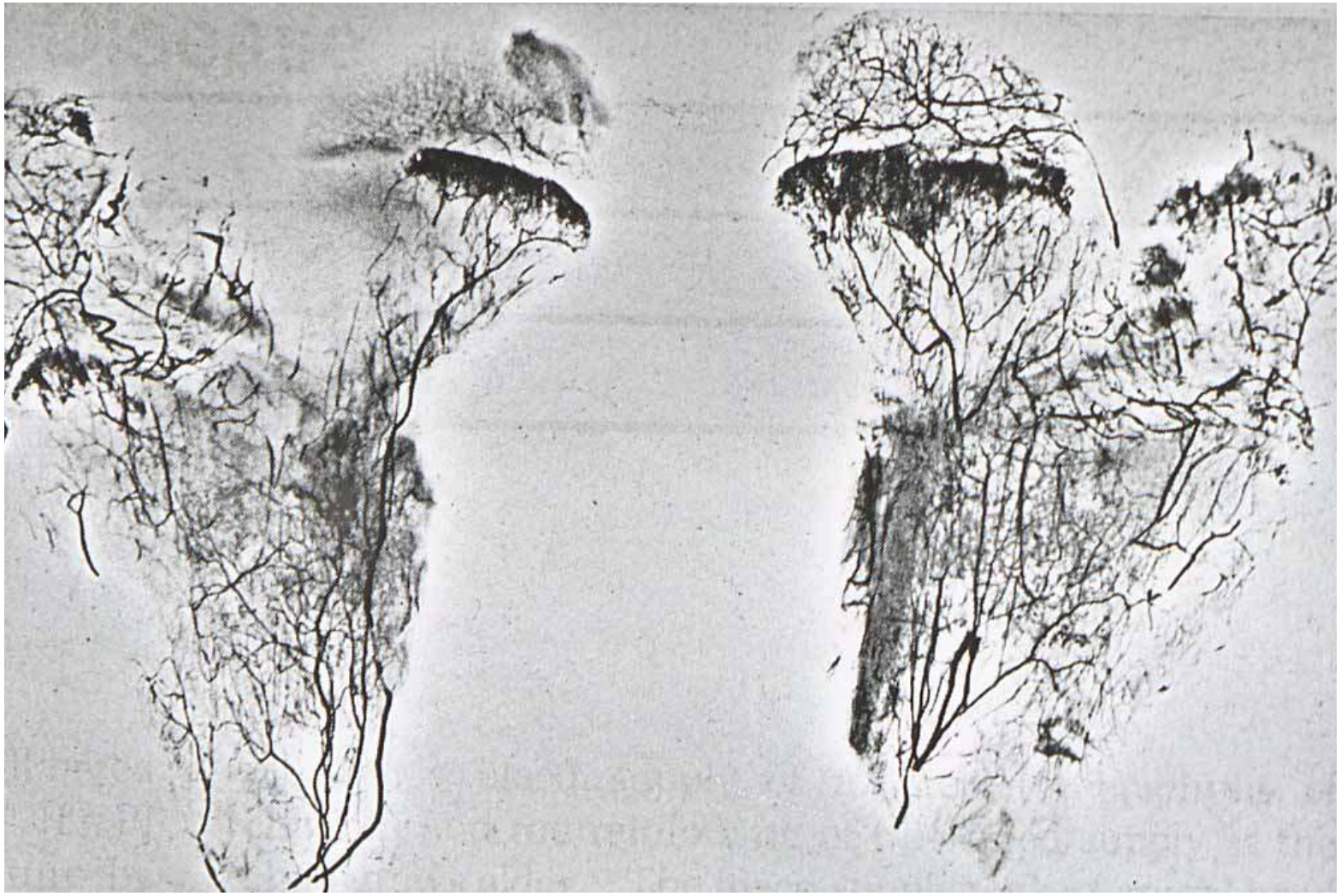
Legg-Calve-Perthes’疾病

- 在4～10個月齡發病
- 單側，有時候雙側
- 大部分的病畜都不使用患肢負重
- 進行性肌肉萎縮（肌肉不使用引起的萎縮）
- 髖關節觸診雖然沒有腫脹，腳往後伸展會引起強烈疼痛
- 捻髮音

Legg-Calve-Perthes’疾病的病理生理學

- 股骨頭缺血性壞死（真正原因不明）
- 股骨頭、骨幹端分布的血管阻塞
- 骨壞死，關節軟骨的破壞，滑膜炎
- 骨頭被破壞，成長板分離
- DJD的進行

Legg-Calve-Perthes’疾病骨頭的微血管造影（Microangiography）



Legg-Calve-Perthes'疾病

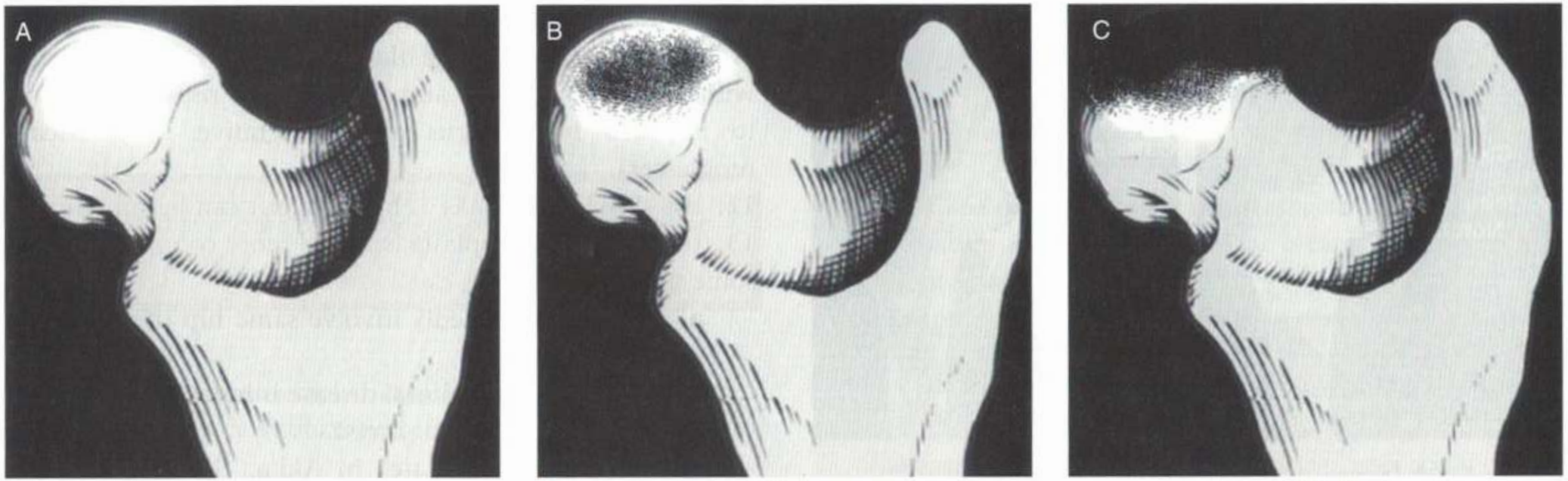


Figure 4.138 Avascular necrosis of femoral head (ventrodorsal radiographs). **A.** Increased opacity of the femoral head. **B.** Demineralization of femoral head. **C.** Collapse of femoral head.

Legg-Calve-Perthes’疾病的診斷

- 跛行的病史，年輕狗，小型犬（馬爾濟斯、約克夏、玩具貴賓等）
- 通常跛行的程度是慢慢的惡化（有些病畜呈現輕度跛行）
- X光診斷：
    - 股骨頭的吸收、變形
    - 股骨頭和髖臼的DJD
    - 股骨頸腫大
    - 股骨頭往外側移動（股骨頭和髖臼的間隔變大-VD的X光攝影）

Legg-Calve-Perthes’疾病

X光攝影





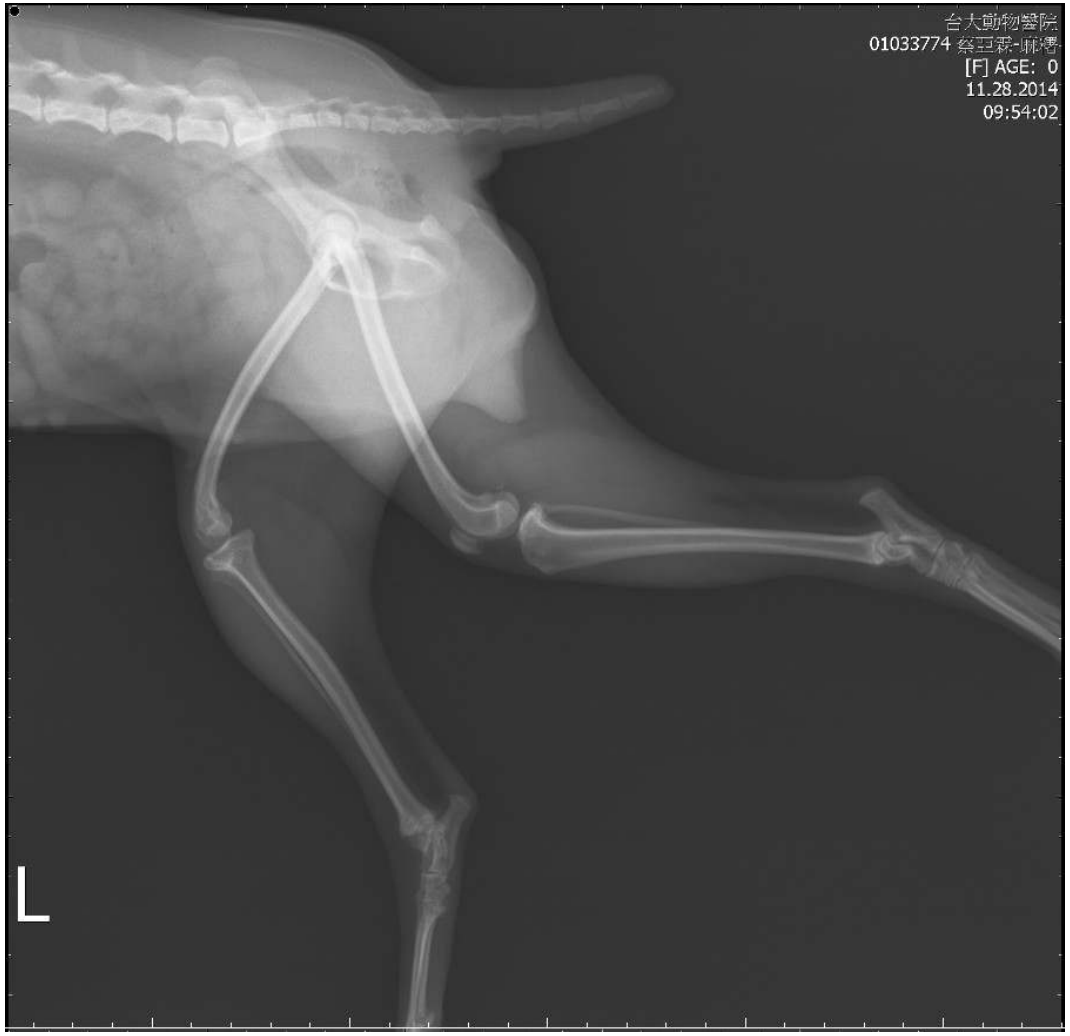
內側膝蓋骨脫臼(MPL)

- 通常是沒有症狀（總是思考造成跛行的真正原因: 臨床獸醫師若發現膝蓋骨異常，傾向將動物跛行的原因診斷為膝蓋骨脫臼。但是若呈現嚴重跛行時，必須思考其他疾病的原因）
- 特徵的描述，跛行的狀況，形式
- 觸診決定級數
- X光攝影
  - 膝蓋骨的位置
  - 股骨和脛骨位置的關係—S形狀的變形，膝關節的關節面的傾斜
  - DJD?

4級MPL-X光攝影

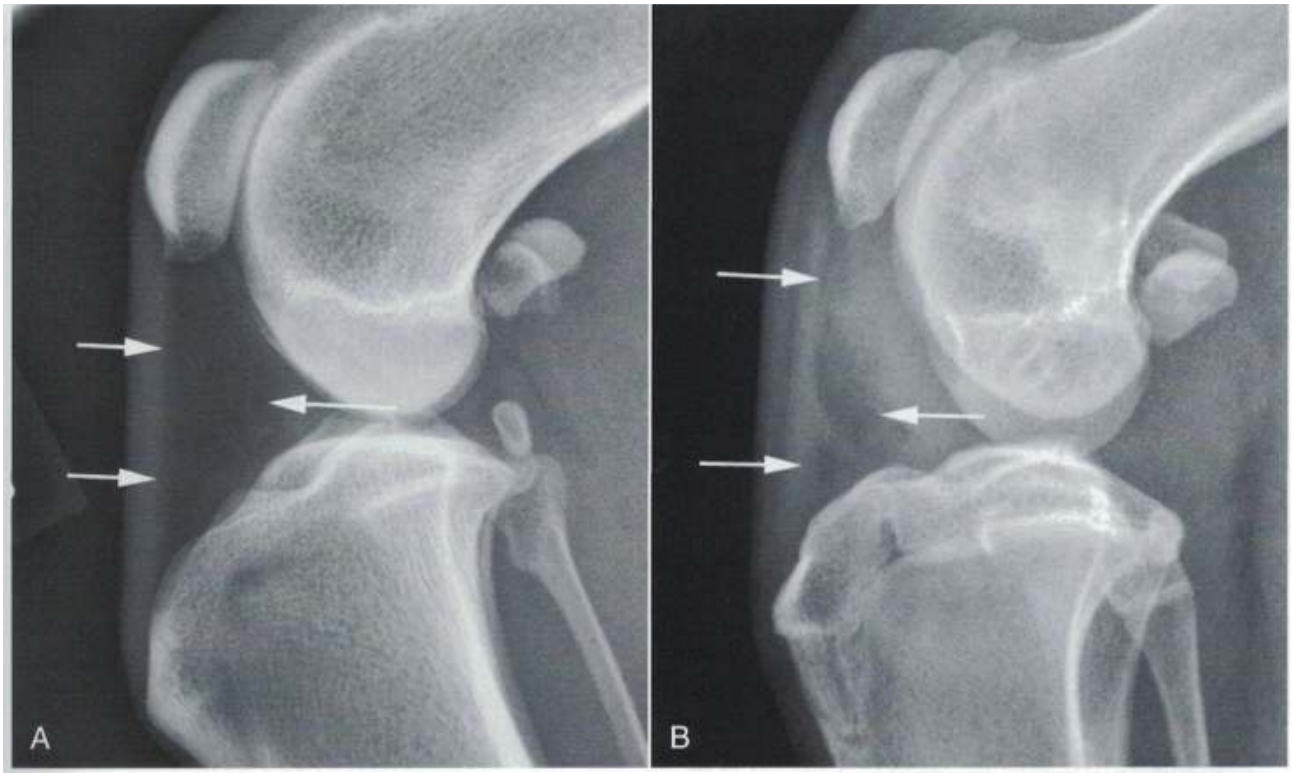


4級MPL-X光攝影


台大動物醫院
01033774 蔡亞霖-麻糬
[F] AGE: 0
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L


台大動物醫院
01033774 蔡亞霖-麻糬
[F] AGE: 0
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RUPTURE OF THE ACL



位於各箭號間的髕下脂肪墊形態改變，即為膝關節滑膜體積無 (A) 或有 (B) 增加的敏感指標。

- Soft tissue and fluid opacity is seen within the joint space displacing the fat pad cranially

RUPTURE OF THE ACL

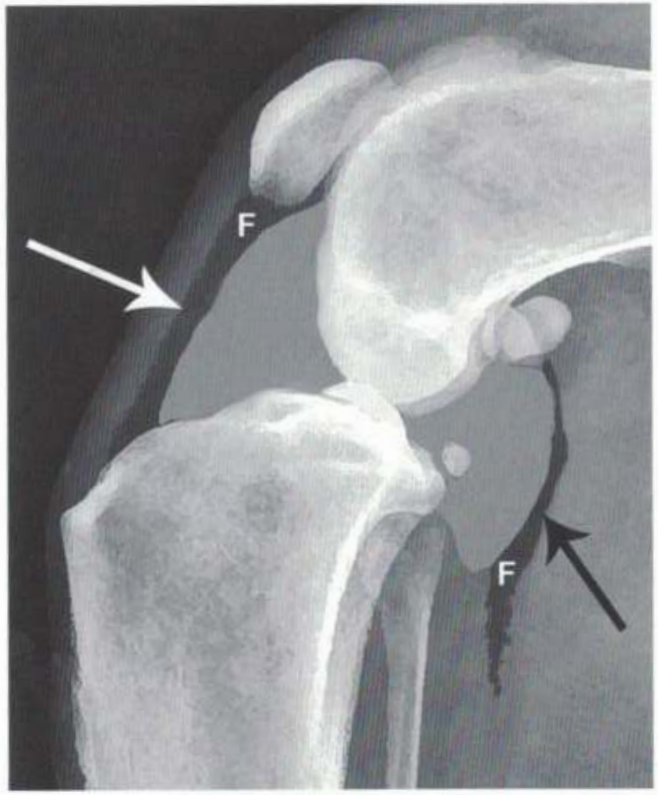


Figure 4.155 Stifle joint effusion (lateral radiograph). Increased soft opacity is present in the femorotibial joint space with partial obliteration of infrapatellar fat pad (white arrow) and caudal distention of joint pouch (black arrow). **F** = fat, caudal to patellar ligament and in fascial plane adjacent to gastrocnemius muscle.

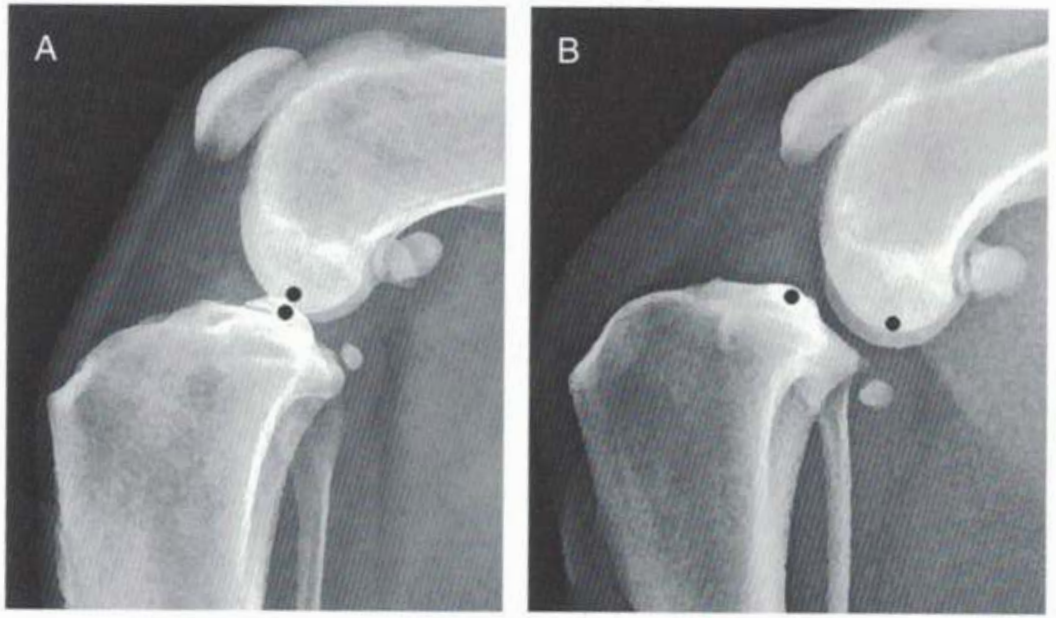
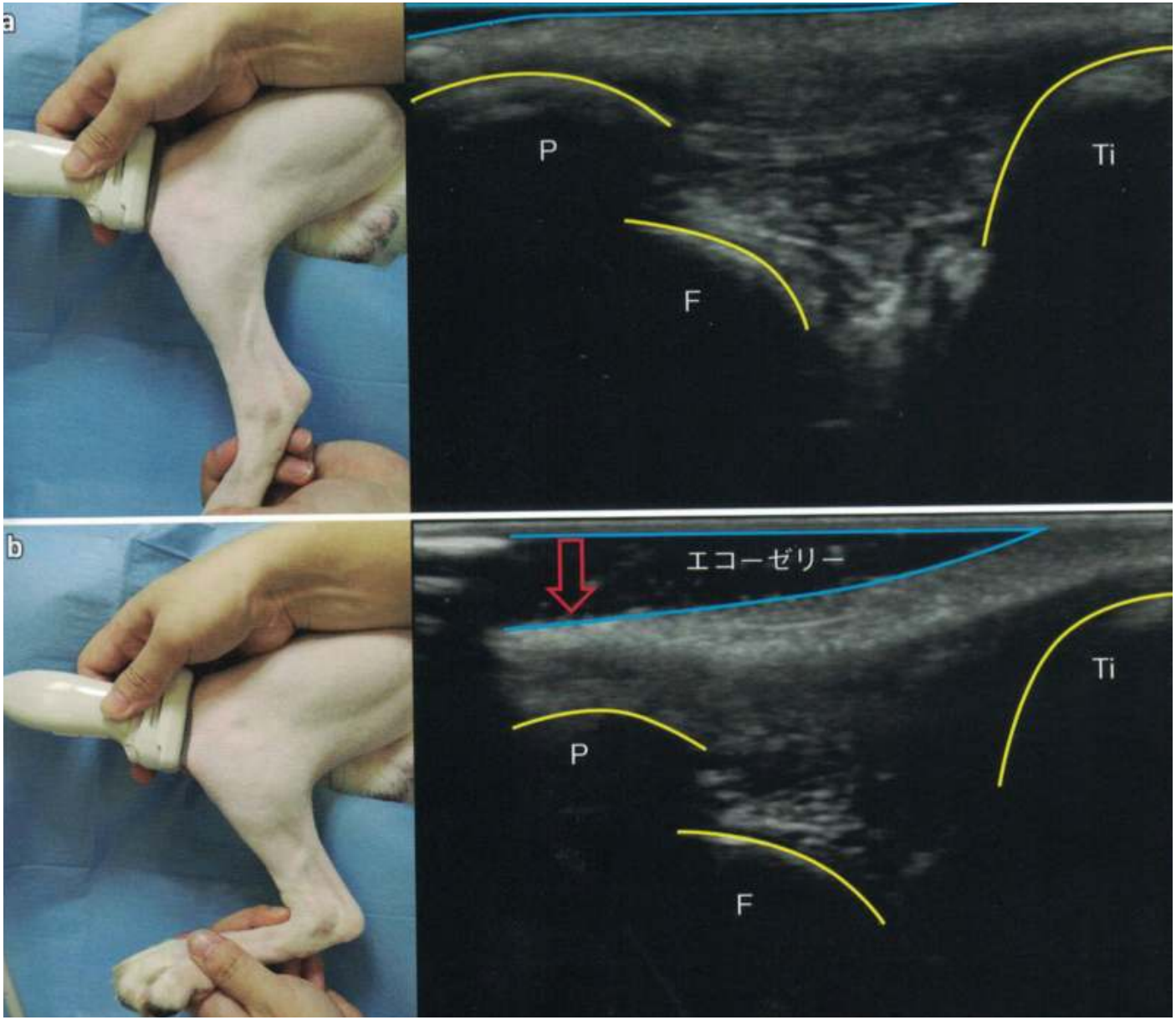


Figure 4.156 Alignment of proximal tibia in relation to distal femur (true lateral radiograph). Black dots designate centers of femoral condyles and intercondylar eminence of tibia. **A.** Normal alignment. **B.** Severe cranial displacement of tibia secondary to damaged cranial cruciate ligament ("tibial thrust" or "cranial drawer sign") with displacement of popliteal sesamoid bone caudally and distally.


a
P
Ti
F
b
エコーゼリー
Ti
P
F


LITTLE WEI
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Page 1/2

IMMUNE MEDIATED ARTHRITIS



圖 18-54 患有免疫性關節病的犬。左前肢 **(A)** 及右前肢 **(B)** 的背 - 掌照 (dorsopalmar)，可見左前臂腕骨關節與右側腕掌關節半脫位。而每隻手掌的遠端指間關節也有半脫位，並有無數個軟骨下骨侵蝕 (lytic) 病灶 (白色箭號)。

退化性關節疾病（degenerative joint disease :DJD）

骨關節病(Osteoarthritis: OA)

- 因為軟骨緩衝的機能降低，外力在軟骨下骨造成直接的壓力
- 由於對骨給予壓力的原因，造成骨硬化及骨變形
- 影響平滑的關節運動(ROM下降)
- 這些病變將會造成關節炎並導致疼痛

退化性關節疾病
（髖關節發育不良）

髖關節發育不良所造成的退化性髖關節



正常髖關節





DJD的病理生理學

- 原發性：年齡，肥胖，每日關節負荷（軟骨損傷）
- 其他關節疾病造成的軟骨損傷（繼發性）
- 一旦發生關節軟骨損傷，就無法再生（纖維軟骨）
- 關節液成份的改變導致軟骨損傷的惡性循環
- 所有關節疾病皆為DJD成因

Shoulder DJD

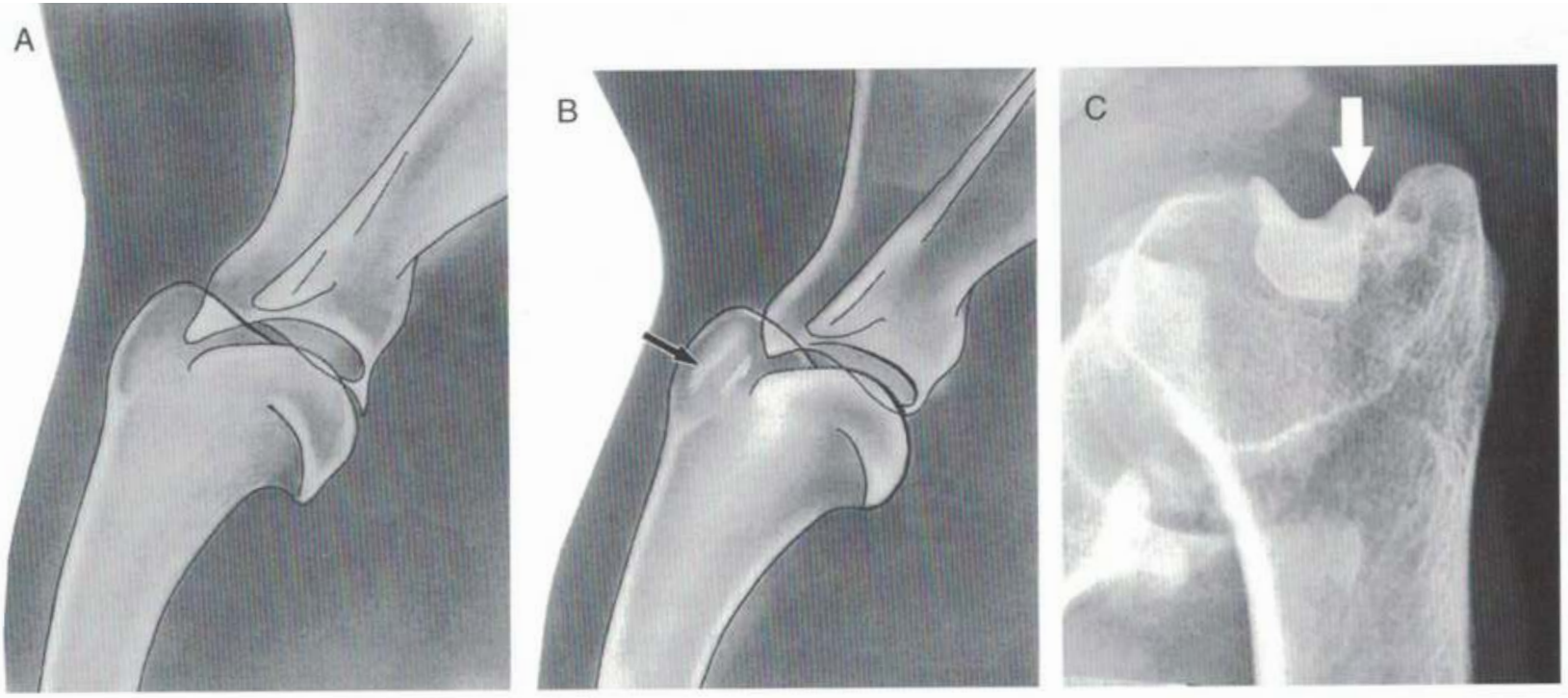


Figure 4.118 Shoulder degenerative joint disease (lateral radiograph). **A.** Osteophytes on caudal articular margins of humeral head and scapular glenoid. **B** and **C.** Osteophytes along bicipital groove (lateral and skyline radiographs). Bicipital groove articular cartilage is highlighted in **C** skyline view with arrow pointing to osteophytes. (A and B adapted from Owens and Biery 1999).

Hip DJD

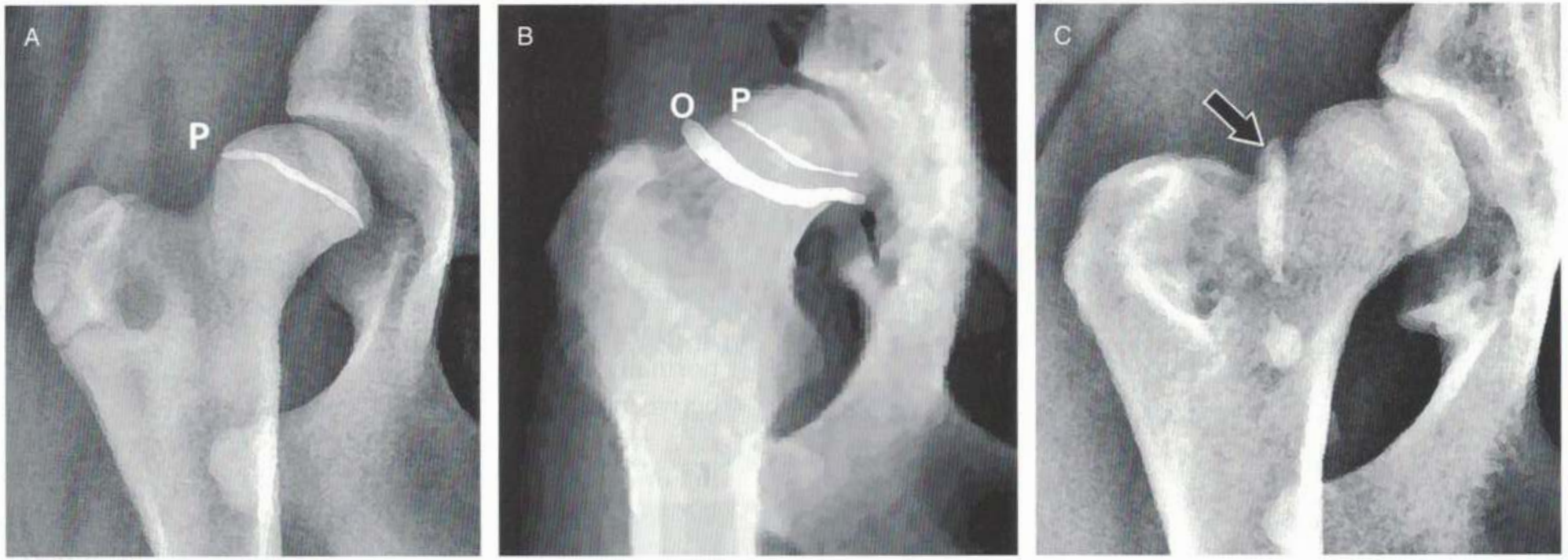


Figure 4.139 Degenerative joint disease in hips (ventrodorsal radiograph). **A.** Coxofemoral subluxation without visible osteophytes or enthesophytes; physeal scar (P). **B.** Osteophytes **(O)** on femoral neck; physeal scar **(P)**. **C.** Enthesophytes at attachment of joint capsule (arrow).

Stifle DJD

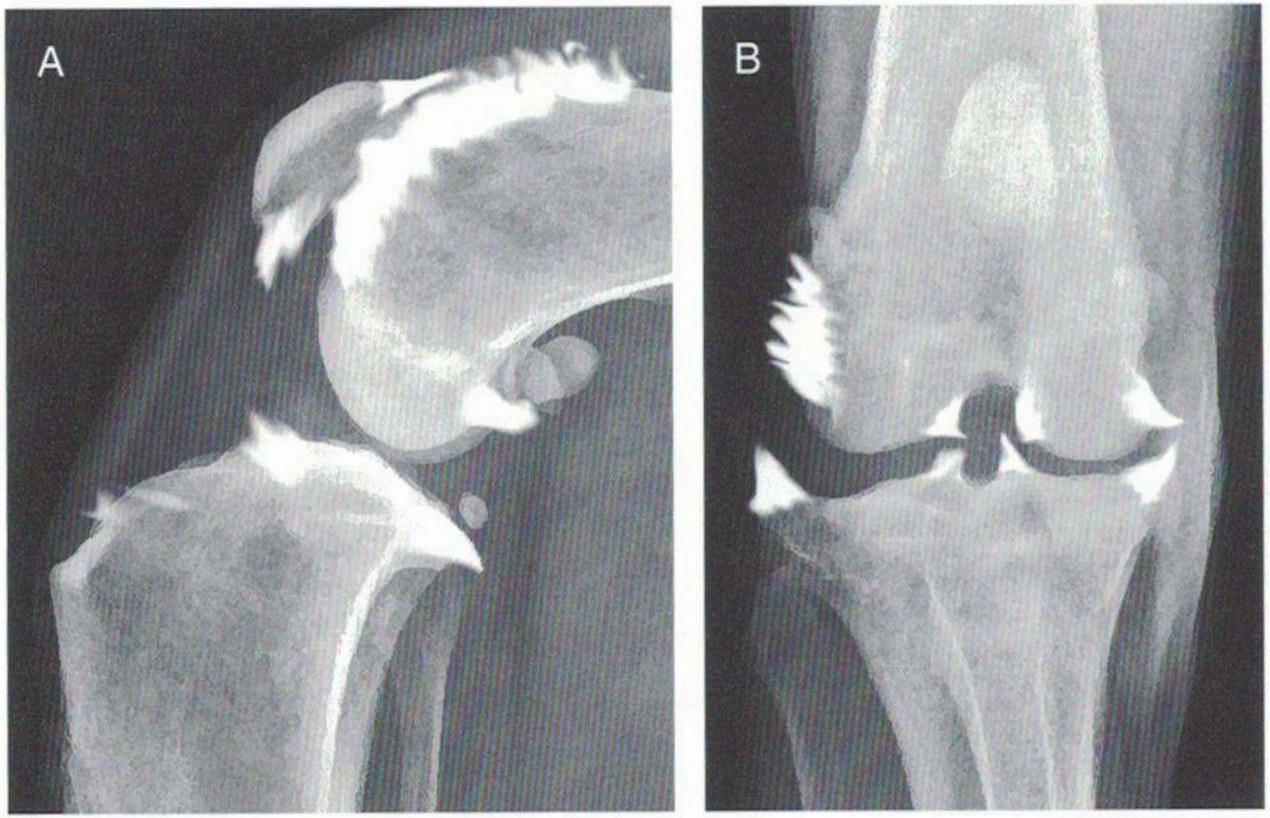


Figure 4.154 Stifle degenerative joint disease. Locations of osteophytes and enthesophytes are highlighted in white. **A.** Lateral radiograph. **B.** Craniocaudal radiograph.

DJD的 X光影像

關節骨變形，不平整的關節骨表面，骨贅形成，軟骨下骨硬化





DJD的X光影像

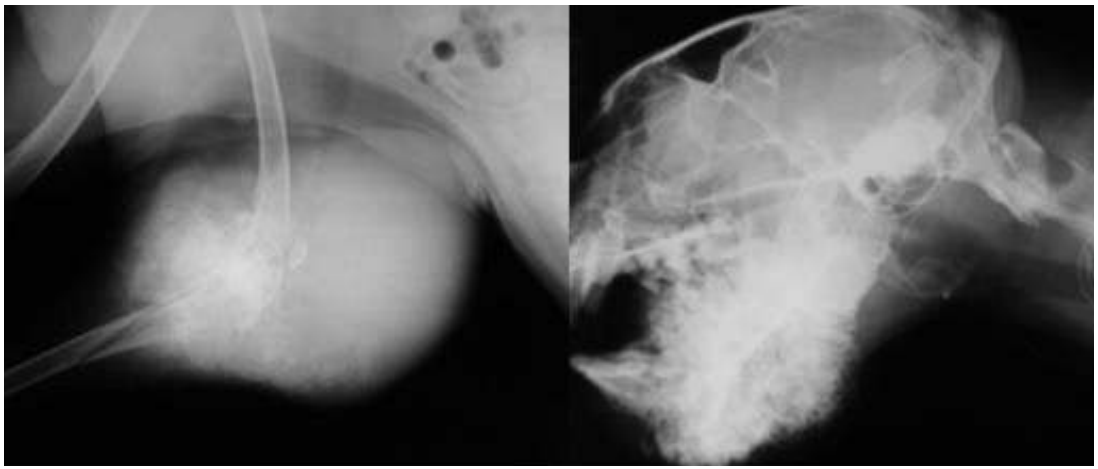
關節周圍骨硬化（有額外的力量作用在關節面，造成骨硬化）

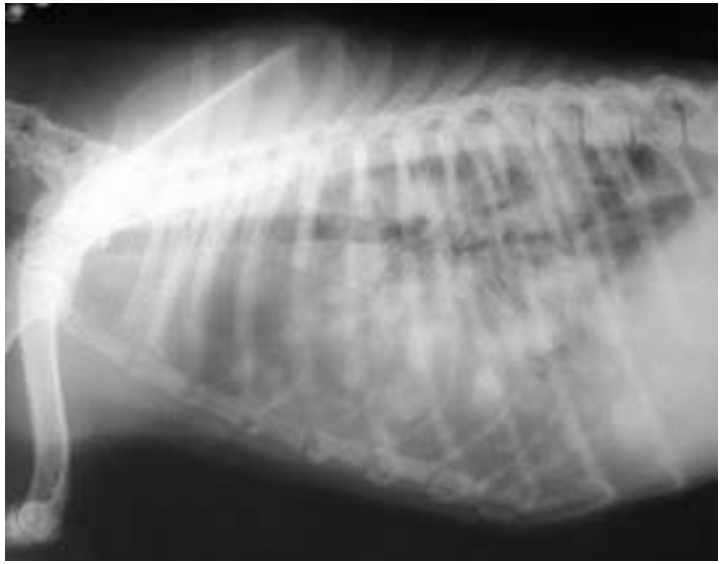




OSTEOSARCOMA AND CHONDROSARCOMA

- **Osteosarcoma**:
  - 長骨末端好發，**close to the stifle, away from elbow**。
  - 50% show sunburst
  - 很少越過關節
  - 大型狗較多
  - 小型狗發病位置較不一定
  - 轉移：肺、肝、腎等
- **Chondrosarcoma**
  - 第二常見之原發骨腫瘤，常為大範圍的溶解
  - 常見於肋骨，有時也發生於scapula, pelvis, skull
  - 發展慢，不易轉移，切除後癒後好。





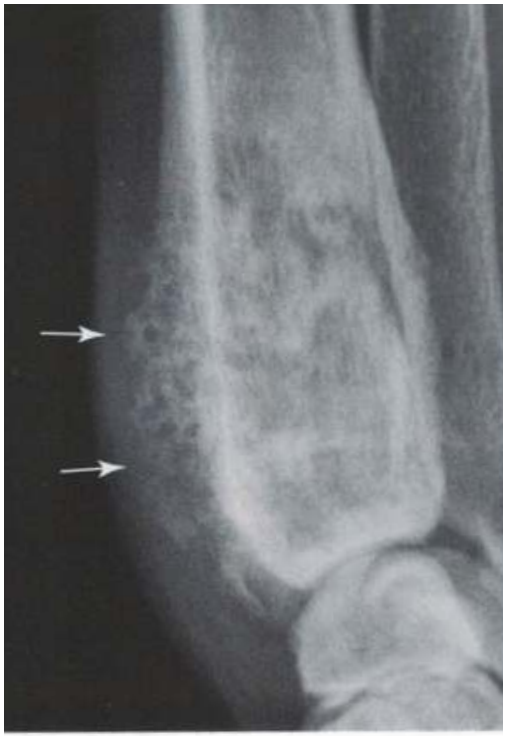
OSTEOMYELITIS

- 早期：骨結構變得hazy (模糊、朦朧)，trabecular 外形變粗糙
- 中期：骨吸收 同時有限度的新骨生成，如骨膜被頂起，也會有一些膜下造骨
- 在異物如骨板或死骨(sequestrum)周圍會形成骨吸收區，這與牙根感染的周圍骨反應相似。
  - Involucrum：在死骨片周圍區域較黑的影像
- Sclerotic margin：病灶處可能會發展來限制感染，常見於慢性病程。
- **惡性腫瘤不容易和骨髓炎區分**。
  - Loss of the normal trabecular pattern:
  - Lysis or destruction
  - Periosteal reaction：感染後7-14天漸漸明顯，也會有新骨生成、Codman's triangle生成。
  - Cortical destruction：感染的地方皮質變薄，漸漸被侵蝕。
- 與骨腫瘤的不同處是：
  - 骨新生較不嚴重(因其基本上是destructive process)
  - 發生在任何地方
  - 會跨過關節
  - 正常與不正常的骨中間界限較明顯，骨髓炎時常形成sclerotic barrier





尺骨具有非侵略性病變的犬，其前臂的



遠端橈骨患有硬骨肉瘤的犬，其前臂遠

表 17-1　小動物四肢骨腫瘤和感染性骨疾病的共同特點 *

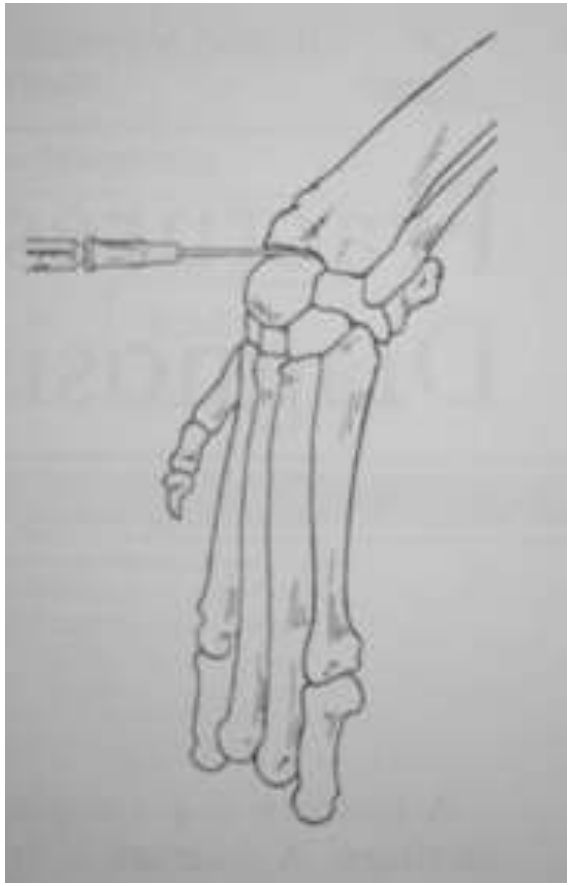
特徵	原發性骨腫瘤	轉移性骨腫瘤	黴菌性骨髓炎	細菌性骨髓炎（血行性）	細菌性骨髓炎（接種）
年齡	年輕或年老	年老	年輕	年輕	年輕或年老
單一骨或多骨	單一骨	多骨 †	多骨 [2]	多骨	單一骨或鄰近的骨
於骨骼中的位置	幹骺端 ‡	幹骺端或骨幹	幹骺端	幹骺端	任何地方
柱狀骨膜反應的可能性	+	–	–	–	+++

* 表列特點只是準則，在各別病畜的結果可能會有所不同。特徵也只根據筆者的經驗，尚未經實證。
† 中軸骨也有可能發生。
‡ 幹骺端和骨幹的腫瘤也會發生。

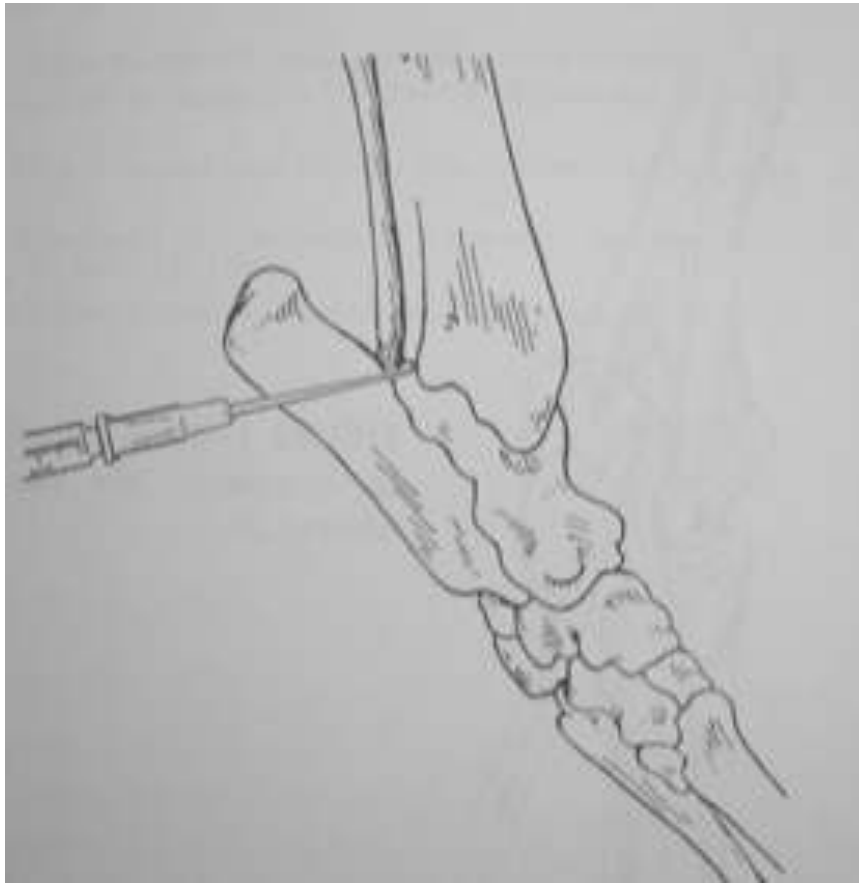
關節液檢查

- 要在無菌的狀態下穿刺（用針刺）（預防感染）
- 若抽取出許多關節液代表關節內積存許多關節液
  - 通常是非常少的
  - 黏稠度如何？觸摸時可以拉出絲的感覺
  - 混濁或清澈？
  - 白血球是否增加？
  - 必要時要做細菌檢查及細胞檢查

ARTHROCENTESIS







Bicipital tenosynovitis

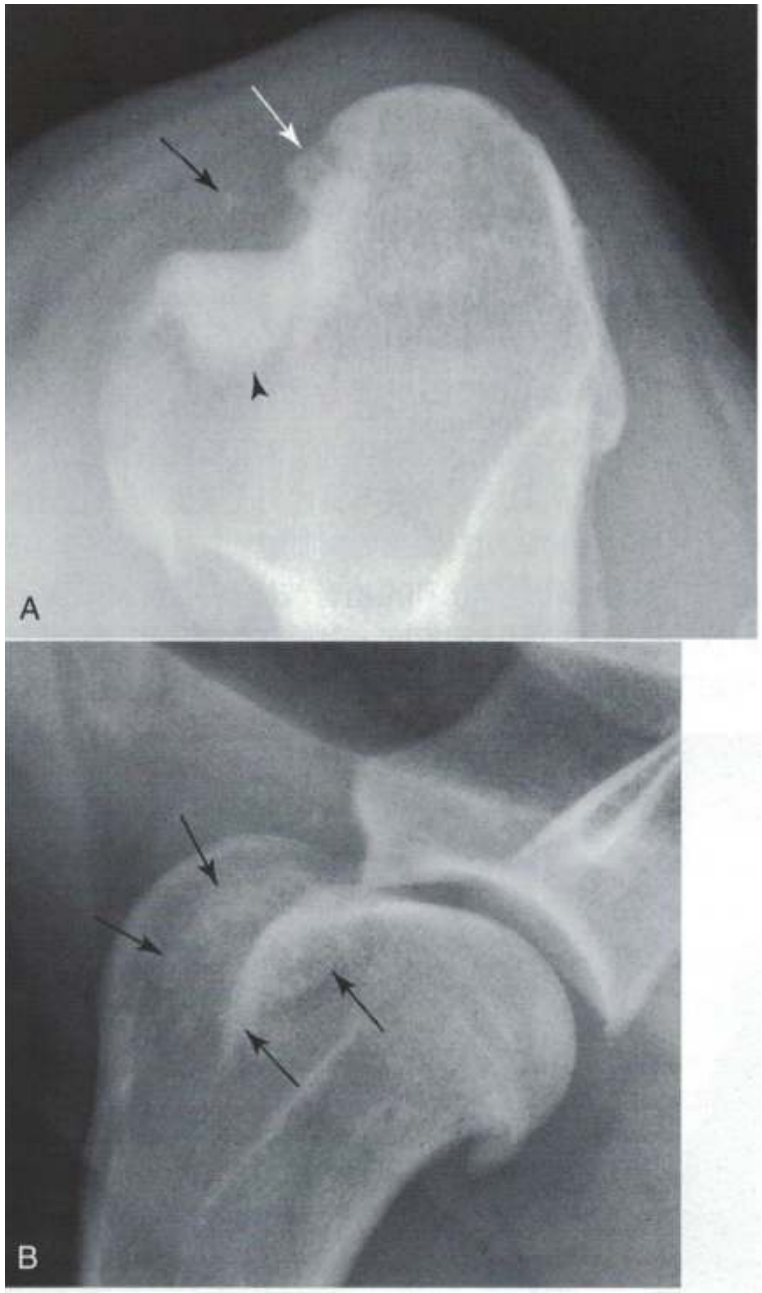


圖 18-41 二頭肌肌腱炎。**B**，增生的新骨與二頭肌溝重疊（黑色箭號），並沿著肱骨大結節的邊緣生長。同時在肱骨頭的後側也有大型的骨贅。**A**，肱骨近端的前近 - 前遠照 crainoproximal-crainodistal view)。在二頭肌溝可見骨贅（白色箭號）、溝的軟骨下硬化（黑色箭頭），及二頭肌肌腱礦化點（黑色箭號）。



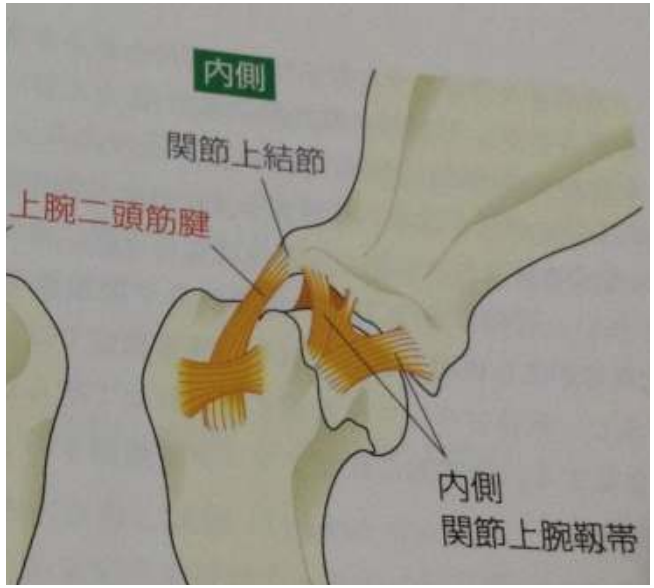
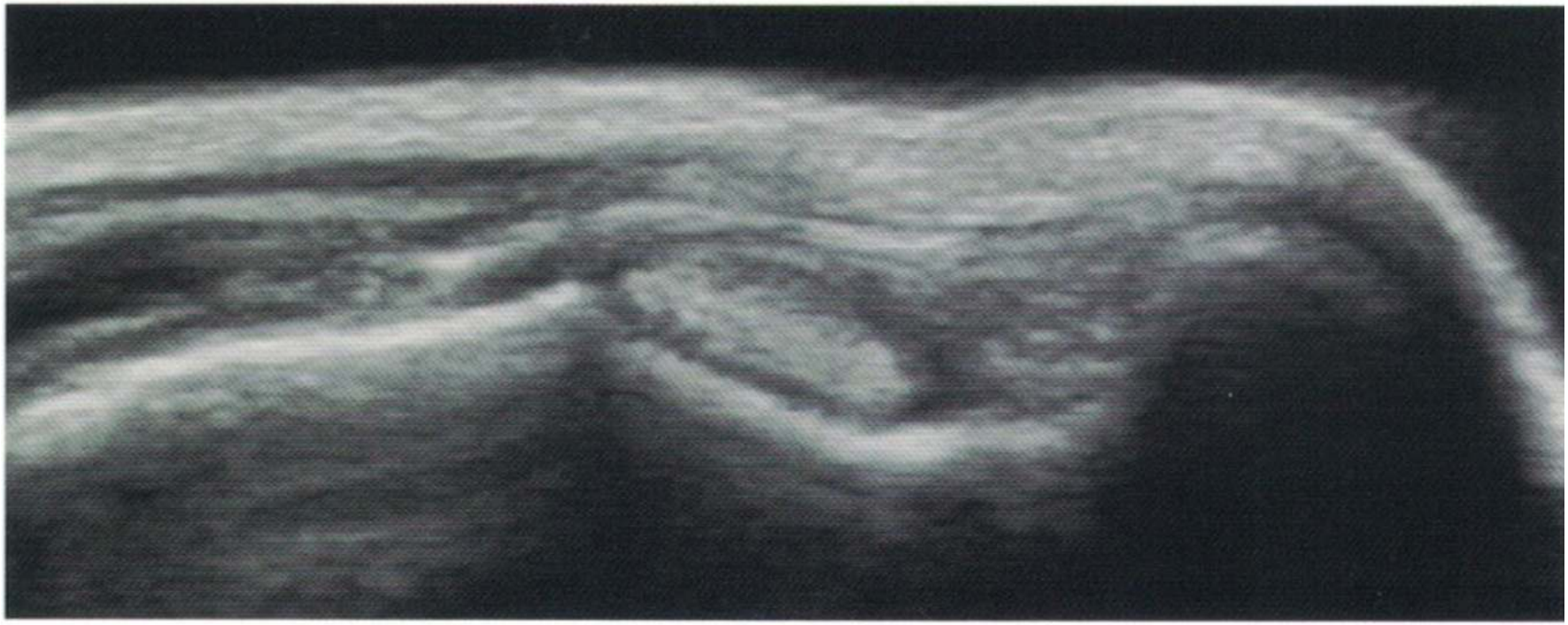




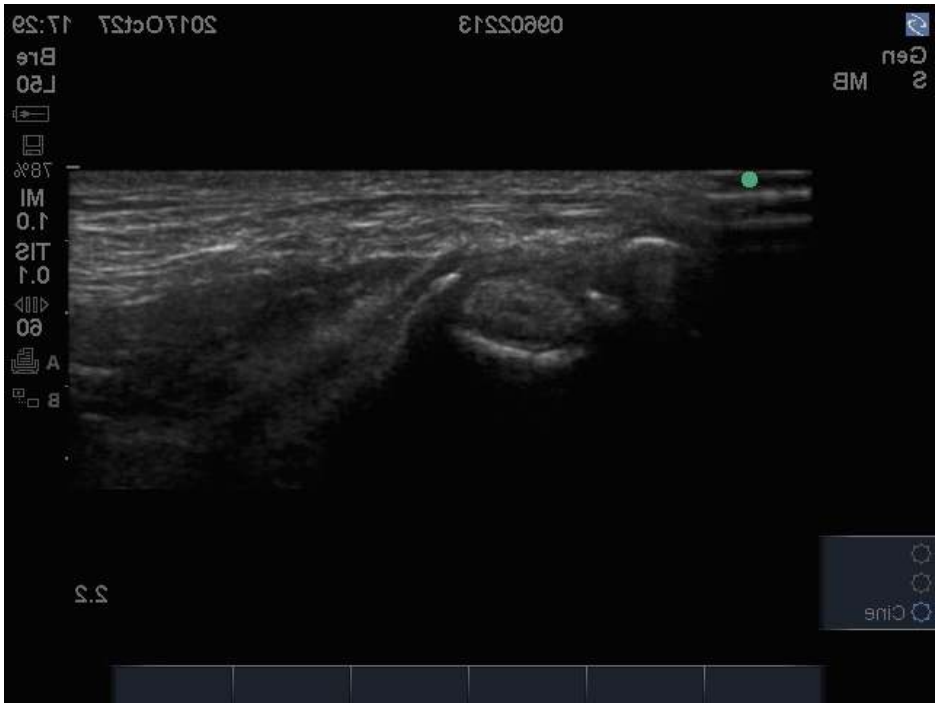


図 10　プローブの当てる

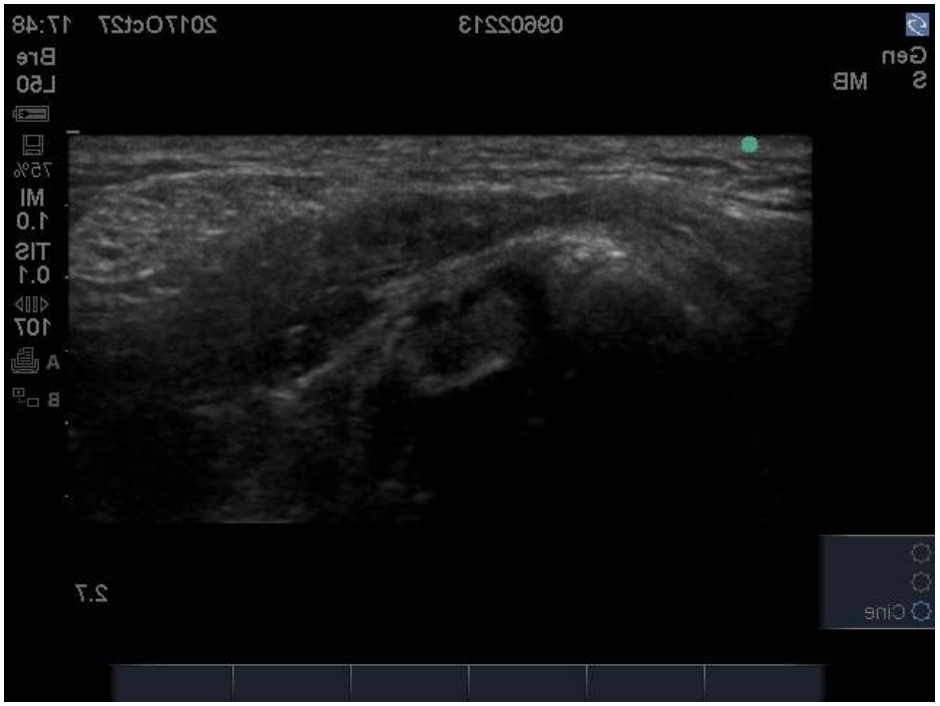


Bicipital tenosynovitis

Left



Right



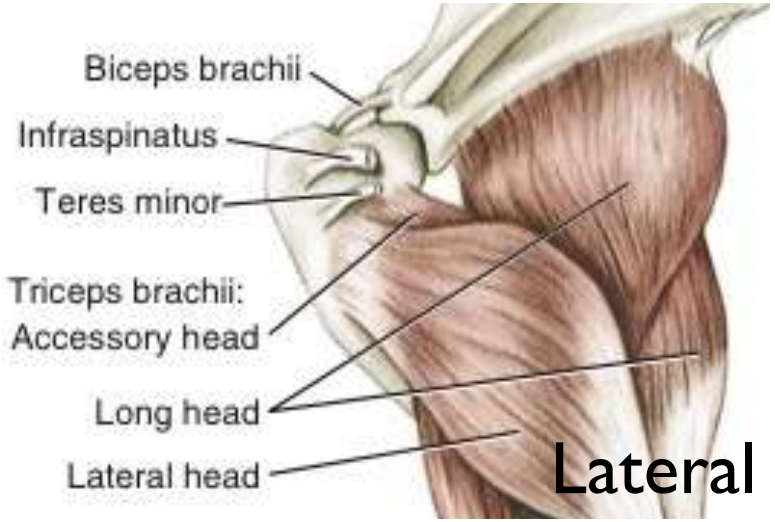
Right bicep tendon



CALCIFICATION OF SUPRASPINATUS TENDON



FIGURE 9–16. Calcification of the supraspinatus tendon. This calcification was asymptomatic in this 2-year-old Rottweiler. Its symptomatic calcified supraspinatus tendon on the opposite side was not as apparent radiographically. Note the numerous pockets of mineralized material located on and superficial to the greater tubercle of the humerus.

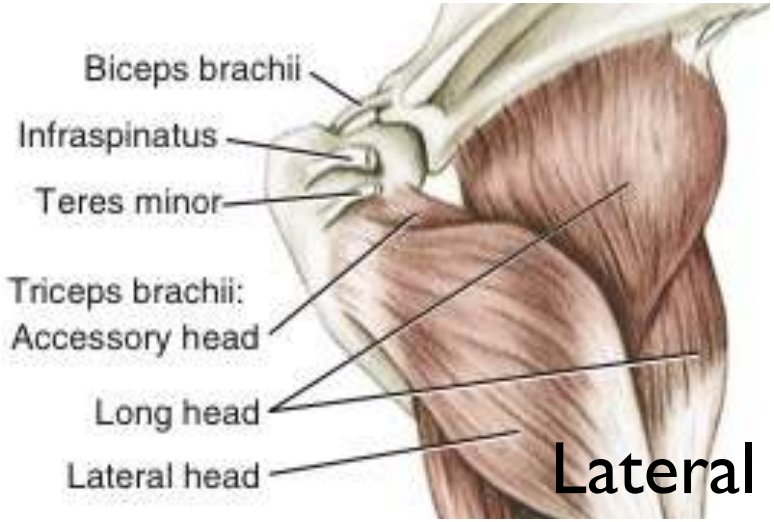




CONTRACTURE OF INFRASPINATUS OR SUPRASPINATUS MUSCLE



Lateral

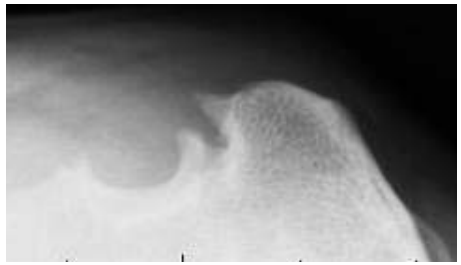


Lateral

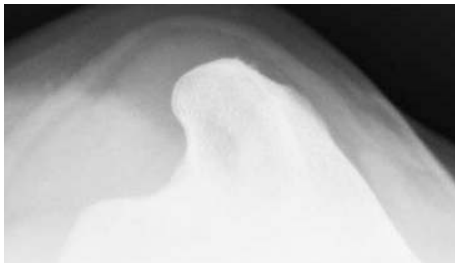


Medial

Left



Right



Left





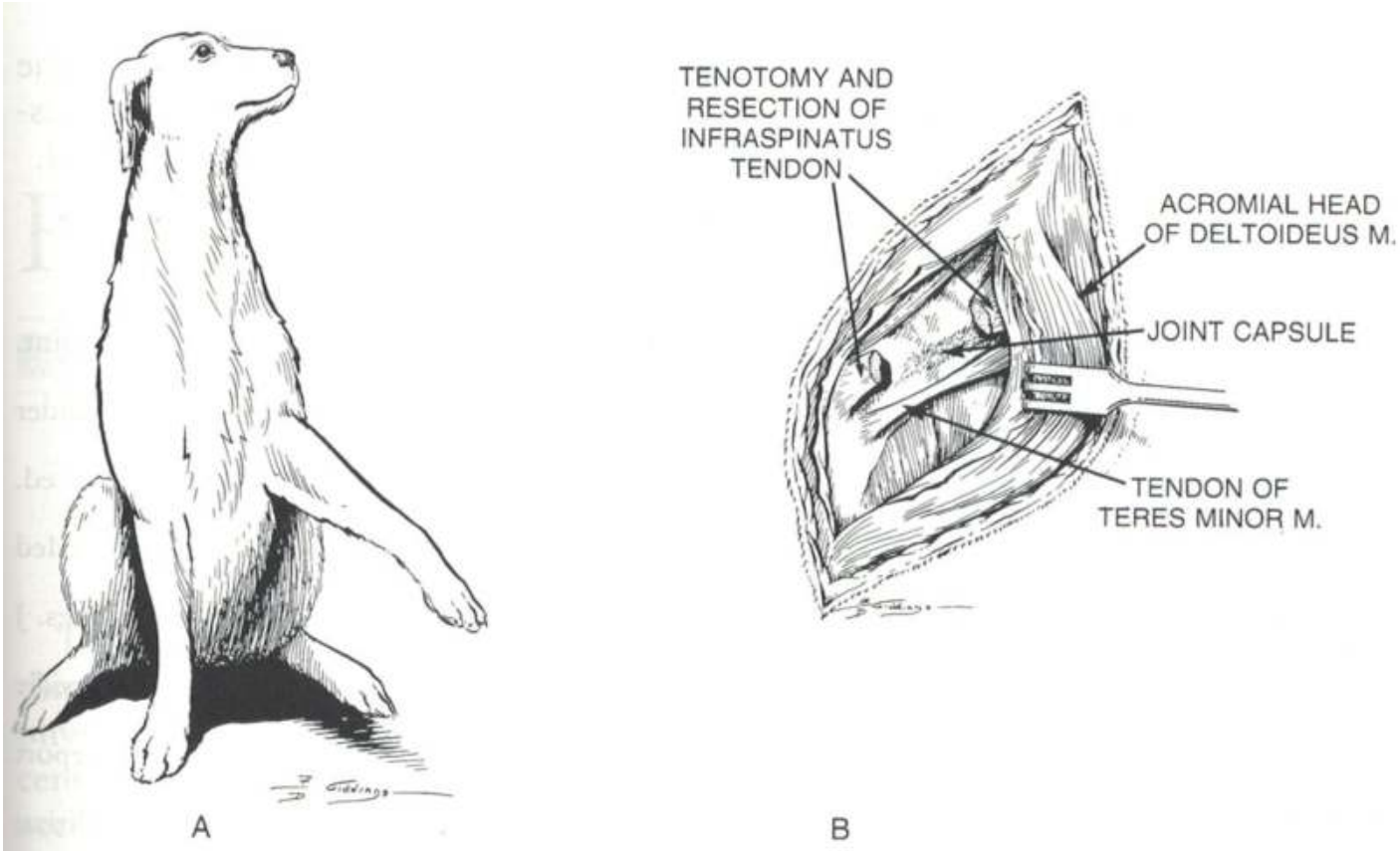


FIGURE 9–27. Fibrotic contracture of the infraspinatus muscle. (*A*) Typical sitting posture of a dog with infraspinatus contracture. The lower limb is permanently externally rotated and therefore shortened. (*B*) The left shoulder has been exposed by the craniolateral approach by tenotomy of the infraspinatus muscle.[3] The tendon is dissected free of the joint capsule until the shoulder moves freely; then about 1 cm of the tendon is excised.





